

APPENDIX I

11,973,181

MULTILAYER NANOPOROUS SEPARATOR

Transaction History

Date	Transaction Description
12-06-2021	Information Disclosure Statement (IDS) Filed
12-06-2021	Patent Term Adjustment - Ready for Examination
12-06-2021	PTO-SB-69-Authorize EPO Access to Search Results
12-06-2021	Applicants have given acceptable permission for participating foreign
12-06-2021	Information Disclosure Statement (IDS) Filed
12-06-2021	Entity Status Set To Undiscounted (Initial Default Setting or Status Change)
12-06-2021	Initial Exam Team nn
12-15-2021	Case Docketed to Examiner in GAU
12-15-2021	Application Dispatched from OIPE
12-15-2021	FITF set to YES - revise initial setting
12-15-2021	Application Is Now Complete
12-15-2021	Filing Receipt
12-16-2021	Application ready for PDX access by participating foreign offices
03-24-2022	PG-Pub Issue Notification
05-19-2022	Information Disclosure Statement (IDS) Filed
05-19-2022	Information Disclosure Statement (IDS) Filed
06-21-2022	Preliminary Amendment
07-20-2022	Filing Receipt - Corrected
07-20-2022	Change in Power of Attorney (May Include Associate POA)
07-21-2022	Filing Receipt - Corrected
09-01-2022	Preliminary Amendment
10-22-2022	Case Docketed to Examiner in GAU
02-24-2023	Restriction-Election Requirement
02-28-2023	Mail Restriction Requirement
04-28-2023	Electronic Information Disclosure Statement
04-28-2023	Response to Election - Restriction Filed
04-28-2023	Information Disclosure Statement (IDS) Filed
05-08-2023	Date Forwarded to Examiner
06-14-2023	Non-Final Rejection
06-14-2023	Information Disclosure Statement considered
06-14-2023	Information Disclosure Statement considered
06-14-2023	Information Disclosure Statement considered
06-15-2023	Mail Non-Final Rejection
10-13-2023	Electronic Information Disclosure Statement
10-13-2023	Electronic Information Disclosure Statement
10-13-2023	Response after Non-Final Action
10-13-2023	Request for Extension of Time - Granted
10-13-2023	Information Disclosure Statement (IDS) Filed
10-17-2023	Date Forwarded to Examiner
10-18-2023	Information Disclosure Statement considered
12-11-2023	Interview Summary - Examiner Initiated - Telephonic
12-27-2023	Notice of Allowance Data Verification Completed
12-28-2023	Mail Notice of Allowance
03-26-2024	Issue Fee Payment Verified
03-26-2024	Issue Fee Payment Received
03-27-2024	Dispatch to FDC
03-27-2024	Application Is Considered Ready for Issue
04-10-2024	Issue Notification Mailed
04-30-2024	Recordation of Patent Grant Mailed
04-30-2024	Patent eGrant Notification
04-30-2024	Recordation of Patent eGrant
04-30-2024	Patent Issue Date Used in PTA Calculation
05-02-2024	Mail Patent eGrant Notification

Issue Classification 	Application/Control No. 17/543,259	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

CPC						
Symbol					Type	Version
H01M	/	10	/	0525	F	2013-01-01
H01M	/	50	/	446	I	2021-01-01
H01M	/	50	/	491	I	2021-01-01
H01M	/	50	/	489	I	2021-01-01
H01M	/	50	/	417	I	2021-01-01
H01M	/	50	/	451	I	2021-01-01
H01M	/	50	/	457	I	2021-01-01

CPC Combination Sets							
Symbol				Type	Set	Ranking	Version
	/		/				

NONE				Total Claims Allowed:	
(Assistant Examiner)		(Date)		12	
/LISA S PARK/		12 December 2023		O.G. Print Claim(s)	O.G. Print Figure
Primary Examiner, Art Unit 1729		(Date)		21	5
(Primary Examiner)					

<i>Issue Classification</i> 	Application/Control No. 17/543,259	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

INTERNATIONAL CLASSIFICATION				
CLAIMED				
H01M	/	50	/	446
H01M	/	10	/	052
H01M	/	50	/	417
H01M	/	50	/	491
H01M	/	50	/	489
H01M	/	50	/	451
H01M	/	10	/	0525
H01M	/	50	/	457
NON-CLAIMED				
	/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	12	
/LISA S PARK/ Primary Examiner, Art Unit 1729 (Primary Examiner)	12 December 2023 (Date)	O.G. Print Claim(s) 21	O.G. Print Figure 5

<i>Issue Classification</i> 	Application/Control No. 17/543,259	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	21	10	30												
2	22	11	42												
3	23	12	43												
4	24														
5	25														
6	26														
7	27														
8	28														
9	29														

NONE	Total Claims Allowed:	
(Assistant Examiner)	(Date)	12
/LISA S PARK/ Primary Examiner, Art Unit 1729 (Primary Examiner)	12 December 2023 (Date)	O.G. Print Claim(s) 21
		O.G. Print Figure 5

<i>Index of Claims</i> 	Application/Control No.	Applicant(s)/Patent Under Reexamination
	17/543,259	Avison et al.
	Examiner	Art Unit
	LISA S PARK	1729

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	06/13/2023								
	21	✓								
	22	✓								
	23	✓								
	24	✓								
	25	✓								
	26	✓								
	27	✓								
	28	✓								
	29	✓								
	30	✓								
	31	N								
	32	N								
	33	N								
	34	N								
	35	N								
	36	N								
	37	N								
	38	N								
	39	N								
	40	N								
	41	N								

<i>Index of Claims</i> 	Application/Control No. 17/543,259	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	06/13/2023	12/17/2023							
	21	✓	=							
	22	✓	=							
	23	✓	=							
	24	✓	=							
	25	✓	=							
	26	✓	=							
	27	✓	=							
	28	✓	=							
	29	✓	=							
	30	✓	=							
	31	N	-							
	32	N	-							
	33	N	-							
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	35	N	-							
	36	N	-							
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	38	N	-							
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	40	N	-							
	41	N	-							
	42		=							
	43		=							

Search Notes 	Application/Control No. 17/543,259	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M10/0525 or H01M50/417 or H01M50/446 or H01M50/451 or H01M50/457 or H01M50/489 or H01M50/491 or Y02E60/10 or H01M10/052 or H01M2300/0025	06/13/2023	/L.S.P./

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

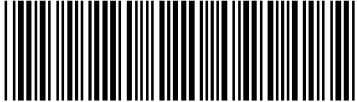
US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PE2E keyword search (see attached search history)	06/13/2023	
PE2E inventor/assignee search	06/14/2023	
PE2E foreign database search	06/14/2023	
Searched prior art in related applications 16755294, 17223845		

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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Search Notes 	Application/Control No. 17/543,259	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M10/0525 or H01M50/417 or H01M50/446 or H01M50/451 or H01M50/457 or H01M50/489 or H01M50/491 or Y02E60/10 or H01M10/052 or H01M2300/0025	06/13/2023, 12/11/2023	/L.S.P./

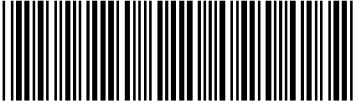
CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PE2E keyword search (see attached search history)	06/13/2023, 12/11/2023	/L.S.P./
PE2E inventor/assignee search	06/14/2023	/L.S.P./
PE2E foreign database search	06/14/2023	/L.S.P./
Searched prior art in related applications 16755294, 17223845	6/14/2023, 12/12/2023	/L.S.P./

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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Search Notes 	Application/Control No. 17/543,259	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H01M10	0525, 052	12/12/2023	/L.S.P./
H01M50	417, 446, 451, 457, 489, 491	12/12/2023	/L.S.P./
Y02E60	10	12/12/2023	/L.S.P./
H01M2300	0025	12/12/2023	/L.S.P./

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	"17543259"	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/13 04:46 PM
L2	1692170	(H01M10/0525 or H01M50/417 or H01M50/446 or H01M50/451 or H01M50/457 or H01M50/489 or H01M50/491 or Y02E60/10 or H01M10/052 or H01M2300/0025).cpc.	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/06/13 04:49 PM
L3	2713	2 AND separator WITH (multilayer ((second first plurality) near3 layer) composite) near4 oxide	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/06/13 04:56 PM
L4	207	2 AND separator near3 (multilayer ((second first plurality) near3 layer) composite) near4 oxide	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/06/13 04:56 PM
L5	0	2 AND separator near3 (multilayer ((second first plurality) near3 layer) composite) near4 beohmite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/06/13 04:58 PM
L6	0	2 AND separator WITH (multilayer ((second first plurality) WITH layer) composite) near4 beohmite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/06/13 04:58 PM
L7	1	2 AND separator WITH (multilayer ((second first plurality) WITH layer) composite) AND beohmite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/06/13 04:58 PM
L9	9	("20160190535" "20170077476" "20190260001" "20110143183" "8405957" "8822082" "8883354" "9166250" "9166251").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/13 05:01 PM
L10	1	("20090067119").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/13 05:02 PM
L12	27	("20110003209" "20110165450" "20160104876" "20160172705" "20170294638" "20170301903" "20180019457" "201	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/13 05:15 PM

		80248164 "6432586 "6723467 "7638241 "7662517 "7709152 "7883799 "8318360 "9293752 "9312527 "9478784 "9484565 "9543077 "9847519 "20170373295 "20160104876 "20160043370 "4024323 "20070190427 "20110003209 "20110165450 "20150155601 "20170301903 "20170012264").PN.					
L13	21	("20160172705 "20170294638 "20110003209 "20110165450 "20160104876 "20160172705 "20170301903 "6723467 "20180019457 "20180248164 "6432586 "7638241 "7662517 "7709152 "7883799 "8318360 "9293752 "9312527 "9478784 "9484565 "9543077 "9847519").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/13 05:16 PM
L14	27	(12 13)	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/13 05:16 PM
L15	11	14 AND coat\$4 near3 (weight load\$4 amount)	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/13 05:21 PM
L16	27	("20160043370" OR "20170373295" OR "4024323" OR "20170012264" OR "20070190427" OR "20110003209" OR "20110165450" OR "20150155601" OR "20160104876" OR "20160172705" OR "20170294638" OR "20170301903" OR "20180019457" OR "20180248164" OR "6432586" OR "6723467" OR "7638241" OR "7662517" OR "7709152" OR "7883799" OR "8318360" OR "9293752" OR "9312527" OR "9478784" OR "9484565" OR "9543077" OR	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/14 05:28 AM

L17	14	"9847519").pn. 16 AND boehmite	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/14 05:29 AM
L18	659	((("LG") near3 ("ENERGY") near3 ("SOLUTION") near3 ("LTD"))).AS,AANM.	(USPAT)	OR	ON	ON	2023/06/14 06:06 AM
L19	9	18 AND boehmite	(USPAT)	OR	ON	ON	2023/06/14 06:06 AM
L20	449	((("AVISON") near3 ("David")) OR (("CARLSON") near3 ("Steven")) OR (("SLOAN") near3 ("Benjamin"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/06/14 06:09 AM
L21	100	20 AND boehmite	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/06/14 06:10 AM
L23	12	21 AND hydrophobic	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/06/14 06:10 AM
L24	5	23 AND coating ADJ weight	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/06/14 06:11 AM
L25	2	("20090067119" "20110003209").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/14 06:18 AM
L26	1	("20100015530").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/06/14 12:00 PM
L27	8088	(h01m50/446.ipc. h01m10/052.ipc. h01m50/417.ipc. h01m50/417.ipc. h01m50/491.ipc. h01m10/0525.ipc. h01m50/489.ipc. h01m50/451.ipc. h01m50/457.ipc.)	(JPO)	OR	ON	ON	2023/06/14 02:40 PM
L28	0	27 AND boehmite AND hydrophobic	(JPO)	OR	ON	ON	2023/06/14 02:40 PM
L29	1	27 AND boehmite AND separator	(JPO)	OR	ON	ON	2023/06/14 02:40 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L30	3	("20090067119" "20110003209" "20110165450").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/12/11 06:13 AM
L31	4	2 AND separator near3 (multilayer ((second first plurality) near3 layer) composite) AND (resin\$4 polymer\$4) ADJ layer WITH (pore porous) WITH (fill\$4 dispos\$4 within) near3 (g/m.sup.2 (gram near4 meter))	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/12/11 06:20 AM
L32	27	("20110003209" "20110 165450" "20160104876 " "20160172705" "2017 0294638" "2017030190 3" "20180019457" "201 80248164" "6432586" " 6723467" "7638241" "7 662517" "7709152" "78 83799" "8318360" "929 3752" "9312527" "9478 784" "9484565" "95430 77" "9847519" "201703 73295" "20160043370" " 4024323" "201501556 01" "20070190427" "20 110003209" "20110165 450" "20160104876" "2 0170301903" "2017001 2264").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/12/11 07:22 AM
L33	4	2 AND separator AND (resin\$4 polymer\$4) ADJ layer WITH (pore porous) WITH (fill\$4 dispos\$4 within) near3 (g/m.sup.2 (gram near4 meter))	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/12/11 07:26 AM
L34	37	2 AND separator AND (pore porous) WITH (fill\$4 dispos\$4 within) near3 (g/m.sup.2 (gram near4 meter))	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/12/11 07:26 AM

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	0	(H01M10/0525 OR H01M50/417 OR	(US-PGPUB; USPAT)	OR	ON	ON	2023/12/12 08:33 AM

		H01M50/446 OR H01M50/451 OR H01M50/457 OR H01M50/489 OR H01M50/491 OR Y02E60/10 OR H01M10/052 OR H01M2300/0025).cpc. AND (hydrophobic AND inorganice ADJ oxide AND pores AND polymeric ADJ (film layer membrane)) AND separator.clm.					
N2	0	(H01M10/0525 OR H01M50/417 OR H01M50/446 OR H01M50/451 OR H01M50/457 OR H01M50/489 OR H01M50/491 OR Y02E60/10 OR H01M10/052 OR H01M2300/0025).cpc. AND (hydrophobic AND inorganice ADJ oxide AND pores AND polymeric ADJ (film layer membrane) AND separator).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2023/12/12 08:33 AM
N3	0	(H01M10/0525 OR H01M50/417 OR H01M50/446 OR H01M50/451 OR H01M50/457 OR H01M50/489 OR H01M50/491 OR Y02E60/10 OR H01M10/052 OR H01M2300/0025).cpc. AND (hydrophobic AND inorganic ADJ oxide AND pores AND polymeric ADJ (film layer membrane) AND separator).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2023/12/12 08:33 AM
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		oxide AND pores AND polymeric ADJ (film layer membrane) AND separator).clm.					
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(12) **United States Patent**
Avison et al.

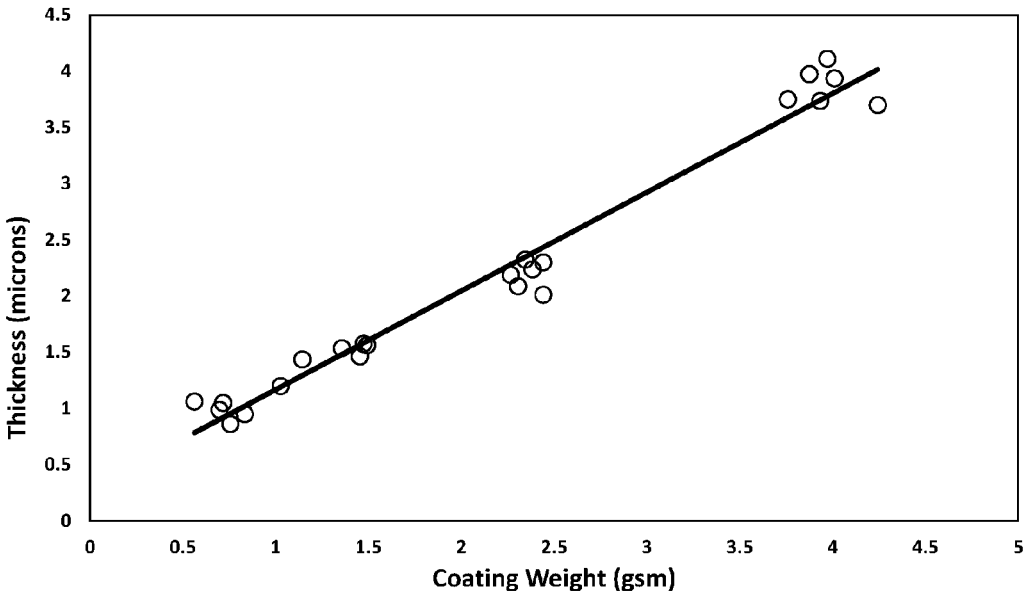
(10) **Patent No.:** **US 11,973,181 B2**
(45) **Date of Patent:** **Apr. 30, 2024**

(54) **MULTILAYER NANOPOROUS SEPARATOR**
(71) Applicant: **LG ENERGY SOLUTION, LTD.**,
Seoul (KR)
(72) Inventors: **David W. Avison**, Boxborough, MA
(US); **Steven A. Carlson**, Cambridge,
MA (US); **Benjamin Sloan**, Exeter, NH
(US)
(73) Assignee: **LG ENERGY SOLUTION, LTD.**,
Seoul (KR)
(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.
(21) Appl. No.: **17/543,259**
(22) Filed: **Dec. 6, 2021**
(65) **Prior Publication Data**
US 2022/0094018 A1 Mar. 24, 2022

(58) **Field of Classification Search**
CPC H01M 10/0525; H01M 50/417; H01M
50/446; H01M 50/451; H01M 50/457;
(Continued)
(56) **References Cited**
U.S. PATENT DOCUMENTS
4,024,323 A 5/1977 Versteegh
6,194,098 B1 * 2/2001 Ying H01M 50/443
429/251
(Continued)
FOREIGN PATENT DOCUMENTS
WO 2016123404 A1 8/2016
OTHER PUBLICATIONS
F. Sola et al., "A new scanning electron microscopy approach to
image aerogels at the nanoscale; A new scanning electron micros-
copy approach to image aerogels at the nanoscale," Nanotechnol-
ogy, IOP, Bristol, GB, vol. 22, No. 17, Mar. 16, 2011, p. 175704,
XP020188888, ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/
175704.
(Continued)

Related U.S. Application Data
(63) Continuation of application No. 17/223,845, filed on
Apr. 6, 2021, which is a continuation of application
(Continued)
(51) **Int. Cl.**
H01M 50/446 (2021.01)
H01M 10/052 (2010.01)
(Continued)
(52) **U.S. Cl.**
CPC **H01M 10/0525** (2013.01); **H01M 50/417**
(2021.01); **H01M 50/446** (2021.01);
(Continued)

Primary Examiner — Lisa S Park
(74) *Attorney, Agent, or Firm* — Amster, Rothstein &
Ebenstein LLP
(57) **ABSTRACT**
A separator for a lithium battery having (a) a porous poly-
meric layer, such as a polyethylene layer; and (b) a nanop-
orous inorganic particle/polymer layer on both sides of the
polymeric layer, the nanoporous layer having an inorganic
oxide and one or more polymers; the volume fraction of the
polymers in the nanoporous layer is about 15% to about
50%, and the crystallite size of the inorganic oxide is 5 nm
to 90 nm.
12 Claims, 6 Drawing Sheets



Related U.S. Application Data			9,312,527 B2	4/2016	Muramatsu et al.
No. 16/755,294, filed as application No. PCT/US2018/055862 on Oct. 15, 2018, now Pat. No. 10,998,536.			9,478,784 B2	10/2016	Iwase
			9,484,565 B2	11/2016	Miyazaki et al.
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Figure 1

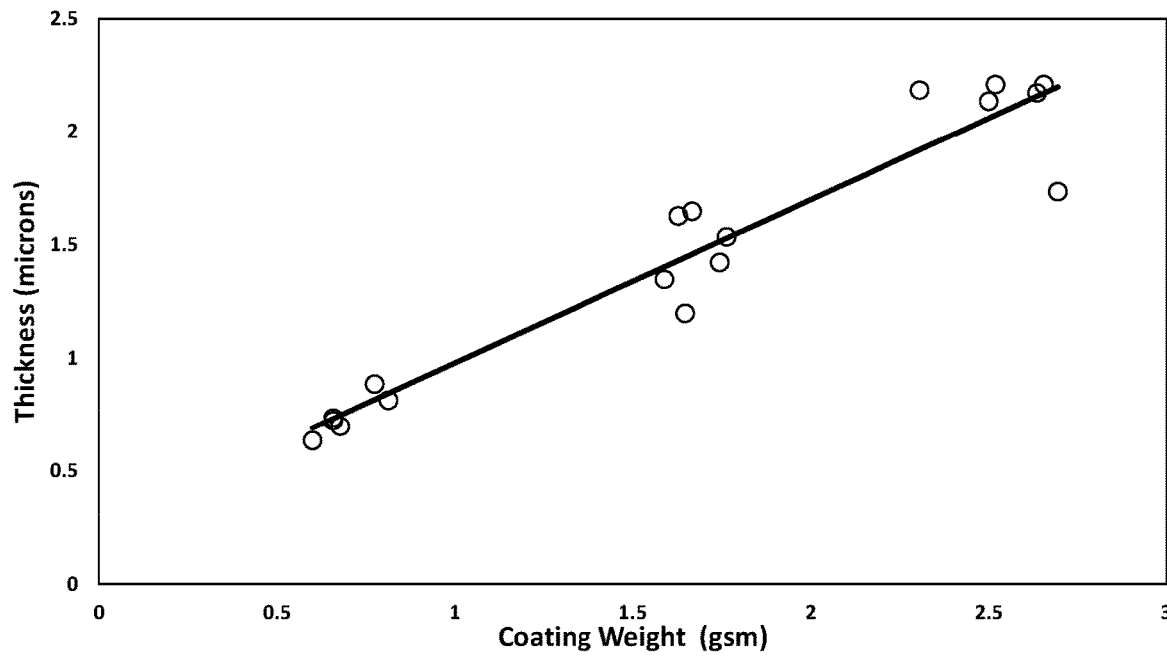


Figure 2

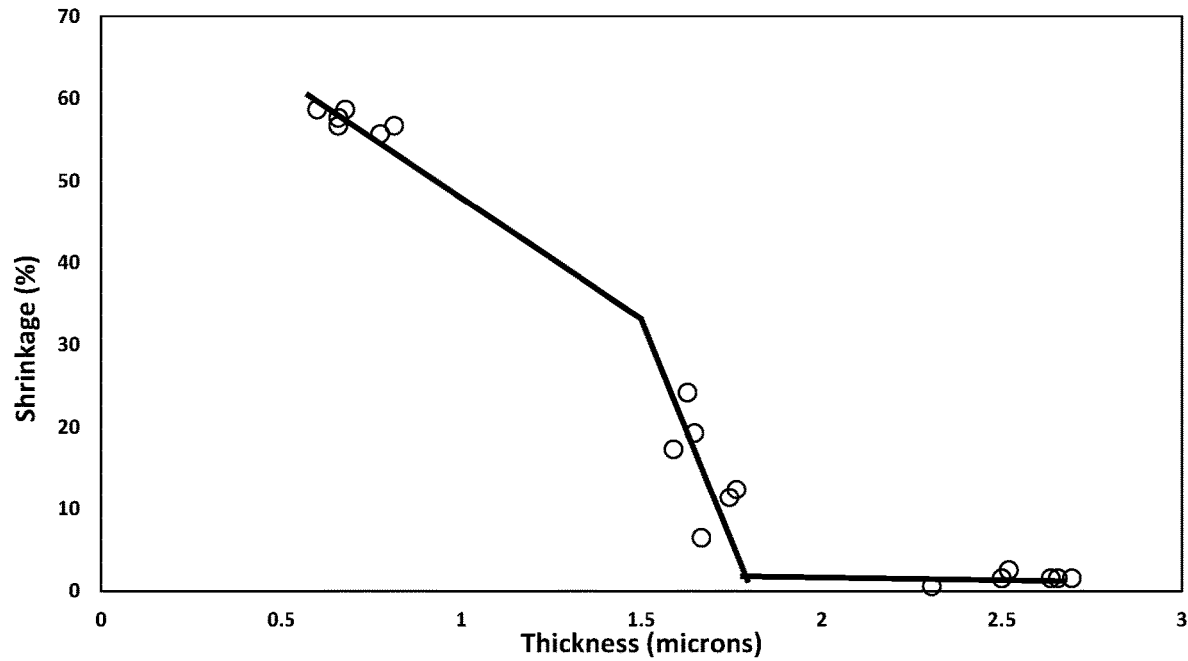


Figure 3

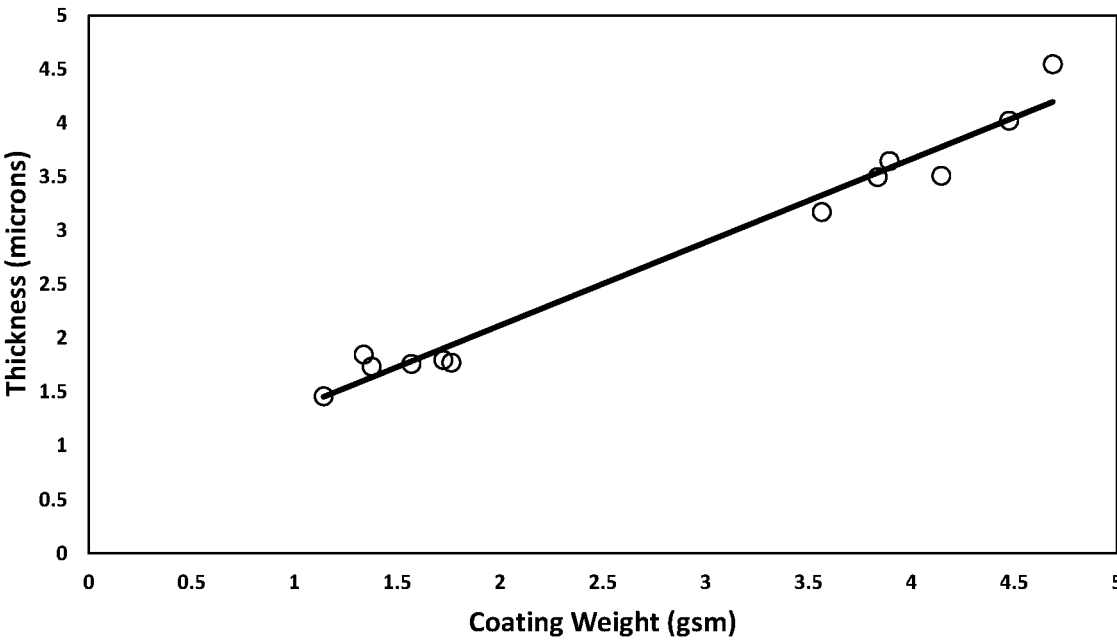


Figure 4

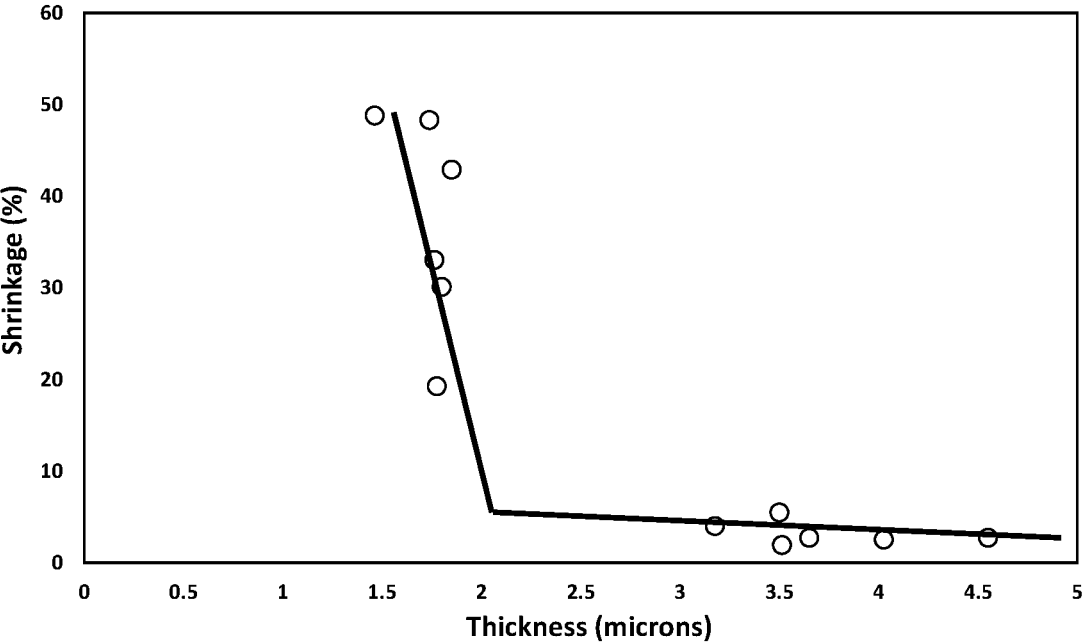


Figure 5

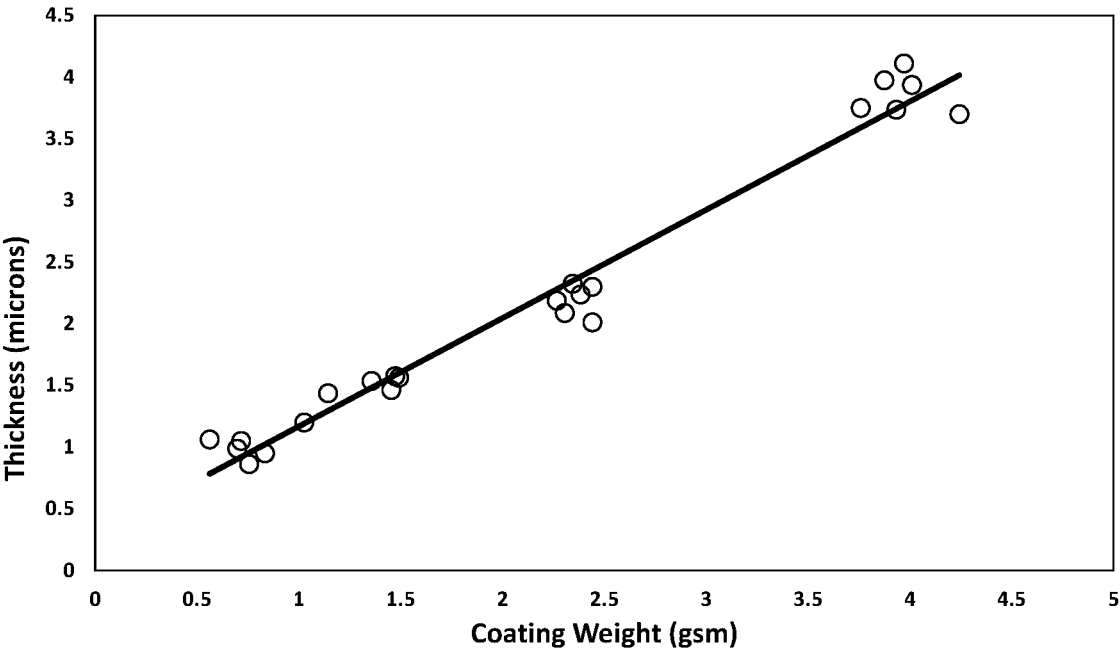


Figure 6



Figure 7

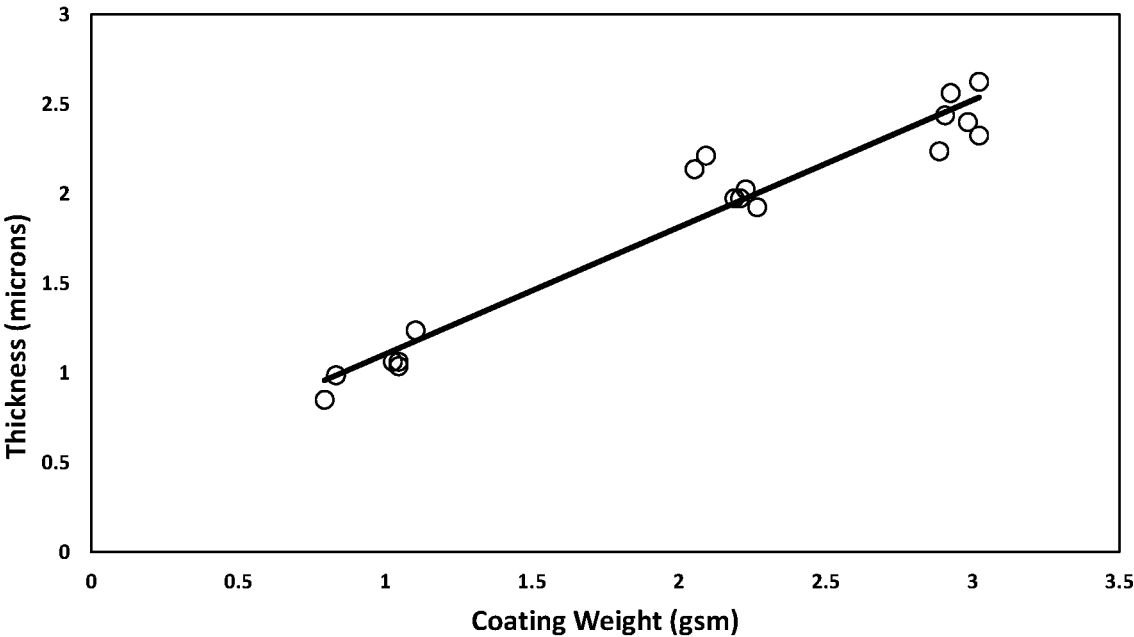


Figure 8

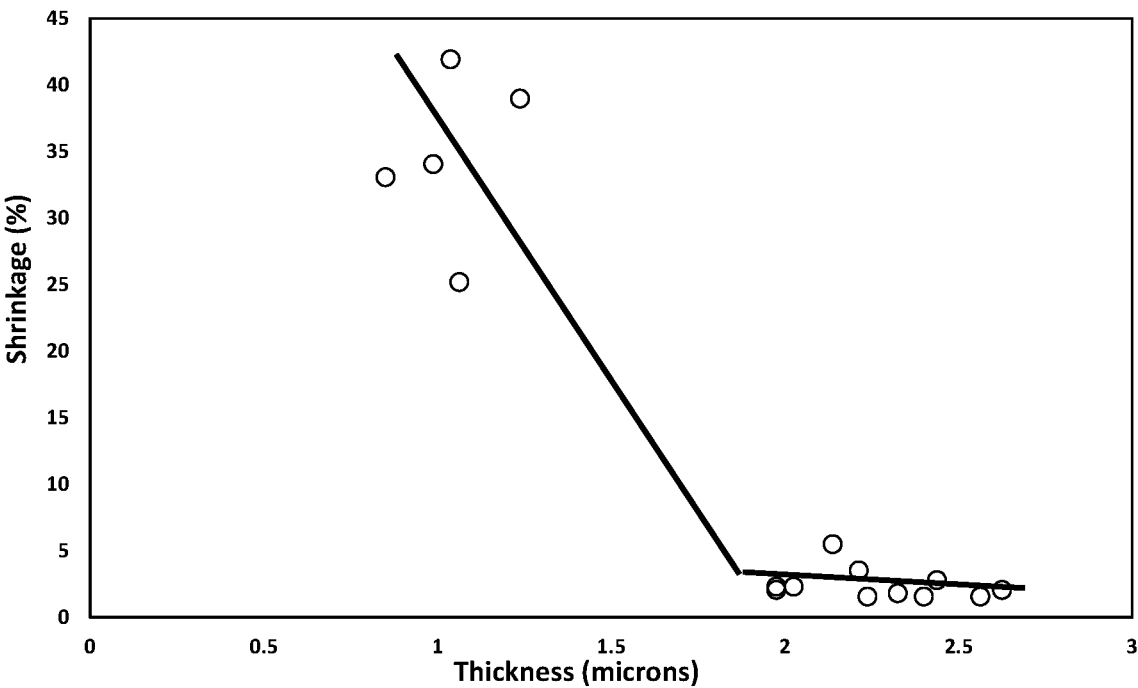


Figure 9

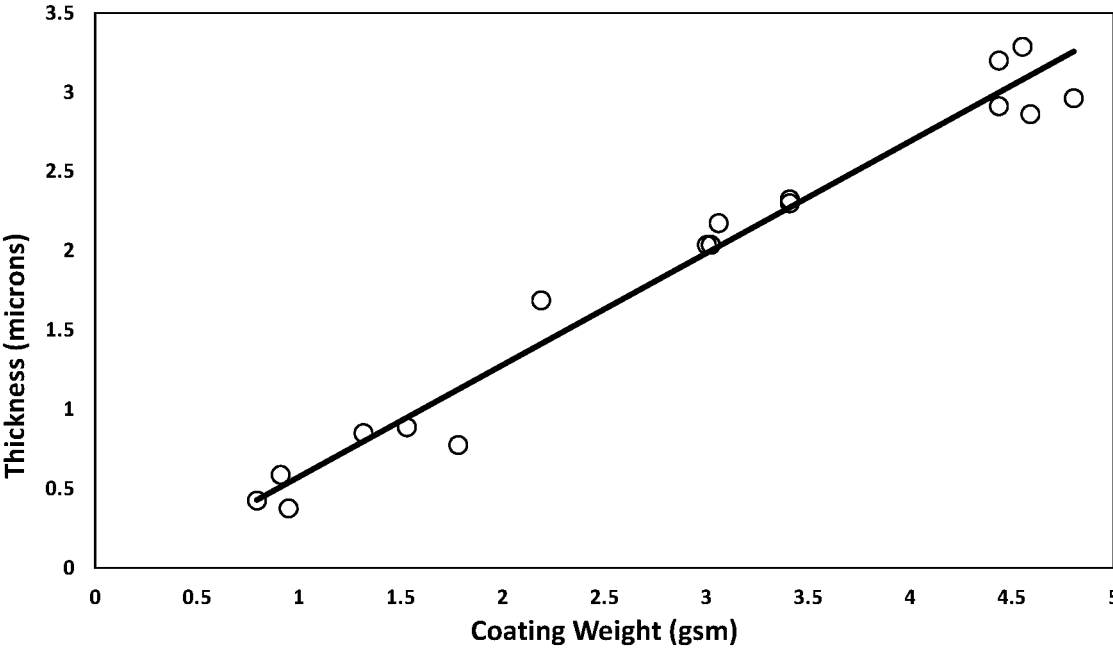


Figure 10



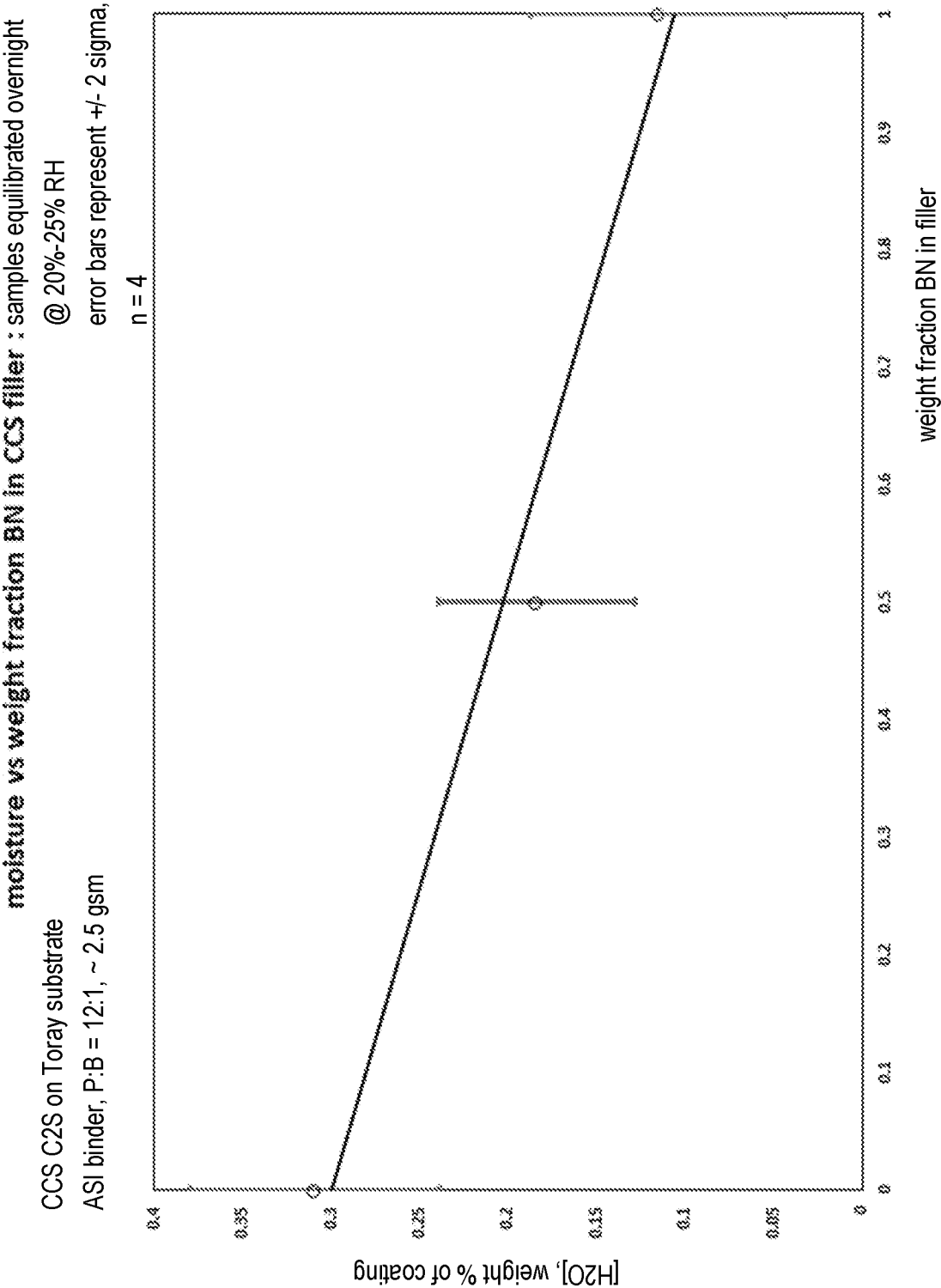


Fig. 11

MULTILAYER NANOPOROUS SEPARATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/223,845, filed on Apr. 6, 2021, which is a continuation of U.S. patent application Ser. No. 16/755,294, filed on Apr. 10, 2020 (now U.S. Pat. No. 10,998,536), which is a U.S. national stage entry under 35 U.S.C. § 371 of PCT International Patent Application No. PCT/US2018/055862, filed on Oct. 15, 2018, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/572,083, filed on Oct. 13, 2017, the entire contents of which are incorporated by reference herein.

FIELD OF THE INVENTION

The present invention relates generally to porous separators for batteries and other electric current producing cells. In particular, it pertains to multilayer porous separators comprising inorganic oxide/polymer layers on a polymeric layer, and to lithium batteries comprising such separators.

BACKGROUND OF THE INVENTION

Lithium batteries are widely used in portable electronics, such as smartphones and portable computers. Among the new applications for lithium batteries are high power batteries for hybrid, plug-in hybrid, and fully electric vehicles. Existing lithium batteries, including rechargeable and non-rechargeable lithium batteries, often utilize multilayer separators. Such separators may include an extruded polyethylene or polypropylene porous film that is coated with an inorganic oxide/polymer coating layer. As the demand for lower cost lithium batteries with increased size and energy density has grown, manufacturers have turned to less safe battery active materials in order to meet these goals. This has resulted in a need for battery separators with increased dimensional stability at temperatures such as 150° C., or higher.

Thus, it would be advantageous if the inorganic oxide/polymer coating layers could provide even greater dimensional stability at elevated temperatures while, at the same time, requiring lower coating weights and coating thicknesses, and providing lower manufacturing costs.

SUMMARY OF THE INVENTION

It is an object of the present invention to provide lighter weight, lower cost, and thinner inorganic oxide/polymer coating layers for polymer-based separators for lithium batteries and other electric current producing cells.

The present invention achieves these goals by providing a separator for a lithium battery comprising: (a) a porous polymeric layer, and (b) nanoporous inorganic oxide/polymer composite layers coated on both sides of the polymeric layer. In one embodiment, the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO₂) and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic oxide particles is 5 nm to 90 nm. This combination of inorganic oxide particles (e.g., boehmite particles) having an extremely small crystallite size, and a high volume of organic polymer, produces a porous heat stable layer that

provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator. This combination was further found to enable a significantly lower coating weight and coating thickness of the heat stable layer.

In another embodiment the nanoporous layer comprises: (a) a blend of inorganic oxide particles (e.g., boehmite particles) and inorganic nitride particles (e.g., boron nitride or aluminum nitride particles), and (b) one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic particles is 5 nm to 90 nm. This combination of: (1) inorganic oxide and nitride particles (e.g., boehmite and boron nitride or aluminum nitride particles) having extremely small crystallite sizes, and (2) a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator.

Preferably, the heat stable inorganic oxide/polymer layer (or inorganic oxide and nitride/polymer layer) is coated on both sides of the porous polymeric layer, such as a porous polyethylene layer or a porous polypropylene layer, as opposed to being on only one side of the polymeric layer. It has been found that this approach provides improved thermal dimensional stability in a thin and low cost separator. However, having the heat stable layer on only one side of the polymeric layer is also useful.

Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO₂) and a polymer, and wherein the crystallite size of the inorganic oxide is 5 nm to 25 nm. It has been found that this smaller crystallite size results in a thinner coating layer with more cohesive strength and with more adhesion to the polymeric layer at an even lower coating weight than comparable layers where the crystallite size for the inorganic oxide is larger such as, for example, about 80 nm.

In another embodiment, the present invention includes lithium batteries that comprise the improved separators of this invention.

A further aspect of the present invention are methods for making separators that comprise coating a solution comprising an inorganic oxide and/or inorganic nitride, and one or more polymers, on one or both surfaces of a porous polymeric layer to form the separators of this invention.

BRIEF DESCRIPTION OF THE DRAWINGS

The features and advantages of the present disclosure will be more fully understood with reference to the following, detailed description when taken in conjunction with the accompanying figures, wherein:

FIG. 1 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

FIG. 2 is a graph of the shrinkage as a function of coating thickness of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

FIG. 3 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

FIG. 4 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

FIG. 5 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

FIG. 6 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

FIG. 7 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

FIG. 8 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

FIG. 9 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

FIG. 10 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

FIG. 11 is a graph of the moisture content as function of the weight fraction of boron nitride in a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 4.

DETAILED DESCRIPTION

The separators of the present invention provide superior safety, lower cost, lower thickness, and other key performance properties for use in lithium batteries and in other electric current producing cells.

One aspect of this invention pertains to a separator for a lithium battery comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on both sides of the polymeric layer, wherein the nanoporous layer comprises boehmite and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to 45%, and wherein the crystallite size of the boehmite is 5 nm to 90 nm. Alternatively, the nanoporous layer may further comprise other inorganic oxide (e.g., SiO_2) or nitride (e.g., BN or AlN) particles, or combinations thereof.

The term “lithium battery” may include, for example, batteries in which the electroactive anode material comprises lithium. This includes, but is not limited to, secondary or rechargeable lithium ion batteries, secondary lithium metal batteries, and primary or non-rechargeable lithium metal batteries.

The term “separator” may include, for example, porous materials in a battery that are interposed between the cathode and the anode of the battery to provide electrical insulation against short circuiting and to provide pores for filling with the electrolyte of the battery.

The term “porous polymeric layer” may include, for example, a layer of a polymer material that is porous such that it has permeability to the flow of air through the polymeric layer. The battery industry typically measures this air permeability using a porosimeter that provides a value for the time in seconds for 100 cubic centimeters (cc) of air to pass through the porous layer. This value is commonly called the Gurley number. The battery industry prefers that the Gurley number for a separator be as low as possible without sacrificing mechanical integrity. Typical Gurley numbers for the air permeability of porous polyethylene or porous polypropylene separators, as used in the battery industry, are 100 seconds/100 cc to 300 seconds/100 cc. For this invention, porous polymer layer does not include any of the non-woven polymeric layers, such as with polyester fibers, that are used as separators in the battery industry.

The term “nanoporous” may include layers having pores up to about 100 nm in diameter, preferably, with at least 90% of the pores being less than 100 nm in diameter. The size and relative number of pores can be measured by mercury porosimetry, or by scanning electron microscope (SEM) or transmission electron microscope (TEM) cross-sections of the layer.

As discussed herein, the nanoporous layer may be present on both sides of a substrate, support, layer, or coating. For example, the nanoporous layer may be present on both sides of the porous polymeric layer, such as a porous polyethylene layer. The general benefits of the nanoporous layers of this invention are also found when the nanoporous layer is present on only one side of the porous polymeric layer, but the overall performance is typically superior when the nanoporous layer is present on both sides of the porous polymeric layer.

The superior performance features of two-side coating of the polymeric layer, compared to one-side coating at the same total coating weight, includes lower shrinkage at high temperatures and improved battery safety. Improved safety is typically the most important performance feature desired in separators for lithium batteries, especially as the batteries become larger and higher in energy density and use less safe battery active materials in order to achieve higher energy densities and lower costs.

Another advantage of two-side coating of the porous polymeric layer with the nanoporous coating layer is that it provides the opportunity to make the coatings on each side different in composition. This enables the nanoporous layer to be optimized for the best performance when it is facing the anode and when it is facing the cathode. In general, by tailoring the formulations and coating thicknesses of the ceramic coatings on either side of the separator, it is possible to optimize: (a) adhesion between separator and a specific electrode, either anode or cathode; (b) interfacial impedance between separator and a specific electrode, either anode or cathode; (c) complexation of migratory species (e.g., solubilized transition metal ions from cathode).

For example, when the coating is facing the cathode, it is particularly useful that the coating be resistant to oxidation, especially at higher charge voltages, e.g., above 4.25 volts. This can be done, for example, by selecting the organic polymer of the nanoporous layer to be resistant to oxidation up to the higher voltage desired, e.g., up to 5.0 volts. In addition, the nanoporous layer on one of the sides can be designed with particularly small pore sizes, such as under 10 nm diameter pores, for example, by using boehmite with a primary crystallite size of less than 10 nm in diameter. It should be understood that the present invention is not limited to the examples provided herein. For example,

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surface-modified alumina particles, including boehmite particles, as described in PCT/IB2015/000272 to Loscutova et al. of Sasol, may be utilized in the separator layers described herein. This patent application discloses the dispersion of these surface-modified aluminas into nano-sized single particle crystallites. This reference also describes that the average crystallite size can be measured by X-ray diffraction on a 120 plane. As an example, the crystallite size (or an average crystallite size) of the inorganic particles in a separator comprising a nanoporous layer according to an embodiment of the invention may be measured by: (i) taking a cross-section sample of the nanoporous layer; (ii) using a TEM or very high power SEM on this sample to measure the diameter/width, in their widest dimension, of 100 single primary (inorganic) particles selected at random in the TEM or SEM image; (iii) excluding the 25 highest and 25 lowest diameter numbers in the 100 diameter/width numbers; and (iv) recording the crystallite size as the average of the middle 50 diameter/width numbers, i.e., an average crystallite size of the measured particles. Since the pore size diameters of ceramic particles in a nanoporous layer usually approximate the crystallite size of the ceramic particles, such as boehmite particles, used to make the nanoporous layer, the pore size diameters may indicate the crystallite size of the ceramic particles for nanoporous layers containing ceramic particles of a narrow crystallite size. However, for blends of ceramic particles of different crystallite sizes in a nanoporous coating, the pore size diameters of the coating layer typically are intermediate, or a blend, among the crystallite size diameters of the different ceramic particles and not individual peaks reflecting the different crystallite sizes. The TEM or SEM method described above for measuring the crystallite size does not have this limitation since it is analyzing single particles, and not blends, and thus is a preferred method to use for this invention. These separator configurations using very small pore sizes are useful when placed against a lithium metal anode, where the very small size of the pores assists in preventing the growth of lithium metal dendrites into and through the separator, which reduces cycle life, capacity, and safety for the cell. Where very small boehmite particles are used and some of these boehmite particles penetrate into the pores of the polymeric separator substrate, this is particularly useful in preventing the growth of lithium metal dendrites in and through the separator.

In another example, boehmite could be modified by reaction with a fluorinated ethylene carbonate for improved cycle life when such nanoporous layer is positioned against an anode comprising silicon or lithium metal. This nanoporous layer with the boehmite covalently reacted with a fluorinated ethylene carbonate could also be used against the cathode to help achieve excellent stability under high voltage operation of the cell, such as at up to 5.0 volts.

In a further example, the nanoporous layer with pores that are less than 10 nm in diameter could be positioned against the cathode to inhibit the diffusion of transition metals, such as nickel ions, from the cathode to the anode, which degrades the cycle life and other performance properties of the cell. Optionally, the very small boehmite crystallite size of less than 10 nm could be combined with reaction of this boehmite with fluorinated ethylene carbonate or another organic carbonate to make a nanoporous layer positioned against the cathode to lessen the diffusion of transition metal ions from the cathode.

The term "Boehmite" is understood in the art. It, for example, may include hydrated aluminum oxide of the empirical chemical formula of $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ where x is 1.0 to 2.0. Another term used interchangeably herein for boehmite

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is AlOOH . Boehmite may be characterized by its distinctive Fourier-transform infrared spectroscopy ("FTIR") fingerprint. Boehmite may also be characterized by its distinctive X-ray diffraction pattern and by its distinctive infrared spectrum of double peaks of hydroxyl groups around 3200 cm^{-1} . Boehmite may have values of x for molecules of water that are above 1.5 and even up to about 2. However, the commercial grades of boehmite of all particle sizes typically have a value of x that is in the range of 1.0 to 1.5. The value of x can be measured by first drying the boehmite at about 120°C . for 1 hour to remove the surface adsorbed water and then measuring the loss of water that occurs when the boehmite sample is heated to 1000°C . for 1 hour and is thereby dehydrated to anhydrous aluminum oxide, Al_2O_3 .

Examples of polymers that may be utilized in the present invention include, but are not limited to, cellulosic polymers such as hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides such as chitosan, polyethylene oxides, and polyvinyl alcohols. Preferably, the polymers, alone or in combination, provide cohesive and adhesive strength to the nanoporous layer, are insoluble in the electrolyte of the lithium battery, and are electrochemically stable over the voltage range of use of the lithium battery. If needed to achieve these properties, one or more of the polymers may be crosslinked.

By the term "volume fraction of the one or more polymers in the solid portion of the nanoporous layer" is meant the fraction expressed as a percentage by volume of the polymer content of the solid portion of the inorganic particle/polymer composite layer. As described hereinafter, the nanoporous layer has a % porosity of between about 30% to about 70%. In other words, much of the volume of the nanoporous layer is air, and not a solid material. The air volume (e.g., pores) is not included in the solid portion for purposes of determining the volume fraction of the one or more polymers in the solid portion of the nanoporous layer. The volume fraction of the polymer and inorganic particle content of the nanoporous layer can be calculated based on the relative densities of the materials. For example, when the inorganic particles are only boehmite with a density of 3.03 g/cc , the organic polymer has a density of 1.3 g/cc , and the weight ratio of the boehmite to the organic polymer is 12:1, the volume fraction of boehmite would be 83.7% and the volume fraction of organic polymer would be 16.3%. Changing the weight ratio of the boehmite to the organic polymer to 4:1 results in a volume fraction of boehmite of 63.2% and of organic polymer of 36.8%. If the density of the organic polymer is lower, such as 1.1 g/cc , then its volume fraction at the same weight ratio of inorganic particle/polymer will be higher than at the higher polymer density of 1.3 g/cc . To determine the volume fraction of one or more organic polymers in the solid portion of the nanoporous layer, for example, the nanoporous layer is removed from the separator, such as by careful scraping of the layer and not removing any of the polymer substrate. Next, a weighed nanoporous layer sample in a crucible is combusted at a sufficient temperature to combust the organic polymer to leave the residue of the ceramic particles. This provides the weight of combusted ceramic particles. When the particles are only boehmite, this weight of the combusted residue is divided by 0.85 in the case of the equivalent of one water molecule per boehmite molecule to obtain the weight of boehmite in the sample prior to the combustion. Although most boehmite particles have a 15% weight loss upon combustion, it should be noted that boehmite particles may have a different weight loss factor, such as 0.84 or 0.82. This same information can be obtained by using a thermogravi-

metric analyzer to record the weight of the nanoporous layer sample as it is heated to 900° C. or higher, until a constant weight is achieved. As noted above, boehmite has a density of 3.03 g/cc. Most organic polymers have a density in the range of 1.1 to 1.4 g/cc, although some polymers, such as polyvinylidene difluoride (PVdF), have a density of about 1.8 g/cc. After the density of the organic polymers is known by analyzing for the identity and weight ratios of the organic polymers, for boehmite-only nanoporous layers, the volume % of the organic polymers in the solid portion of the nanoporous layer can be calculated. For example, if the weight percent of boehmite is 88.9%, and the weight percent of the organic polymers is 11.1% and its density is 1.3 g/cc—with a pigment:binder ratio of 8:1 by weight—dividing the weight percents of boehmite and the organic polymers by their densities gives relative volume percents of 29.34% for boehmite and 8.54% for the organic polymers. Normalizing these volume percents to a total volume percent of 100% for the solid portion of the nanoporous layer gives a volume percent of 22.54% for the organic polymers. If the nanoporous layer is a blend of boehmite and other ceramic particles or if the density of the organic polymers is not known, further analysis, as known in the art, may be undertaken to measure the volume percent of the organic polymers.

The term “crystallite size” of the particles refers to the average size of the primary or single crystals of the inorganic particles. This is typically measured using X-ray diffraction, a transmission electron microscope (TEM), or a very high resolution scanning electron microscope (SEM). PCT/IB2015/000272 to Loscutova et al. of Sasol cited above provides examples of using X-ray diffraction to determine an average size of the crystals of the inorganic particles, such as alumina and boehmite. As described earlier, a TEM or SEM method may be used to determine the crystallite size. Many inorganic oxide particles, such as boehmite particles, are nearly cubic in shape, although they can have other shapes such as platelets and needles. In the dry solid state and in solution, boehmite particles are typically agglomerated into larger particles comprised of many primary boehmite particles, such as, for example, 10 to 20 primary particles. The inorganic particles in the nanoporous layers of this invention result in features, such as pore size diameter, that are related to the crystallite size of the particles.

In one embodiment of the separators of the present invention, the ratio of the total thickness of the nanoporous layer on both sides of the polymeric layer to the total thickness of the separator is 10% to 80%. The thickness of these layers may be measured by SEM analysis of cross-sections of the separator. Because of the nanoporous pore sizes and the high volume fraction of organic polymer in the nanoporous layer, it has been found that a lower limit for the proportional thickness of the nanoporous layer of about 10% of the total thickness of the separator, rather than 15% or more typically 25%, is able to provide an excellent improvement in the dimensional stability of the separator at high temperatures, which is needed for improved safety. It has been found that the combination of nanoporous pore sizes and very high volume fractions of organic polymer in the nanoporous layer (e.g., from about 10% to about 50%) provides acceptable % porosity (e.g., from about 30% to about 70%) with good ionic conductivity of the electrolyte. It has also been found that the separator of this invention functions well with the total thickness of the nanoporous layer as high as 80% of the total thickness of the separator.

In one embodiment, the heat shrinkage of the separator at 150° C. for 1 hour is less than 5%. This heat shrinkage may

be measured by the change in the area of the separator after heating in an oven for 1 hour. This measurement of the heat shrinkage may be done by hanging a large sheet of the separator in the oven after marking and measuring the dimensions and area of a square or rectangle such as a 10 cm by 10 cm square, on a smaller section of the large sheet. After the heating in the oven, the marked area is measured for the % change or % shrinkage of the dimensions and the area.

In one embodiment, the heat shrinkage of the separator at 200° C. for 1 hour is less than 5%. The nanoporous separators of the present invention have been found to retain their dimensional stability even when heated well above the melting point of the polymeric layer, such as a porous polyethylene layer.

In one embodiment of the separators of this invention, the total coating weight of the nanoporous layer on both sides is less than 4.5 grams per square meter (gsm), while providing a heat shrinkage of the separator at 150° C. for 1 hour of less than 5%. It should be noted that the use of boehmite with its lower density of about 3.03 g/cc, as compared to the higher density anhydrous aluminum oxide or alumina (which has a density of about 4 g/cc), helps to further lower the coating weight of the heat stabilizing layer. The lower density of the inorganic oxide particle proportionately lowers the weight of coating needed for a specific thickness and % porosity.

For example, a 4 micron thick alumina/polymer layer with a 45% porosity for polyethylene-based separators typically weighs about 8 grams per square meter (gsm). However, switching to boehmite (or a blend of boehmite and other inorganic particles of a similar or lower density) at the same thickness and % porosity lowers the coating weight to about 6 gsm or less. The high volume fraction of polymer in the nanoporous layers of the separators of this invention, combined with the use of boehmite with a crystallite size of 5 nm to 90 nm, provides increased dimensional stability for a specific thickness of the separator layer. Although it is possible to still make the total thickness of the inorganic particle/polymer layer in the separator 4 microns or higher, it is desirable for lower cost and for lower total thickness of the separator to make the thickness of the inorganic particle/polymer layer as thin as possible, such as about 3 microns total, while still achieving an excellent dimensional stability at high temperature. In one embodiment, the heat shrinkage of the separator at 200° C. for 1 hour is less than 5%.

In one embodiment, other inorganic oxides and/or inorganic nitrides with a density of about 3.0 g/cc or lower, such as silicon dioxide and boron nitride, may be blended with the boehmite pigment. As discussed further below, this provides lower total coating weight of the nanoporous layer and other desirable properties, such as lower water content.

In another embodiment of the separators of this invention, the total coating weight of the nanoporous layer of both sides is less than 3.5 grams per square meter (gsm), while providing a heat shrinkage of the separator at 150° C. for 1 hour of less than 5%. In one embodiment, the heat shrinkage of the separator at 200° C. for 1 hour is less than 5%.

In one embodiment of the separators of the present invention, the crystallite size of the inorganic particles may be about 30 nm to about 50 nm, preferably about 5 nm to about 25 nm, and more preferably about 5 nm to about 10 nm. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of about 30 nm to about 50 nm. In one embodiment, the crystallite size of the boehmite may be about 5 nm to about 10 nm, and preferably about 5 nm to about 8 nm.

In another embodiment of this invention, the % porosity of the nanoporous layer is between 30% and 55%. The % porosity of the nanoporous layer may be calculated from the density in g/cc of the materials in the dry coating assuming it is a 100% solid coating with 0% porosity and comparing it to the actual density of the coating. The actual density of the nanoporous layer may be calculated by dividing its coat weight in gsm by its coating thickness to provide the density in g/cc of the nanoporous layer. Dividing the density of the nanoporous layer by the calculated density of the same materials in a non-porous layer and multiplying by 100% provides the % solid fraction by volume in the nanoporous layer. Subtracting this value of the % solid fraction from 100% gives the % pore volume or % porosity of the nanoporous layer. The % porosity can also be measured by mercury porosimetry analysis.

In one embodiment, the pore volume of the nanoporous layer is between 0.8 cc/g and 1.0 cc/g. The pore volume can be calculated by measuring the coat weight of the nanoporous layer in gsm and measuring the thickness of the nanoporous layer to determine the total cc in a square meter. The pore volume in cc/g is then calculated by dividing the total cc in a square meter by the grams of the gsm measurement. In one embodiment, the % porosity of the nanoporous layer is 55% to 70%. Above this level of % porosity, it is difficult to obtain the cohesive and adhesive strength of the nanoporous layer that are preferred for long term, safer lithium battery separator applications.

In one embodiment of the separators of the present invention, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 15% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 20% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 25% to 45%.

In one embodiment of the separators of this invention, the nanoporous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of the pores have a diameter less than 80 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of the pores have a diameter less than 50 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 25 nm or less and at least 70% of the pores have a diameter less than 25 nm.

In one embodiment of the separators of this invention, the ratio of the air permeability of the separator to the air permeability of the polymeric layer without any nanoporous layer present is 1.0 to 1.4. As described hereinabove, this air permeability is typically measured on an instrument which provides a Gurley number of the number of seconds for 100 cc of air to pass through the separator. A low level of increase of the air permeability is an important indicator of an expected low decrease in the ionic conductivity of electrolyte in the separator due to adding the inorganic oxide/polymer layer. If the ionic conductivity decreases too much, this can interfere with the cycling rate capability of the lithium battery particularly its power rate properties, and also interfere with the cycling lifetime of the lithium battery.

In one embodiment of the separators of the present invention, the polymeric layer comprises a polyolefin. Suitable polyolefins include, but are not limited to, polyethylenes and polypropylenes. Other examples of the polyolefin polymers include, but are not limited to, homopolymers of 1-butene, 1-pentene, 1-hexene, and 1-octene, as well as copolymers and terpolymers thereof, used alone or in com-

bination with one or more types of polymers. In one embodiment, the polymeric layer comprises polyethylene. The polymeric layer may be of the "wet process" type which is produced by extrusion followed by solvent extraction to provide the porosity or of the "dry process" type which is produced by extrusion followed by stretching to provide the porosity. Various molecular weights, densities, and stereochemistries of polyethylene and the other polymers, with various melt flow and mechanical properties, may be utilized in the porous polymeric layers.

For the purposes of better uniform wetting of the coating fluid of the nanoporous layer and of better adhesion of the nanoporous layer to the substrate, the porous polymeric layer may have a coating layer on the surface to which the nanoporous layer is applied.

In one embodiment of the separators comprising polyethylene in the polymeric layer, the heat shrinkage of the separator at 120° C. for 1 hour is less than 1.0%. Without the nanoporous layer of the separators of this invention, the heat shrinkage of a polyethylene separator of comparable total thickness is typically greater than 1.0% at 105° C. In one embodiment, the ratio of the air permeability of the separator after the heat shrinkage at 120° C., for 1 hour to the air permeability of the separator before the heat shrinkage at 120° C. is 0.8 to 1.2. This is an important feature for allowing vacuum drying of the polyethylene-based separators of this invention at a higher temperature of 120° C., without any negative effect on the air permeability and ionic conductivity and, in some cases, with a positive effect on ionic conductivity due to the heat treatment. Vacuum drying is beneficial for lowering the moisture content of separators for increased lithium battery cycling lifetimes and capacities and for higher lithium battery cycling rate capabilities. Vacuum drying is typically done for 4 to 24 hours at about 80° C. for polyethylene separators and at about 90° C. for polyethylene separators with one or more inorganic oxide/polymer layers. The capability of using a higher temperature of 120° C. and a shorter drying time of 1 to 3 hours for the polyethylene-based separators of this invention is useful.

In another embodiment of the separators of the present invention, the nanoporous layer is a laminated layer on the polymeric layer. By the term "laminated layer" is meant that the nanoporous layer is not initially formed on the polymeric layer, such as by direct coating, but is instead laminated to the polymeric layer in a subsequent step. For example, the nanoporous layer is coated on a release substrate, and then this nanoporous layer is transferred by lamination onto the polymeric layer. This lamination may optionally include the use of heat and/or the use of some organic solvent to adhere the nanoporous layer to the polymeric layer and to aid in the subsequent delamination of the release substrate without interfering with the quality of the laminated multilayer separator.

Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic particle/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite), or a blend of an inorganic oxide and an inorganic nitride (e.g., BN or AlN), and a polymer, and wherein the crystallite size of the inorganic particles is about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of about 25 nm or less and at least about 70% of the pores have a diameter of about 25 nm or less.

A further aspect of the separators for a lithium battery of the present invention pertains to a separator comprising (a) a porous polyethylene layer, and (b) a nanoporous inorganic particles/polymer composite layer on both sides of the polyethylene layer, wherein the nanoporous layer comprises boehmite, or a blend of boehmite and boron nitride, and one or more polymers, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is about 10% to about 50%, the total coating weight of the nanoporous layer on both sides is 5.5 grams per square meter or less, and the heat shrinkage of the separator at 150° C. for 1 hour is less than 5%. In one embodiment, the total coating weight of the nanoporous layer on both sides is 3.5 grams per square meter or less and the thickness of the polyethylene layer is between 5 microns and 16 microns.

In another embodiment, this invention is lithium batteries comprising a separator of the present invention.

Another aspect of the present invention are methods for making the separator of this invention, which method comprises coating a solution comprising inorganic particles and one or more polymers on one or both surfaces of a porous polymeric layer. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of 5 nm to 90 nm. In one embodiment, the boehmite particles have a crystallite size of about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the coating after drying is about 10% to about 50%, preferably about 12% to about 45%, and more preferably about 15% to about 45%.

EXAMPLES

Several embodiments of the present invention are described in the following examples, which are offered by way of illustration and not by way of limitation.

Example 1

A boehmite slurry was prepared by adding 12 parts of boehmite (DISPAL 10F4 supplied by Sasol Inc., Houston, Texas) and 1 part of chitosan (deacetylated poly(D-glucosamine), medium molecular weight, of a polysaccharide supplied by Sigma-Aldrich, St. Louis, MO) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 40 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 1.2 grams per square meter (gsm) to about 5.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70° C. for 2 minutes in a convection oven. The boehmite coating was then applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

The 12:1 by weight ratio of boehmite pigment to chitosan binder is calculated to have 15.3% by volume of chitosan binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.4 g/cc for chitosan.

The coating thickness was measured using a Dorsey gauge on the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the thickness values before and after the coating is the coating thickness in microns. An alternative method for

measuring the coating thickness is using the Dorsey gauge on the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

The coating weight was measured by weighing a 10 cm×10 cm sample of the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the weights before and after the coating multiplied by 1,000 is the coat weight in grams per square meter (gsm). An alternative method for measuring the coating weight was weighing a 10 cm×10 cm sample of the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

FIG. 1 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.6 gsm to about 2.6 gsm. For two sided coatings, these numbers are doubled to give a total coating weight range of about 1.2 gsm to about 5.2 gsm. The total boehmite coating thickness for both sides varies from about 1.2 microns to about 4.4 microns. If a more accurate measurement of the coating thickness is desired, scanning electron microscope (SEM) analysis of cross-sections of the boehmite-coated plastic separator can be utilized.

Using a SEM analysis for the total coating thickness showed a density of about 1.06 gsm per micron of coating thickness. From this, the % porosity for this 12:1 pigment to binder ratio is estimated as 62%.

One objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150° C. FIG. 2 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 1.9 microns or thicker on the 12 micron polyethylene substrate of this example. For coatings on two sides, this number is doubled to give a total coating thickness of about 3.8 microns or thicker. Based on the density of the boehmite coating determined by SEM analysis, a total coating thickness of about 3.8 microns on both sides equates to about 4.0 gsm of coating weight.

For better wetting and coating adhesion of the boehmite shiny, a corona treatment of the polyethylene substrate is preferred. Besides wire wound coating rods, a variety of other coating application methods may be used, such as, but not limited to, gravure coating, screen printer coating, reverse roll coating, blade coating, and slot die coating.

Example 2

A boehmite slurry was prepared by adding 12 parts of boehmite (DISPERAL D60 supplied by Sasol Inc., Hamburg, Germany) and 1 part of a vinyl pyrrolidone copolymer (Soteras CCS grade supplied by Ashland Inc., Wilmington, DE) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 75 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 2.4 grams per square meter (gsm) to about 9.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70° C. for 2 minutes in a convection oven. Then, the boehmite coating was applied to the reverse uncoated

side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

The 12:1 by weight ratio of boehmite pigment to vinyl pyrrolidone copolymer binder is calculated to have 18.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.1 g/cc for the vinyl pyrrolidone copolymer.

FIG. 3 shows a plot of the average boehmite coating thickness and coating weight per side of about 1.2 gsm to about 4.6 gsm. For two sided coating, these numbers are doubled to give a total coating weight range of about 2.4 gsm to about 9.2 gsm. The total boehmite coating thickness for both sides varies from about 2.8 microns to about 9.0 microns.

As discussed, one objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150° C. for 1 hour. FIG. 4 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 2.0 microns or thicker on the 12 micron polyethylene substrate of this example. For the coatings on two sides, this number is doubled to give a total coating thickness of about 4.0 microns or thicker. Based on FIG. 3, a total coating thickness of about 4.0 microns on both sides equates to about 3.9 gsm of coating weight.

A boehmite slurry of an 8 to 1 ratio by weight of the DISPERAL D60 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was also coated using wire wound rods on the 12 micron polyethylene substrate and dried, as for the 12:1 boehmite D60:polymer coating in this example. It was calculated to have a volume percent of polymer binder of 25.6%, FIG. 5 shows a plot of the average boehmite coating weight and coating thickness per side of about 0.8 gsm to about 4.2 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a range of about 1.6 gsm to 8.4 gsm. The corresponding total boehmite coating thicknesses for both sides varies from about 1.6 microns to 8.2 microns. FIG. 6 shows that a less than 5% shrinkage at 150° C. for 1 hour is obtained with an average coating thickness per side of about 2.2 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 4.4 microns or thicker. Based on FIG. 5, a total coating thickness of about 4.4 microns equates to about 4.8 gsm of coating weight.

Example 3

Using the procedure of Example 2, a boehmite slurry of a 12:1 by weight of DISPAL 25F4 boehmite and the vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was prepared. The DISPAL 25F4 is supplied by Sasol, Lake Charles, LA, and has a primary crystallite particle size of 8 nm. This boehmite slurry was coated on the EP12 12 micron polyethylene substrate from ENTEK and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 18.7%. FIG. 7 shows a plot of the average boehmite coating thickness and the coating weight per side of about 0.8 gsm to about 3.0 gsm. For the coatings on two sides for this boehmite coating with a 12:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to about 6.0 gsm. FIG. 8 shows that a less than 5%

shrinkage at 150° C. for 1 hour is obtained with an average coating thickness per side of about 1.8 microns or thicker. For two sided coatings, this number is doubled to give a total coating thickness on both sides of about 3.6 microns or thicker. Based on FIG. 7, a total coating thickness of about 3.6 microns equates to about 3.6 gsm.

A boehmite slurry of an 8 to 1 ratio by weight of the DISPAL 25F4 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol was also coated on the 12 micron polyethylene substrate and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 25.6%. FIG. 9 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.8 gsm to about 4.7 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to 9.4 gsm. FIG. 10 shows that a less than 5% shrinkage at 150° C. for 1 hour is obtained with an average coating thickness per side of about 0.9 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 1.8 microns or thicker. Based on FIG. 9, a total coating thickness of about 1.8 microns equates to about 2.9 gsm of coating weight. For a thickness of about 1.8 microns of the boehmite coating, the coating weight should have been about 2.0 gsm based on the density of the boehmite coating of about 1.1 g/cc shown in Example 1, while factoring in the lower boehmite weight % of the 8:1 boehmite pigment:vinyl pyrrolidone copolymer and the higher vinyl pyrrolidone copolymer weight % and lower density of the vinyl pyrrolidone copolymer compared to chitosan. This shows that about 0.9 gsm of the boehmite coating penetrated into and dried in the pores of the polyethylene substrate to produce a boehmite/polyethylene substrate with a boehmite coating on its surface. At the 8:1 pigment:binder (P:B) ratio of this coating, the about 0.9 gsm of boehmite coating in the pores equates to about 0.8 gsm of boehmite particles in the pores of the boehmite/polyethylene substrate or layer. Since this boehmite coating was applied on both sides of the polyethylene substrate, about one-half, or about 0.4 gsm of boehmite, would have penetrated and dried in the pores on each side of the polyethylene substrate to form the boehmite/polyethylene substrate or layer that resulted in a boehmite-coated boehmite/polyethylene substrate.

A similar 8:1 boehmite coating with Dispal 25F4, with chitosan substituted for the vinyl pyrrolidone copolymer and a 12 micron thick polyethylene substrate from Shenzhen Senior, Shenzhen, China, used instead of the polyethylene substrate from Entek, was made using the procedure of this Example to give a coat weight of 1.8 gsm. This coating had a heat shrinkage of less than 5% when heated at 150° C. for 1 hour. When the procedure of removing the boehmite coating with an adhesive tape was used, even with very careful tape pulls, it was not possible to remove the boehmite coating from the 10×10 cm samples without pulling off some of the polyethylene substrate. This was in contrast to the tape pull results of easy and complete removal of the boehmite coatings from the polyethylene substrate for Examples 1 and 2 and Comparative Examples 1 and 2. This result was consistent with some of the boehmite coating penetrating into the pores of the polyethylene layer and, upon drying, providing increased adhesive and cohesive strength to the boehmite coating such that the point of failure in the tape pull was below the surface of the polyethylene layer at most areas, rather than at the interface of the polyethylene surface and the boehmite coating on the poly-

ethylene surface. After the tape pull, some areas were found where the polyethylene layer had not been disturbed. From these areas, the added thickness of the boehmite layer was found to be about 1.8 microns.

Since the tape pulls were not able to cleanly remove the boehmite coating from the polyethylene layer, a methanol-saturated soft cloth was used to wash and rub the coating off. The 10×10 cm sample after washing had a weight of 0.0757 grams (g). Two more methanol rubs did not change this weight of the washed and rubbed sample. The polyethylene substrate used in this Example contained no inorganic pigment, such as alumina or boehmite. A furnace combustion of the methanol-washed sample in a crucible, at a temperature of 1000° C. for 70 minutes showed 0.0045 g of anhydrous aluminum oxide after dehydrating the boehmite pigment with a weight loss of 15% of the boehmite pigment weight. Using this boehmite to aluminum oxide conversion factor by dividing the aluminum oxide weight by 0.85, this gives a boehmite content of about 0.0053, or at least about 0.53 gsm of boehmite pigment in the pores of the polyethylene layer. Since the wet rubbing to remove the boehmite coating on the surface may have removed a small amount of the boehmite coating in the upper pores of the polyethylene layer, it is possible that the weight of boehmite pigment in the pores was somewhat greater than 0.53 gsm. Since there were two coatings on the polyethylene layer, the amount of boehmite in the pores from a single coating is about one-half of this total, or about 0.27 gsm. As further evidence for penetration of the boehmite coating into the pores of the polyethylene substrate, the dimensional stability of this “washed” polyethylene with about 0.53 gsm of boehmite in the pores after heating at 150° C. for 1 hour was 54% and 36.5% in the two directions (—machine direction (MD) and cross direction (CD)). By contrast, the polyethylene substrate before coating showed heat shrinkage of 70% and 75.5%. This increased heat stability with boehmite pigment in the pores of the polyethylene substrate is one explanation for the greater dimensional stability at 150° C. with a much lower coat weight and coating thickness added to the surface of the polyethylene layer. For analyzing the presence and amount of boehmite particles in the pores of the polymeric layer, the removal of the surface coating of the nanoporous layer and subsequent combustion of all of the organic polymer, as described above, in a combustion and/or thermogravimetric method may be used for the case of only boehmite particles in the nanoporous layer and a polymer layer that contained no inorganic or boehmite particles before the application of the boehmite coating. Another method is to use the TEM or SEM method used for measuring the average crystallite size to make images of the presence of boehmite particles in the pores of the polymeric layer and to determine their crystallite size at the same time. Combining this method with the combustion and/or thermogravimetric method above to quantify the weight of the boehmite, or other ceramic particles in the pores, if any, would provide the weight in gsm of boehmite particles in the separator. The known analytical techniques for boehmite particles, such as FTIR, X-ray diffraction, thermogravimetric analysis (TGA), high resolution photoelectron spectroscopy (ESCA), and ²⁷Al nuclear magnetic resonance (nmr) can be utilized to further confirm the presence and quantity of boehmite particles in the pores of the polymeric layer, if needed. An alternative approach for collecting the ceramic particles, such as boehmite particles, in the pores of the polymeric layer, after removing any surface coating from the polymeric layer, would be to do a Soxhlet extraction of the polymeric layer with a hot solvent that dissolves the polymer from the

polymeric layer to isolate the insoluble inorganic particles. Besides providing the weight of inorganic particles, this could be combined with the above analytical methods, combustion and/or thermogravimetric analysis, and TEM or SEM method, to confirm the presence and quantity of boehmite particles. One benefit of this penetration of nano-sized boehmite particles would be the ability to make and sell boehmite/polyethylene substrates, with no or minimal surface boehmite or other coating, for various applications, including for manufacturing CCS, where the boehmite/polyethylene substrate would have multiple advantages over polyethylene substrates, including, but not limited to, a much greater dimensional stability at high heats, a modified and smaller pore size distribution, and other benefits, such as ease of wetting by subsequent coatings. One example of a particularly useful application for boehmite/polymeric substrates would be filtration applications, where the very small nano-sized boehmite particles, with and without various surface treatments, in the pores of the polymer could provide the desired nanofiltration and ultrafiltration properties not available from the polymeric substrate without boehmite particles in its pores, and could be adapted for desalination of sea water. This benefit would also be available for other boehmite/polymeric substrates besides polyethylene, such as for boehmite/polypropylene and boehmite/other polyolefin substrates.

Example 4

The procedure of Example 1 was used to make a nanoporous separator comprising a 50:50 blend by weight of boehmite and boron nitride pigments, except that a 12 micron thick porous polyethylene separator from Toray Industries, Tokyo, Japan, was substituted for the Entek polyethylene separator, and the vinyl pyrrolidone polymer, Soteras CCS, was substituted for the chitosan. The boron nitride pigment was NX1, supplied by Momentive Performance Materials. The total coating weight on both sides was 2.5 gsm. FIG. 11 shows the reduced water content of this separator compared to the control with only boehmite pigment and no boron nitride. Specifically, as shown in FIG. 11, at a 50:50 by weight ratio of boehmite and boron nitride, the water content at 20% RH was reduced by one-third from its value with boehmite pigment only.

Comparative Example 1

An alumina mix was prepared in water by adding 20 parts of alumina pigment (HJA-0719 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the alumina was greater than 400 nm. The alumina mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 4.4 gsm. The alumina coating was first applied to one side of the polyethylene separator and dried at 70° C. for 2 minutes in a convection oven. Then, the

alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

The 20:1 by weight ratio of alumina pigment to hydroxypropyl cellulose polymer binder is calculated to have 13.6% by volume of polymer hinder in the solid portion of the porous alumina coating layer. This calculation is based on a specific gravity of 4.0 g/cc for alumina and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating weight of about 4.4 gsm, this alumina coating had a shrinkage of about 60% at 150° C. for 1 hour.

Comparative Example 2

A boehmite mix was prepared in water by adding 20 parts of boehmite pigment (BG-613 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Mucci H grade supplied by Ashland, Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the boehmite was greater than 400 nm. The boehmite mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 7.0 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70° C. for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

The 20:1 by weight ratio of boehmite pigment to hydroxypropyl cellulose polymer binder is calculated to have 10.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating weight of about 7.0 gsm, this boehmite coating had a shrinkage of about 60% at 150° C. for 1 hour.

Table 1 shows a summary of the data from Examples 1 to 3.

TABLE 1

Example	Boehmite Type	Primary Particle Size (nm)	Pigment: Polymer Binder Weight Ratio	Volume % Polymer	Total Coat Weight for <5% Shrinkage at 150° C. for 1 hour
1	10F4	40	12:1	15.3%	≥4.0 gsm
2	D60	75	12:1	18.7%	≥3.9 gsm
2	D60	75	8:1	25.6%	≥4.8 gsm
3	25F4	8	12:1	18.7%	≥3.6 gsm
3	25F4	8	8:1	25.6%	≥2.9 gsm

One challenge in using nano-sized ceramic particles, such as nano-sized boehmite particles, is their increased tendency to absorb water because of their much larger surface areas compared to micro-sized ceramic particles, such as micro-sized alumina and micro-sized boehmite particles. For lithium ion cells, lithium metal cells, and other electrochemical cells with sensitivity to water content, the separa-

tor needs to have a low water content, such as under 600 ppm of water in the separator, preferably less than 300 ppm of water, and more preferably under 100 ppm of water as typically found in porous polymer only separators with no ceramic materials in them. Although vacuum drying of the separator at up to 80° C. can help to lower this moisture content in separators with nano-sized ceramic particles, and the separators of this invention have increased thermal stability that permits even higher vacuum drying temperatures of up to 90° C. to about 110° C., some users of separators may prefer not to perform this extra step of vacuum drying. It would be preferable if the greater heat stability, lower cost from the use of a lower amount of ceramic coating materials, and lower ceramic coating thickness, including from the penetration of the ceramic coating and ceramic particles into the porous polymeric separator layer, of the separators of this invention would include a lower moisture content in the separators from the use of more hydrophobic ceramic particles and organic polymers. Having a lower amount of ceramic particles from the lower coating weights of the invention is a significant benefit for lowering the moisture content of the separator, but it would be further useful if the ceramic particles were more hydrophobic and less prone to moisture pickup.

One approach which provides this lower moisture absorption is to treat the surface of the ceramic particles with a hydrophobic material, such as by using hydrophobically-treated boehmite particles, particularly if the hydrophobic material is chemically reacted with the boehmite particles and thus is permanently bonded to the boehmite particles. Suitable hydrophobic materials for chemical modification of boehmite pigments include, but are not limited to, organic sulfonic acids, organic carbonates, and polymers with hydrophobic groups and with reactive groups, such as carboxylic acid and hydroxyl groups, that can be made to react with hydroxyl groups of the boehmite or other ceramic particles.

Examples of the organic sulfonic acid-treated boehmite particles include Dispal® 25SR, available from Sasol in Lake Charles, LA, a boehmite pigment chemically reacted with p-toluene sulfonic acid through the sulfonic acid groups. The reaction leaves the hydrophobic p-toluene group on the outer surface of the boehmite particle to make it less prone to pick up water. An even more hydrophobic example from Sasol is p-dodecylbenzene sulfonic acid surface reacted to boehmite particles through the sulfonic acid group and having the very hydrophobic dodecyl group on the outer surface of the boehmite pigment. This makes the boehmite particles so hydrophobic that this treated boehmite can be dispersed in hydrocarbon solvents. The use of boehmite particles surface treated with organic sulfonic acids in boehmite coatings for separators is disclosed in U.S. Patent Application Publication No. 2013/0171500 by Xu et al.

Examples of the organic carbonate-treated boehmite particles include, but are not limited to, those with ethylene carbonate and fluoroethylene carbonate as the organic carbonates. Besides the option of choosing an organic carbonate that is very hydrophobic in the part of the molecule that will face away from the surface of the boehmite, the choice can also be made with a view to picking an organic carbonate which will provide increased ionic conductivity to the separator. The use of boehmite particles surface treated with ethylene carbonate to provide separators with increased conductivity is described in U.S. Pat. No. 8,883,354 to Carlson et al.

The use of boehmite particles surface treated with organic polymers to provide separators is described in U.S. Pat. No.

9,871,239, to Carlson et al. As the primary crystallite size and resulting pore size become smaller in the boehmite separators and the ionic conductivity of the separator becomes lower because of the smaller pore sizes, it is useful to have surface treatments, such as the three types described above, that can be utilized to increase the ionic conductivity of the separator, as well as to decrease the water content of the separator.

The preparation of a ceramic mix or slurry with the surface-treated hydrophobic materials can be done conveniently by adding a precursor, such as an organic carbonate, to the slurry containing the untreated boehmite particles and heating the slurry to a temperature, such as 80° C., for a sufficient time, such as 1 hour, to react the surface treatment material, such as an organic carbonate, with the boehmite particles to the desired level of surface treatment. There is no need to isolate the surface-treated boehmite particles by a drying process, but rather the slurry can be used as is, usually after some dilution with water and/or solvents to the desired % solids level for the coating of the porous polymer substrate. If it is desired to further reduce the particle size of the ceramic particles, such as boehmite particles, and/or to homogenize the slurry for more uniformity and perhaps a lower viscosity at the desired % solids level for coating, the slurry, before and/or after any surface treatment, can be further mixed in an impingement mill or similar milling device. Examples of the use of an impingement mill for reducing and homogenizing coating mixes containing pigments is described in U.S. Pat. Nos. 5,210,114 and 5,292,588, to Katsen.

While the invention has been described in detail and with reference to specific and general embodiments thereof, it will be apparent to one skilled in the art that various changes and modifications can be made therein without departing from the spirit and scope thereof.

The invention claimed is:

1. A separator for a lithium battery comprising: (a) a porous polymeric layer, and (b) a second porous layer

adjacent said porous polymeric layer, wherein said second porous layer comprises hydrophobic inorganic oxide particles and one or more polymers, wherein 0.27 grams, or greater, per square meter of hydrophobic inorganic oxide particles are disposed within the pores of the porous polymeric layer, and wherein the amount of water in the separator is no more than 600 parts per million.

2. The separator of claim 1, wherein the hydrophobic inorganic oxide particles comprise boehmite.

3. The separator of claim 2, wherein the second porous layer comprises an organic carbonate treated boehmite.

4. The separator of claim 3, wherein the organic carbonate reacts with hydroxyl groups of the boehmite.

5. The separator of claim 3, wherein the organic carbonate is ethylene carbonate or fluoroethylene carbonate.

6. The separator of claim 1, wherein the inorganic oxide particles have a crystallite size of about 30 nm to about 90 nm.

7. The separator of claim 2, wherein the boehmite has a crystallite size of about 30 nm to about 50 nm.

8. The separator of claim 1, wherein the amount of water in the separator is no more than 300 parts per million of water.

9. The separator of claim 1, wherein the amount of water in the separator is no more than 100 parts per million of water.

10. The separator of claim 1, wherein the second porous layer is a nanoporous layer.

11. The separator of claim 1, wherein the total coating weight of said second porous layer is less than 4.5 grams per square meter.

12. The separator of claim 1, wherein the wherein an average crystallite size of said inorganic particles disposed within the pores of said second porous polymeric layer is between 5 nm and 90 nm.

* * * * *

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(57) Abstract: A separator for a lithium battery having (a) a porous polymeric layer, such as a polyethylene layer; and (b) a nanoporous inorganic particle/polymer layer on both sides of the polymeric layer, the nanoporous layer having an inorganic oxide and one or more polymers; the volume fraction of the polymers in the nanoporous layer is about 15% to about 50%, and the crystallite size of the inorganic oxide is 5 nm to 90 nm.

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MULTILAYER NANOPOROUS SEPARATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of and priority to U.S. Provisional Patent Application No. 62/572,083, filed on October 13, 2017, the entire contents of which are incorporated by reference herein

FIELD OF THE INVENTION

[0002] The present invention relates generally to porous separators for batteries and other electric current producing cells. In particular, it pertains to multilayer porous separators comprising inorganic oxide/polymer layers on a polymeric layer, and to lithium batteries comprising such separators.

BACKGROUND OF THE INVENTION

[0003] Lithium batteries are widely used in portable electronics, such as smartphones and portable computers. Among the new applications for lithium batteries are high power batteries for hybrid, plug-in hybrid, and fully electric vehicles. Existing lithium batteries, including rechargeable and non-rechargeable lithium batteries, often utilize multilayer separators. Such separators may include an extruded polyethylene or polypropylene porous film that is coated with an inorganic oxide/polymer coating layer. As the demand for lower cost lithium batteries with increased size and energy density has grown, manufacturers have turned to less safe battery active materials in order to meet these goals. This has resulted in a need for battery separators with increased dimensional stability at temperatures such as 150°C, or higher.

[0004] Thus, it would be advantageous if the inorganic oxide/polymer coating layers could provide even greater dimensional stability at elevated temperatures while, at the same time, requiring lower coating weights and coating thicknesses, and providing lower manufacturing costs.

SUMMARY OF THE INVENTION

[0005] It is an object of the present invention to provide lighter weight, lower cost, and thinner inorganic oxide/polymer coating layers for polymer-based separators for lithium batteries and other electric current producing cells.

[0006] The present invention achieves these goals by providing a separator for a lithium battery comprising: (a) a porous polymeric layer, and (b) nanoporous inorganic oxide/polymer composite layers coated on both sides of the polymeric layer. In one embodiment, the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO_2) and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic oxide particles is 5 nm to 90 nm. This combination of inorganic oxide particles (e.g., boehmite particles) having an extremely small crystallite size, and a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator. This combination was further found to enable a significantly lower coating weight and coating thickness of the heat stable layer.

[0007] In another embodiment the nanoporous layer comprises: (a) a blend of inorganic oxide particles (e.g., boehmite particles) and inorganic nitride particles (e.g., boron nitride or aluminum nitride particles), and (b) one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic particles is 5 nm to 90 nm. This combination of: (1) inorganic oxide and nitride particles (e.g., boehmite and boron nitride or aluminum nitride particles) having extremely small crystallite sizes, and (2) a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator.

[0008] Preferably, the heat stable inorganic oxide/polymer layer (or inorganic oxide and nitride/polymer layer) is coated on both sides of the porous polymeric layer, such as a porous polyethylene layer or a porous polypropylene layer, as opposed to being on only one side of the polymeric layer. It has been found that this approach provides improved thermal dimensional stability in a thin and low cost separator. However, having the heat stable layer on only one side of the polymeric layer is also useful.

[0009] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO_2) and a polymer, and wherein the crystallite size of the inorganic oxide is 5 nm to 25 nm. It has been found that this smaller crystallite size results in a thinner coating layer with more cohesive strength and with more adhesion to the polymeric layer at an even lower coating weight than comparable layers where the crystallite size for the inorganic oxide is larger such as, for example, about 80 nm.

[0010] In another embodiment, the present invention includes lithium batteries that comprise the improved separators of this invention.

[0011] A further aspect of the present invention are methods for making separators that comprise coating a solution comprising an inorganic oxide and/or inorganic nitride, and one or more polymers, on one or both surfaces of a porous polymeric layer to form the separators of this invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The features and advantages of the present disclosure will be more fully understood with reference to the following, detailed description when taken in conjunction with the accompanying figures, wherein:

[0013] FIG. 1 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0014] FIG. 2 is a graph of the shrinkage as a function of coating thickness of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0015] FIG. 3 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0016] FIG. 4 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0017] FIG. 5 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0018] FIG. 6 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0019] FIG. 7 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0020] FIG. 8 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0021] FIG. 9 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0022] FIG. 10 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0023] FIG. 11 is a graph of the moisture content as function of the weight fraction of boron nitride in a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 4.

DETAILED DESCRIPTION

[0024] The separators of the present invention provide superior safety, lower cost, lower thickness, and other key performance properties for use in lithium batteries and in other electric current producing cells.

[0025] One aspect of this invention pertains to a separator for a lithium battery comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on both sides of the polymeric layer, wherein the nanoporous layer comprises boehmite and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to 45%, and wherein the crystallite size of the boehmite is 5 nm to 90 nm. Alternatively, the nanoporous layer may further comprise other inorganic oxide (e.g., SiO₂) or nitride (e.g., BN or AlN) particles, or combinations thereof.

[0026] The term “lithium battery” may include, for example, batteries in which the electroactive anode material comprises lithium. This includes, but is not limited to, secondary or rechargeable lithium ion batteries, secondary lithium metal batteries, and primary or non-rechargeable lithium metal batteries.

[0027] The term “separator” may include, for example, porous materials in a battery that are interposed between the cathode and the anode of the battery to provide electrical insulation against short circuiting and to provide pores for filling with the electrolyte of the battery.

[0028] The term “porous polymeric layer” may include, for example, a layer of a polymer material that is porous such that it has permeability to the flow of air through the polymeric layer. The battery industry typically measures this air permeability using a porosimeter that provides a value for the time in seconds for 100 cubic centimeters (cc) of air to pass through the porous layer. This value is commonly called the Gurley number. The battery industry prefers that the Gurley number for a separator be as low as possible without sacrificing mechanical integrity. Typical Gurley numbers for the air permeability of porous polyethylene or porous polypropylene separators, as used in the battery industry, are 100 seconds/100 cc to 300 seconds/100 cc. For this invention, porous polymer layer does not include any of the non-woven polymeric layers, such as with polyester fibers, that are used as separators in the battery industry.

[0029] The term “nanoporous” may include layers having pores up to about 100 nm in diameter, preferably, with at least 90% of the pores being less than 100 nm in diameter. The size and relative number of pores can be measured by mercury porosimetry, or by scanning electron microscope (SEM) or transmission electron microscope (TEM) cross-sections of the layer.

[0030] As discussed herein, the nanoporous layer may be present on both sides of a substrate, support, layer, or coating. For example, the nanoporous layer may be present on both sides of the porous polymeric layer, such as a porous polyethylene layer. The general benefits of the nanoporous layers of this invention are also found when the nanoporous layer is present on only one side of the porous polymeric layer, but the overall performance is typically superior when the nanoporous layer is present on both sides of the porous polymeric layer.

[0031] The superior performance features of two-side coating of the polymeric layer, compared to one-side coating at the same total coating weight, includes lower shrinkage at

high temperatures and improved battery safety. Improved safety is typically the most important performance feature desired in separators for lithium batteries, especially as the batteries become larger and higher in energy density and use less safe battery active materials in order to achieve higher energy densities and lower costs.

[0032] Another advantage of two-side coating of the porous polymeric layer with the nanoporous coating layer is that it provides the opportunity to make the coatings on each side different in composition. This enables the nanoporous layer to be optimized for the best performance when it is facing the anode and when it is facing the cathode. In general, by tailoring the formulations and coating thicknesses of the ceramic coatings on either side of the separator, it is possible to optimize: (a) adhesion between separator and a specific electrode, either anode or cathode; (b) interfacial impedance between separator and a specific electrode, either anode or cathode; (c) complexation of migratory species (e.g., solubilized transition metal ions from cathode).

[0033] For example, when the coating is facing the cathode, it is particularly useful that the coating be resistant to oxidation, especially at higher charge voltages, e.g., above 4.25 volts. This can be done, for example, by selecting the organic polymer of the nanoporous layer to be resistant to oxidation up to the higher voltage desired, e.g., up to 5.0 volts. In addition, the nanoporous layer on one of the sides can be designed with particularly small pore sizes, such as under 10 nm diameter pores, for example, by using boehmite with a primary crystallite size of less than 10 nm in diameter. It should be understood that the present invention is not limited to the examples provided herein. For example, surface-modified alumina particles, including boehmite particles, as described in PCT/IB2015/000272 to Loscutova *et al.* of Sasol, may be utilized in the separator layers described herein. This patent application discloses the dispersion of these surface-modified aluminas into nano-sized single particle crystallites. This reference also describes that the average crystallite size can be measured by X-ray diffraction on a 120 plane. As an example, the crystallite size (or an average crystallite size) of the inorganic particles in a separator comprising a nanoporous layer according to an embodiment of the invention may be measured by: (i) taking a cross-section sample of the nanoporous layer; (ii) using a TEM or very high power SEM on this sample to measure the diameter/width, in their widest dimension, of 100 single primary (inorganic) particles selected at random in the TEM or SEM image; (iii) excluding the 25 highest and 25 lowest diameter numbers in the 100 diameter/width numbers; and (iv) recording the crystallite size as the average of the middle

50 diameter/width numbers, i.e., an average crystallite size of the measured particles. Since the pore size diameters of ceramic particles in a nanoporous layer usually approximate the crystallite size of the ceramic particles, such as boehmite particles, used to make the nanoporous layer, the pore size diameters may indicate the crystallite size of the ceramic particles for nanoporous layers containing ceramic particles of a narrow crystallite size. However, for blends of ceramic particles of different crystallite sizes in a nanoporous coating, the pore size diameters of the coating layer typically are intermediate, or a blend, among the crystallite size diameters of the different ceramic particles and not individual peaks reflecting the different crystallite sizes. The TEM or SEM method described above for measuring the crystallite size does not have this limitation since it is analyzing single particles, and not blends, and thus is a preferred method to use for this invention. These separator configurations using very small pore sizes are useful when placed against a lithium metal anode, where the very small size of the pores assists in preventing the growth of lithium metal dendrites into and through the separator, which reduces cycle life, capacity, and safety for the cell. Where very small boehmite particles are used and some of these boehmite particles penetrate into the pores of the polymeric separator substrate, this is particularly useful in preventing the growth of lithium metal dendrites in and through the separator.

[0034] In another example, boehmite could be modified by reaction with a fluorinated ethylene carbonate for improved cycle life when such nanoporous layer is positioned against an anode comprising silicon or lithium metal. This nanoporous layer with the boehmite covalently reacted with a fluorinated ethylene carbonate could also be used against the cathode to help achieve excellent stability under high voltage operation of the cell, such as at up to 5.0 volts.

[0035] In a further example, the nanoporous layer with pores that are less than 10 nm in diameter could be positioned against the cathode to inhibit the diffusion of transition metals, such as nickel ions, from the cathode to the anode, which degrades the cycle life and other performance properties of the cell. Optionally, the very small boehmite crystallite size of less than 10 nm could be combined with reaction of this boehmite with fluorinated ethylene carbonate or another organic carbonate to make a nanoporous layer positioned against the cathode to lessen the diffusion of transition metal ions from the cathode.

[0036] The term “Boehmite” is understood in the art. It, for example, may include hydrated aluminum oxide of the empirical chemical formula of $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ where x is 1.0 to 2.0. Another term used interchangeably herein for boehmite is AlOOH . Boehmite may be

characterized by its distinctive Fourier-transform infrared spectroscopy (“FTIR”) fingerprint. Boehmite may also be characterized by its distinctive X-ray diffraction pattern and by its distinctive infrared spectrum of double peaks of hydroxyl groups around 3200 cm^{-1} . Boehmite may have values of x for molecules of water that are above 1.5 and even up to about 2. However, the commercial grades of boehmite of all particle sizes typically have a value of x that is in the range of 1.0 to 1.5. The value of x can be measured by first drying the boehmite at about 120°C for 1 hour to remove the surface adsorbed water and then measuring the loss of water that occurs when the boehmite sample is heated to 1000°C for 1 hour and is thereby dehydrated to anhydrous aluminum oxide, Al_2O_3 .

[0037] Examples of polymers that may be utilized in the present invention include, but are not limited to, cellulosic polymers such as hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides such as chitosan, polyethylene oxides, and polyvinyl alcohols. Preferably, the polymers, alone or in combination, provide cohesive and adhesive strength to the nanoporous layer, are insoluble in the electrolyte of the lithium battery, and are electrochemically stable over the voltage range of use of the lithium battery. If needed to achieve these properties, one or more of the polymers may be crosslinked.

[0038] By the term “volume fraction of the one or more polymers in the solid portion of the nanoporous layer” is meant the fraction expressed as a percentage by volume of the polymer content of the solid portion of the inorganic particle/polymer composite layer. As described hereinafter, the nanoporous layer has a % porosity of between about 30% to about 70%. In other words, much of the volume of the nanoporous layer is air, and not a solid material. The air volume (e.g., pores) is not included in the solid portion for purposes of determining the volume fraction of the one or more polymers in the solid portion of the nanoporous layer. The volume fraction of the polymer and inorganic particle content of the nanoporous layer can be calculated based on the relative densities of the materials. For example, when the inorganic particles are only boehmite with a density of 3.03 g/cc , the organic polymer has a density of 1.3 g/cc , and the weight ratio of the boehmite to the organic polymer is 12:1, the volume fraction of boehmite would be 83.7% and the volume fraction of organic polymer would be 16.3%. Changing the weight ratio of the boehmite to the organic polymer to 4:1 results in a volume fraction of boehmite of 63.2% and of organic polymer of 36.8%. If the density of the organic polymer is lower, such as 1.1 g/cc , then its volume fraction at the same weight ratio of inorganic particle/polymer will be higher than at the higher polymer density of 1.3 g/cc . To determine the volume fraction of one or more organic

polymers in the solid portion of the nanoporous layer, for example, the nanoporous layer is removed from the separator, such as by careful scraping of the layer and not removing any of the polymer substrate. Next, a weighed nanoporous layer sample in a crucible is combusted at a sufficient temperature to combust the organic polymer to leave the residue of the ceramic particles. This provides the weight of combusted ceramic particles. When the particles are only boehmite, this weight of the combusted residue is divided by 0.85 in the case of the equivalent of one water molecule per boehmite molecule to obtain the weight of boehmite in the sample prior to the combustion. Although most boehmite particles have a 15% weight loss upon combustion, it should be noted that boehmite particles may have a different weight loss factor, such as 0.84 or 0.82. This same information can be obtained by using a thermogravimetric analyzer to record the weight of the nanoporous layer sample as it is heated to 900 °C or higher, until a constant weight is achieved. As noted above, boehmite has a density of 3.03 g/cc. Most organic polymers have a density in the range of 1.1 to 1.4 g/cc, although some polymers, such as polyvinylidene difluoride (PVdF), have a density of about 1.8 g/cc. After the density of the organic polymers is known by analyzing for the identity and weight ratios of the organic polymers, for boehmite-only nanoporous layers, the volume % of the organic polymers in the solid portion of the nanoporous layer can be calculated. For example, if the weight percent of boehmite is 88.9%, and the weight percent of the organic polymers is 11.1% and its density is 1.3 g/cc--with a pigment:binder ratio of 8:1 by weight--dividing the weight percents of boehmite and the organic polymers by their densities gives relative volume percents of 29.34% for boehmite and 8.54% for the organic polymers. Normalizing these volume percents to a total volume percent of 100% for the solid portion of the nanoporous layer gives a volume percent of 22.54% for the organic polymers. If the nanoporous layer is a blend of boehmite and other ceramic particles or if the density of the organic polymers is not known, further analysis, as known in the art, may be undertaken to measure the volume percent of the organic polymers.

[0039] The term “crystallite size” of the particles refers to the average size of the primary or single crystals of the inorganic particles. This is typically measured using X-ray diffraction, a transmission electron microscope (TEM), or a very high resolution scanning electron microscope (SEM). PCT/IB2015/000272 to Loscutova *et al.* of Sasol cited above provides examples of using X-ray diffraction to determine an average size of the crystals of the inorganic particles, such as alumina and boehmite. As described earlier, a TEM or SEM method may be used to determine the crystallite size. Many inorganic oxide particles, such

as boehmite particles, are nearly cubic in shape, although they can have other shapes such as platelets and needles. In the dry solid state and in solution, boehmite particles are typically agglomerated into larger particles comprised of many primary boehmite particles, such as, for example, 10 to 20 primary particles. The inorganic particles in the nanoporous layers of this invention result in features, such as pore size diameter, that are related to the crystallite size of the particles.

[0040] In one embodiment of the separators of the present invention, the ratio of the total thickness of the nanoporous layer on both sides of the polymeric layer to the total thickness of the separator is 10% to 80%. The thickness of these layers may be measured by SEM analysis of cross-sections of the separator. Because of the nanoporous pore sizes and the high volume fraction of organic polymer in the nanoporous layer, it has been found that a lower limit for the proportional thickness of the nanoporous layer of about 10% of the total thickness of the separator, rather than 15% or more typically 25%, is able to provide an excellent improvement in the dimensional stability of the separator at high temperatures, which is needed for improved safety. It has been found that the combination of nanoporous pore sizes and very high volume fractions of organic polymer in the nanoporous layer (e.g., from about 10% to about 50%) provides acceptable % porosity (e.g., from about 30% to about 70%) with good ionic conductivity of the electrolyte. It has also been found that the separator of this invention functions well with the total thickness of the nanoporous layer as high as 80% of the total thickness of the separator.

[0041] In one embodiment, the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. This heat shrinkage may be measured by the change in the area of the separator after heating in an oven for 1 hour. This measurement of the heat shrinkage may be done by hanging a large sheet of the separator in the oven after marking and measuring the dimensions and area of a square or rectangle such as a 10 cm by 10 cm square, on a smaller section of the large sheet. After the heating in the oven, the marked area is measured for the % change or % shrinkage of the dimensions and the area.

[0042] In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%. The nanoporous separators of the present invention have been found to retain their dimensional stability even when heated well above the melting point of the polymeric layer, such as a porous polyethylene layer.

[0043] In one embodiment of the separators of this invention, the total coating weight of the nanoporous layer on both sides is less than 4.5 grams per square meter (gsm), while

providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. It should be noted that the use of boehmite with its lower density of about 3.03 g/cc, as compared to the higher density anhydrous aluminum oxide or alumina (which has a density of about 4 g/cc), helps to further lower the coating weight of the heat stabilizing layer. The lower density of the inorganic oxide particle proportionately lowers the weight of coating needed for a specific thickness and % porosity.

[0044] For example, a 4 micron thick alumina/polymer layer with a 45% porosity for polyethylene-based separators typically weighs about 8 grams per square meter (gsm). However, switching to boehmite (or a blend of boehmite and other inorganic particles of a similar or lower density) at the same thickness and % porosity lowers the coating weight to about 6 gsm or less. The high volume fraction of polymer in the nanoporous layers of the separators of this invention, combined with the use of boehmite with a crystallite size of 5 nm to 90 nm, provides increased dimensional stability for a specific thickness of the separator layer. Although it is possible to still make the total thickness of the inorganic particle/polymer layer in the separator 4 microns or higher, it is desirable for lower cost and for lower total thickness of the separator to make the thickness of the inorganic particle/polymer layer as thin as possible, such as about 3 microns total, while still achieving an excellent dimensional stability at high temperature. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0045] In one embodiment, other inorganic oxides and/or inorganic nitrides with a density of about 3.0 g/cc or lower, such as silicon dioxide and boron nitride, may be blended with the boehmite pigment. As discussed further below, this provides lower total coating weight of the nanoporous layer and other desirable properties, such as lower water content.

[0046] In another embodiment of the separators of this invention, the total coating weight of the nanoporous layer of both sides is less than 3.5 grams per square meter (gsm), while providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0047] In one embodiment of the separators of the present invention, the crystallite size of the inorganic particles may be about 30 nm to about 50 nm, preferably about 5 nm to about 25 nm, and more preferably about 5 nm to about 10 nm. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of about 30 nm to about 50 nm. In one embodiment, the crystallite size of the boehmite may be about 5 nm to about 10 nm, and preferably about 5 nm to about 8 nm.

[0048] In another embodiment of this invention, the % porosity of the nanoporous layer is between 30% and 55%. The % porosity of the nanoporous layer may be calculated from the density in g/cc of the materials in the dry coating assuming it is a 100% solid coating with 0% porosity and comparing it to the actual density of the coating. The actual density of the nanoporous layer may be calculated by dividing its coat weight in gsm by its coating thickness to provide the density in g/cc of the nanoporous layer. Dividing the density of the nanoporous layer by the calculated density of the same materials in a non-porous layer and multiplying by 100% provides the % solid fraction by volume in the nanoporous layer. Subtracting this value of the % solid fraction from 100% gives the % pore volume or % porosity of the nanoporous layer. The % porosity can also be measured by mercury porosimetry analysis.

[0049] In one embodiment, the pore volume of the nanoporous layer is between 0.8 cc/g and 1.0 cc/g. The pore volume can be calculated by measuring the coat weight of the nanoporous layer in gsm and measuring the thickness of the nanoporous layer to determine the total cc in a square meter. The pore volume in cc/g is then calculated by dividing the total cc in a square meter by the grams of the gsm measurement. In one embodiment, the % porosity of the nanoporous layer is 55% to 70%. Above this level of % porosity, it is difficult to obtain the cohesive and adhesive strength of the nanoporous layer that are preferred for long term, safer lithium battery separator applications.

[0050] In one embodiment of the separators of the present invention, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 15% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 20% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 25% to 45%.

[0051] In one embodiment of the separators of this invention, the nanoporous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of the pores have a diameter less than 80 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of the pores have a diameter less than 50 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 25 nm or less and at least 70% of the pores have a diameter less than 25 nm.

[0052] In one embodiment of the separators of this invention, the ratio of the air permeability of the separator to the air permeability of the polymeric layer without any

nanoporous layer present is 1.0 to 1.4. As described hereinabove, this air permeability is typically measured on an instrument which provides a Gurley number of the number of seconds for 100 cc of air to pass through the separator. A low level of increase of the air permeability is an important indicator of an expected low decrease in the ionic conductivity of electrolyte in the separator due to adding the inorganic oxide/polymer layer. If the ionic conductivity decreases too much, this can interfere with the cycling rate capability of the lithium battery particularly its power rate properties, and also interfere with the cycling lifetime of the lithium battery.

[0053] In one embodiment of the separators of the present invention, the polymeric layer comprises a polyolefin. Suitable polyolefins include, but are not limited to, polyethylenes and polypropylenes. Other examples of the polyolefin polymers include, but are not limited to, homopolymers of 1-butene, 1-pentene, 1-hexene, and 1-octene, as well as copolymers and terpolymers thereof, used alone or in combination with one or more types of polymers. In one embodiment, the polymeric layer comprises polyethylene. The polymeric layer may be of the “wet process” type which is produced by extrusion followed by solvent extraction to provide the porosity or of the “dry process” type which is produced by extrusion followed by stretching to provide the porosity. Various molecular weights, densities, and stereochemistries of polyethylene and the other polymers, with various melt flow and mechanical properties, may be utilized in the porous polymeric layers.

[0054] For the purposes of better uniform wetting of the coating fluid of the nanoporous layer and of better adhesion of the nanoporous layer to the substrate, the porous polymeric layer may have a coating layer on the surface to which the nanoporous layer is applied.

[0055] In one embodiment of the separators comprising polyethylene in the polymeric layer, the heat shrinkage of the separator at 120°C for 1 hour is less than 1.0%. Without the nanoporous layer of the separators of this invention, the heat shrinkage of a polyethylene separator of comparable total thickness is typically greater than 1.0% at 105°C. In one embodiment, the ratio of the air permeability of the separator after the heat shrinkage at 120°C for 1 hour to the air permeability of the separator before the heat shrinkage at 120°C is 0.8 to 1.2. This is an important feature for allowing vacuum drying of the polyethylene-based separators of this invention at a higher temperature of 120°C without any negative effect on the air permeability and ionic conductivity and, in some cases, with a positive effect on ionic conductivity due to the heat treatment. Vacuum drying is beneficial for lowering the moisture

content of separators for increased lithium battery cycling lifetimes and capacities and for higher lithium battery cycling rate capabilities. Vacuum drying is typically done for 4 to 24 hours at about 80°C for polyethylene separators and at about 90°C for polyethylene separators with one or more inorganic oxide/polymer layers. The capability of using a higher temperature of 120°C and a shorter drying time of 1 to 3 hours for the polyethylene-based separators of this invention is useful.

[0056] In another embodiment of the separators of the present invention, the nanoporous layer is a laminated layer on the polymeric layer. By the term “laminated layer” is meant that the nanoporous layer is not initially formed on the polymeric layer, such as by direct coating, but is instead laminated to the polymeric layer in a subsequent step. For example, the nanoporous layer is coated on a release substrate, and then this nanoporous layer is transferred by lamination onto the polymeric layer. This lamination may optionally include the use of heat and/or the use of some organic solvent to adhere the nanoporous layer to the polymeric layer and to aid in the subsequent delamination of the release substrate without interfering with the quality of the laminated multilayer separator.

[0057] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic particle/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite), or a blend of an inorganic oxide and an inorganic nitride (e.g., BN or AlN), and a polymer, and wherein the crystallite size of the inorganic particles is about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of about 25 nm or less and at least about 70% of the pores have a diameter of about 25 nm or less.

[0058] A further aspect of the separators for a lithium battery of the present invention pertains to a separator comprising (a) a porous polyethylene layer, and (b) a nanoporous inorganic particles/polymer composite layer on both sides of the polyethylene layer, wherein the nanoporous layer comprises boehmite, or a blend of boehmite and boron nitride, and one or more polymers, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is about 10% to about 50%, the total coating weight of the nanoporous layer on both sides is 5.5 grams per square meter or less, and the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. In one embodiment, the total coating weight of the

nanoporous layer on both sides is 3.5 grams per square meter or less and the thickness of the polyethylene layer is between 5 microns and 16 microns.

[0059] In another embodiment, this invention is lithium batteries comprising a separator of the present invention.

[0060] Another aspect of the present invention are methods for making the separator of this invention, which method comprises coating a solution comprising inorganic particles and one or more polymers on one or both surfaces of a porous polymeric layer. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of 5 nm to 90 nm. In one embodiment, the boehmite particles have a crystallite size of about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the coating after drying is about 10% to about 50%, preferably about 12% to about 45 %, and more preferably about 15% to about 45%.

[0061] **EXAMPLES**

[0062] Several embodiments of the present invention are described in the following examples, which are offered by way of illustration and not by way of limitation.

[0063] **Example 1**

[0064] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPAL 10F4 supplied by Sasol Inc., Houston, Texas) and 1 part of chitosan (deacetylated chitin, poly(D-glucosamine), medium molecular weight, of a polysaccharide supplied by Sigma-Aldrich, St. Louis, MO) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 40 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 1.2 grams per square meter (gsm) to about 5.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. The boehmite coating was then applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0065] The 12:1 by weight ratio of boehmite pigment to chitosan binder is calculated to have 15.3% by volume of chitosan binder in the solid portion of the porous boehmite

coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.4 g/cc for chitosan.

[0066] The coating thickness was measured using a Dorsey gauge on the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the thickness values before and after the coating is the coating thickness in microns. An alternative method for measuring the coating thickness is using the Dorsey gauge on the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0067] The coating weight was measured by weighing a 10 cm x 10 cm sample of the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the weights before and after the coating multiplied by 1,000 is the coat weight in grams per square meter (gsm). An alternative method for measuring the coating weight was weighing a 10 cm x 10 cm sample of the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0068] Figure 1 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.6 gsm to about 2.6 gsm. For two sided coatings, these numbers are doubled to give a total coating weight range of about 1.2 gsm to about 5.2 gsm. The total boehmite coating thickness for both sides varies from about 1.2 microns to about 4.4 microns. If a more accurate measurement of the coating thickness is desired, scanning electron microscope (SEM) analysis of cross-sections of the boehmite-coated plastic separator can be utilized.

[0069] Using a SEM analysis for the total coating thickness showed a density of about 1.06 gsm per micron of coating thickness. From this, the % porosity for this 12:1 pigment to binder ratio is estimated as 62%.

[0070] One objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C. Figure 2 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 1.9 microns or thicker on the 12 micron polyethylene substrate of this example. For coatings on two sides, this number is doubled to give a total coating thickness of about 3.8 microns or thicker. Based on the density of the

boehmite coating determined by SEM analysis, a total coating thickness of about 3.8 microns on both sides equates to about 4.0 gsm of coating weight.

[0071] For better wetting and coating adhesion of the boehmite slurry, a corona treatment of the polyethylene substrate is preferred. Besides wire wound coating rods, a variety of other coating application methods may be used, such as, but not limited to, gravure coating, screen printer coating, reverse roll coating, blade coating, and slot die coating.

[0072] **Example 2**

[0073] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPERAL D60 supplied by Sasol Inc., Hamburg, Germany) and 1 part of a vinyl pyrrolidone copolymer (Soteras CCS grade supplied by Ashland Inc., Wilmington, DE) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 75 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 2.4 grams per square meter (gsm) to about 9.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the boehmite coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0074] The 12:1 by weight ratio of boehmite pigment to vinyl pyrrolidone copolymer binder is calculated to have 18.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.1 g/cc for the vinyl pyrrolidone copolymer.

[0075] Figure 3 shows a plot of the average boehmite coating thickness and coating weight per side of about 1.2 gsm to about 4.6 gsm. For two sided coating, these numbers are doubled to give a total coating weight range of about 2.4 gsm to about 9.2 gsm. The total boehmite coating thickness for both sides varies from about 2.8 microns to about 9.0 microns.

[0076] As discussed, one objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C for 1 hour. Figure 4 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 2.0 microns or thicker on the 12 micron polyethylene substrate of this example. For the coatings on two sides, this number is doubled to give a total coating thickness of about 4.0 microns or thicker.

Based on Figure 3, a total coating thickness of about 4.0 microns on both sides equates to about 3.9 gsm of coating weight.

[0077] A boehmite slurry of an 8 to 1 ratio by weight of the DISPERAL D60 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was also coated using wire wound rods on the 12 micron polyethylene substrate and dried, as for the 12:1 boehmite D60:polymer coating in this example. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 5 shows a plot of the average boehmite coating weight and coating thickness per side of about 0.8 gsm to about 4.2 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a range of about 1.6 gsm to 8.4 gsm. The corresponding total boehmite coating thicknesses for both sides varies from about 1.6 microns to 8.2 microns. Figure 6 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 2.2 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 4.4 microns or thicker. Based on Figure 5, a total coating thickness of about 4.4 microns equates to about 4.8 gsm of coating weight.

[0078] **Example 3**

[0079] Using the procedure of Example 2, a boehmite slurry of a 12:1 by weight of DISPAL 25F4 boehmite and the vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was prepared. The DISPAL 25F4 is supplied by Sasol, Lake Charles, LA, and has a primary crystallite particle size of 8 nm. This boehmite slurry was coated on the EP12 12 micron polyethylene substrate from ENTEK and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 18.7%. Figure 7 shows a plot of the average boehmite coating thickness and the coating weight per side of about 0.8 gsm to about 3.0 gsm. For the coatings on two sides for this boehmite coating with a 12:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to about 6.0 gsm. Figure 8 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 1.8 microns or thicker. For two sided coatings, this number is doubled to give a total coating thickness on both sides of about 3.6 microns or thicker. Based on Figure 7, a total coating thickness of about 3.6 microns equates to about 3.6 gsm.

[0080] A boehmite slurry of an 8 to 1 ratio by weight of the DISPAL 25F4 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol was also coated on the 12 micron polyethylene substrate and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 9 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.8 gsm to about 4.7 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to 9.4 gsm. Figure 10 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 0.9 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 1.8 microns or thicker. Based on Figure 9, a total coating thickness of about 1.8 microns equates to about 2.9 gsm of coating weight. For a thickness of about 1.8 microns of the boehmite coating, the coating weight should have been about 2.0 gsm based on the density of the boehmite coating of about 1.1 g/cc shown in Example 1, while factoring in the lower boehmite weight % of the 8:1 boehmite pigment:vinyl pyrrolidone copolymer and the higher vinyl pyrrolidone copolymer weight % and lower density of the vinyl pyrrolidone copolymer compared to chitosan. This shows that about 0.9 gsm of the boehmite coating penetrated into and dried in the pores of the polyethylene substrate to produce a boehmite/polyethylene substrate with a boehmite coating on its surface. At the 8:1 pigment:binder (P:B) ratio of this coating, the about 0.9 gsm of boehmite coating in the pores equates to about 0.8 gsm of boehmite particles in the pores of the boehmite/polyethylene substrate or layer. Since this boehmite coating was applied on both sides of the polyethylene substrate, about one-half, or about 0.4 gsm of boehmite, would have penetrated and dried in the pores on each side of the polyethylene substrate to form the boehmite/polyethylene substrate or layer that resulted in a boehmite-coated boehmite/polyethylene substrate.

[0081] A similar 8:1 boehmite coating with Dispall 25F4, with chitosan substituted for the vinyl pyrrolidone copolymer and a 12 micron thick polyethylene substrate from Shenzhen Senior, Shenzhen, China, used instead of the polyethylene substrate from Entek, was made using the procedure of this Example to give a coat weight of 1.8 gsm. This coating had a heat shrinkage of less than 5% when heated at 150°C for 1 hour. When the procedure of removing the boehmite coating with an adhesive tape was used, even with very careful tape pulls, it was not possible to remove the boehmite coating from the 10 x 10 cm samples without pulling off

some of the polyethylene substrate. This was in contrast to the tape pull results of easy and complete removal of the boehmite coatings from the polyethylene substrate for Examples 1 and 2 and Comparative Examples 1 and 2. This result was consistent with some of the boehmite coating penetrating into the pores of the polyethylene layer and, upon drying, providing increased adhesive and cohesive strength to the boehmite coating such that the point of failure in the tape pull was below the surface of the polyethylene layer at most areas, rather than at the interface of the polyethylene surface and the boehmite coating on the polyethylene surface. After the tape pull, some areas were found where the polyethylene layer had not been disturbed. From these areas, the added thickness of the boehmite layer was found to be about 1.8 microns.

[0082] Since the tape pulls were not able to cleanly remove the boehmite coating from the polyethylene layer, a methanol-saturated soft cloth was used to wash and rub the coating off. The 10 x 10 cm sample after washing had a weight of 0.0757 grams (g). Two more methanol rubs did not change this weight of the washed and rubbed sample. The polyethylene substrate used in this Example contained no inorganic pigment, such as alumina or boehmite. A furnace combustion of the methanol-washed sample in a crucible, at a temperature of 1000 °C for 70 minutes showed 0.0045 g of anhydrous aluminum oxide after dehydrating the boehmite pigment with a weight loss of 15% of the boehmite pigment weight. Using this boehmite to aluminum oxide conversion factor by dividing the aluminum oxide weight by 0.85, this gives a boehmite content of about 0.0053, or at least about 0.53 gsm of boehmite pigment in the pores of the polyethylene layer. Since the wet rubbing to remove the boehmite coating on the surface may have removed a small amount of the boehmite coating in the upper pores of the polyethylene layer, it is possible that the weight of boehmite pigment in the pores was somewhat greater than 0.53 gsm. Since there were two coatings on the polyethylene layer, the amount of boehmite in the pores from a single coating is about one-half of this total, or about 0.27 gsm. As further evidence for penetration of the boehmite coating into the pores of the polyethylene substrate, the dimensional stability of this “washed” polyethylene with about 0.53 gsm of boehmite in the pores after heating at 150°C for 1 hour was 54% and 36.5% in the two directions(--machine direction (MD) and cross direction (CD)). By contrast, the polyethylene substrate before coating showed heat shrinkage of 70% and 75.5%. This increased heat stability with boehmite pigment in the pores of the polyethylene substrate is one explanation for the greater dimensional stability at 150°C with a much lower coat weight and coating thickness added to the surface of the

polyethylene layer. For analyzing the presence and amount of boehmite particles in the pores of the polymeric layer, the removal of the surface coating of the nanoporous layer and subsequent combustion of all of the organic polymer, as described above, in a combustion and/or thermogravimetric method may be used for the case of only boehmite particles in the nanoporous layer and a polymer layer that contained no inorganic or boehmite particles before the application of the boehmite coating. Another method is to use the TEM or SEM method used for measuring the average crystallite size to make images of the presence of boehmite particles in the pores of the polymeric layer and to determine their crystallite size at the same time. Combining this method with the combustion and/or thermogravimetric method above to quantify the weight of the boehmite, or other ceramic particles in the pores, if any, would provide the weight in gsm of boehmite particles in the separator. The known analytical techniques for boehmite particles, such as FTIR, X-ray diffraction, thermogravimetric analysis (TGA), high resolution photoelectron spectroscopy (ESCA), and ²⁷Al nuclear magnetic resonance (nmr) can be utilized to further confirm the presence and quantity of boehmite particles in the pores of the polymeric layer, if needed. An alternative approach for collecting the ceramic particles, such as boehmite particles, in the pores of the polymeric layer, after removing any surface coating from the polymeric layer, would be to do a Soxhlet extraction of the polymeric layer with a hot solvent that dissolves the polymer from the polymeric layer to isolate the insoluble inorganic particles. Besides providing the weight of inorganic particles, this could be combined with the above analytical methods, combustion and/or thermogravimetric analysis, and TEM or SEM method, to confirm the presence and quantity of boehmite particles. One benefit of this penetration of nano-sized boehmite particles would be the ability to make and sell boehmite/polyethylene substrates, with no or minimal surface boehmite or other coating, for various applications, including for manufacturing CCS, where the boehmite/polyethylene substrate would have multiple advantages over polyethylene substrates, including, but not limited to, a much greater dimensional stability at high heats, a modified and smaller pore size distribution, and other benefits, such as ease of wetting by subsequent coatings. One example of a particularly useful application for boehmite/polymeric substrates would be filtration applications, where the very small nano-sized boehmite particles, with and without various surface treatments, in the pores of the polymer could provide the desired nanofiltration and ultrafiltration properties not available from the polymeric substrate without boehmite particles in its pores, and could be adapted for desalination of sea water. This benefit would also be available for other

boehmite/polymeric substrates besides polyethylene, such as for boehmite/polypropylene and boehmite/other polyolefin substrates.

[0083] Example 4

[0084] The procedure of Example 1 was used to make a nanoporous separator comprising a 50:50 blend by weight of boehmite and boron nitride pigments, except that a 12 micron thick porous polyethylene separator from Toray Industries, Tokyo, Japan, was substituted for the Entek polyethylene separator, and the vinyl pyrrolidone polymer, Soteras CCS, was substituted for the chitosan. The boron nitride pigment was NX1, supplied by Momentive Performance Materials. The total coating weight on both sides was 2.5 gsm. Figure 11 shows the reduced water content of this separator compared to the control with only boehmite pigment and no boron nitride. Specifically, as shown in Figure 11, at a 50:50 by weight ratio of boehmite and boron nitride, the water content at 20% RH was reduced by one-third from its value with boehmite pigment only.

[0085] Comparative Example 1

[0086] An alumina mix was prepared in water by adding 20 parts of alumina pigment (HJA-0719 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the alumina was greater than 400 nm. The alumina mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 4.4 gsm. The alumina coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0087] The 20:1 by weight ratio of alumina pigment to hydroxypropyl cellulose polymer binder is calculated to have 13.6% by volume of polymer binder in the solid portion of the porous alumina coating layer. This calculation is based on a specific gravity of 4.0 g/cc for alumina and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating

weight of about 4.4 gsm, this alumina coating had a shrinkage of about 60% at 150°C for 1 hour.

[0088] **Comparative Example 2**

[0089] A boehmite mix was prepared in water by adding 20 parts of boehmite pigment (BG-613 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the boehmite was greater than 400 nm. The boehmite mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 7.0 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0090] The 20:1 by weight ratio of boehmite pigment to hydroxypropyl cellulose polymer binder is calculated to have 10.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating weight of about 7.0 gsm, this boehmite coating had a shrinkage of about 60% at 150°C for 1 hour.

[0091] Table 1 shows a summary of the data from Examples 1 to 3.

Table 1

Example	Boehmite Type	Primary Particle Size (nm)	Pigment:Polymer Binder Weight Ratio	Volume % Polymer	Total Coat Weight for <5% Shrinkage at 150°C for 1 hour
1	10F4	40	12:1	15.3%	≥4.0 gsm
2	D60	75	12:1	18.7%	≥3.9 gsm
2	D60	75	8:1	25.6%	≥4.8 gsm

3	25F4	8	12:1	18.7%	≥3.6 gsm
3	25F4	8	8:1	25.6%	≥2.9 gsm

[0092] One challenge in using nano-sized ceramic particles, such as nano-sized boehmite particles, is their increased tendency to absorb water because of their much larger surface areas compared to micro-sized ceramic particles, such as micro-sized alumina and micro-sized boehmite particles. For lithium ion cells, lithium metal cells, and other electrochemical cells with sensitivity to water content, the separator needs to have a low water content, such as under 600 ppm of water in the separator, preferably less than 300 ppm of water, and more preferably under 100 ppm of water as typically found in porous polymer only separators with no ceramic materials in them. Although vacuum drying of the separator at up to 80 °C can help to lower this moisture content in separators with nano-sized ceramic particles, and the separators of this invention have increased thermal stability that permits even higher vacuum drying temperatures of up to 90 °C to about 110 °C, some users of separators may prefer not to perform this extra step of vacuum drying. It would be preferable if the greater heat stability, lower cost from the use of a lower amount of ceramic coating materials, and lower ceramic coating thickness, including from the penetration of the ceramic coating and ceramic particles into the porous polymeric separator layer, of the separators of this invention would include a lower moisture content in the separators from the use of more hydrophobic ceramic particles and organic polymers. Having a lower amount of ceramic particles from the lower coating weights of the invention is a significant benefit for lowering the moisture content of the separator, but it would be further useful if the ceramic particles were more hydrophobic and less prone to moisture pickup.

[0093] One approach which provides this lower moisture absorption is to treat the surface of the ceramic particles with a hydrophobic material, such as by using hydrophobically-treated boehmite particles, particularly if the hydrophobic material is chemically reacted with the boehmite particles and thus is permanently bonded to the boehmite particles. Suitable hydrophobic materials for chemical modification of boehmite pigments include, but are not limited to, organic sulfonic acids, organic carbonates, and polymers with hydrophobic groups and with reactive groups, such as carboxylic acid and hydroxyl groups, that can be made to react with hydroxyl groups of the boehmite or other ceramic particles.

[0094] Examples of the organic sulfonic acid-treated boehmite particles include Dispal® 25SR, available from Sasol in Lake Charles, LA, a boehmite pigment chemically reacted with

p-toluene sulfonic acid through the sulfonic acid groups. The reaction leaves the hydrophobic p-toluene group on the outer surface of the boehmite particle to make it less prone to pick up water. An even more hydrophobic example from Sasol is p-dodecylbenzene sulfonic acid surface reacted to boehmite particles through the sulfonic acid group and having the very hydrophobic dodecyl group on the outer surface of the boehmite pigment. This makes the boehmite particles so hydrophobic that this treated boehmite can be dispersed in hydrocarbon solvents. The use of boehmite particles surface treated with organic sulfonic acids in boehmite coatings for separators is disclosed in U.S. Patent Application Publication No. 2013/0171500 by Xu et al.

[0095] Examples of the organic carbonate-treated boehmite particles include, but are not limited to, those with ethylene carbonate and fluoroethylene carbonate as the organic carbonates. Besides the option of choosing an organic carbonate that is very hydrophobic in the part of the molecule that will face away from the surface of the boehmite, the choice can also be made with a view to picking an organic carbonate which will provide increased ionic conductivity to the separator. The use of boehmite particles surface treated with ethylene carbonate to provide separators with increased conductivity is described in U.S. Pat. No. 8,883,354 to Carlson et al.

[0096] The use of boehmite particles surface treated with organic polymers to provide separators is described in U.S. Pat. No. 9,871,239, to Carlson et al. As the primary crystallite size and resulting pore size become smaller in the boehmite separators and the ionic conductivity of the separator becomes lower because of the smaller pore sizes, it is useful to have surface treatments, such as the three types described above, that can be utilized to increase the ionic conductivity of the separator, as well as to decrease the water content of the separator.

[0097] The preparation of a ceramic mix or slurry with the surface-treated hydrophobic materials can be done conveniently by adding a precursor, such as an organic carbonate, to the slurry containing the untreated boehmite particles and heating the slurry to a temperature, such as 80 °C, for a sufficient time, such as 1 hour, to react the surface treatment material, such as an organic carbonate, with the boehmite particles to the desired level of surface treatment. There is no need to isolate the surface-treated boehmite particles by a drying process, but rather the slurry can be used as is, usually after some dilution with water and/or solvents to the desired % solids level for the coating of the porous polymer substrate. If it is desired to further reduce the particle size of the ceramic particles, such as boehmite particles,

and/or to homogenize the slurry for more uniformity and perhaps a lower viscosity at the desired % solids level for coating, the slurry, before and/or after any surface treatment, can be further mixed in an impingement mill or similar milling device. Examples of the use of an impingement mill for reducing and homogenizing coating mixes containing pigments is described in U.S. Pat. Nos. 5,210,114 and 5,292,588, to Katsen.

[0098] While the invention has been described in detail and with reference to specific and general embodiments thereof, it will be apparent to one skilled in the art that various changes and modifications can be made therein without departing from the spirit and scope thereof.

CLAIMS

1. A separator for use in a lithium battery comprising:
 - (a) a porous polymeric layer, which porous polymeric layer comprises a plurality of pores, and
 - (b) 0.27 grams, or greater, per square meter of boehmite particles disposed within the plurality of pores of the porous polymeric layer.
2. The separator of claim 1, wherein the boehmite particles comprise hydrophobic boehmite particles.
3. The separator of claim 2, wherein the hydrophobic boehmite particles comprise a reaction product of an organic carbonate with boehmite.
4. The separator of claim 3, wherein the organic carbonate is ethylene carbonate.
5. The separator of claim 3, wherein the organic carbonate is fluoroethylene carbonate.
6. The separator of claim 2, wherein the hydrophobic boehmite particles comprise an organic polymer chemically reacted to at least a portion of the boehmite of said boehmite particles.

7. The separator of claim 1, wherein an average crystallite size of said boehmite particles disposed within the pores of said porous polymeric layer is between 5 nm and 25 nm.
8. The separator of claim 1, wherein an average crystallite size of said boehmite particles disposed within the pores of said porous polymeric layer is between 5 nm and 8 nm.
9. The separator of claim 1, wherein the separator further comprises a nanoporous layer adjacent at least one of two opposing sides of the porous polymeric layer, wherein said nanoporous layer comprises boehmite particles and one or more organic polymer binders.
10. The separator of claim 9, wherein the boehmite particles of said nanoporous layer comprise hydrophobic boehmite particles.
11. The separator of claim 10, wherein the hydrophobic boehmite particles of said nanoporous layer comprise a reaction product of an organic carbonate with boehmite.
12. The separator of claim 11, wherein the organic carbonate is ethylene carbonate.
13. The separator of claim 11, wherein the organic carbonate is fluoroethylene carbonate.

14. The separator of claim 10, wherein the hydrophobic boehmite particles of said nanoporous layer comprise an organic polymer chemically reacted to at least a portion of the boehmite of said boehmite particles.
15. The separator of claim 9, wherein the boehmite particles of said nanoporous layer have an average crystallite size between 5 nm and 90 nm.
16. The separator of claim 9, wherein the boehmite particles of said nanoporous layer have an average crystallite size between 30 nm and 50 nm.
17. The separator of claim 9, wherein a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.
18. The separator of claim 9, wherein a heat shrinkage of said separator at 200°C for 1 hour is less than 5%.
19. The separator of claim 9, wherein a heat shrinkage of said separator at 120°C for 1 hour is less than 1.0%.
20. A lithium battery comprising
 - (a) an anode;
 - (b) a cathode;
 - (c) an organic electrolyte comprising a lithium salt; and
 - (d) a separator interposed between said anode and said cathode, wherein said separator comprises:

- (i) a porous polymeric layer, which porous polymeric layer comprises a plurality of pores, and
- (ii) 0.27, or greater, grams per square meter of boehmite particles disposed within the plurality of pores of the porous polymeric layer.

Figure 1

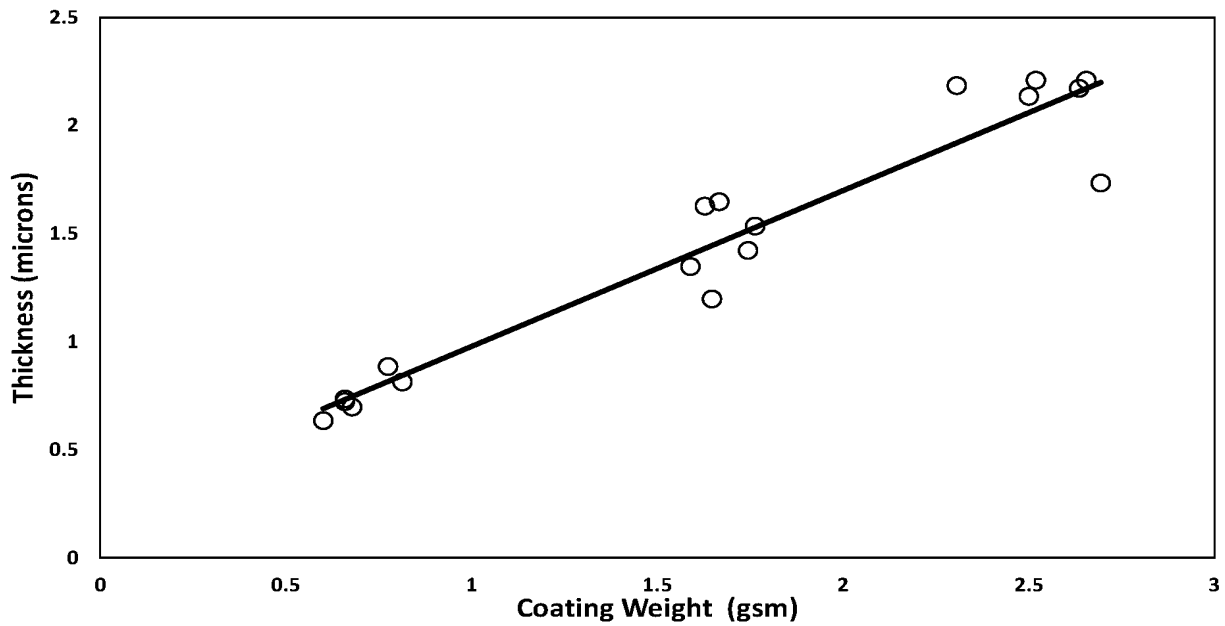


Figure 2

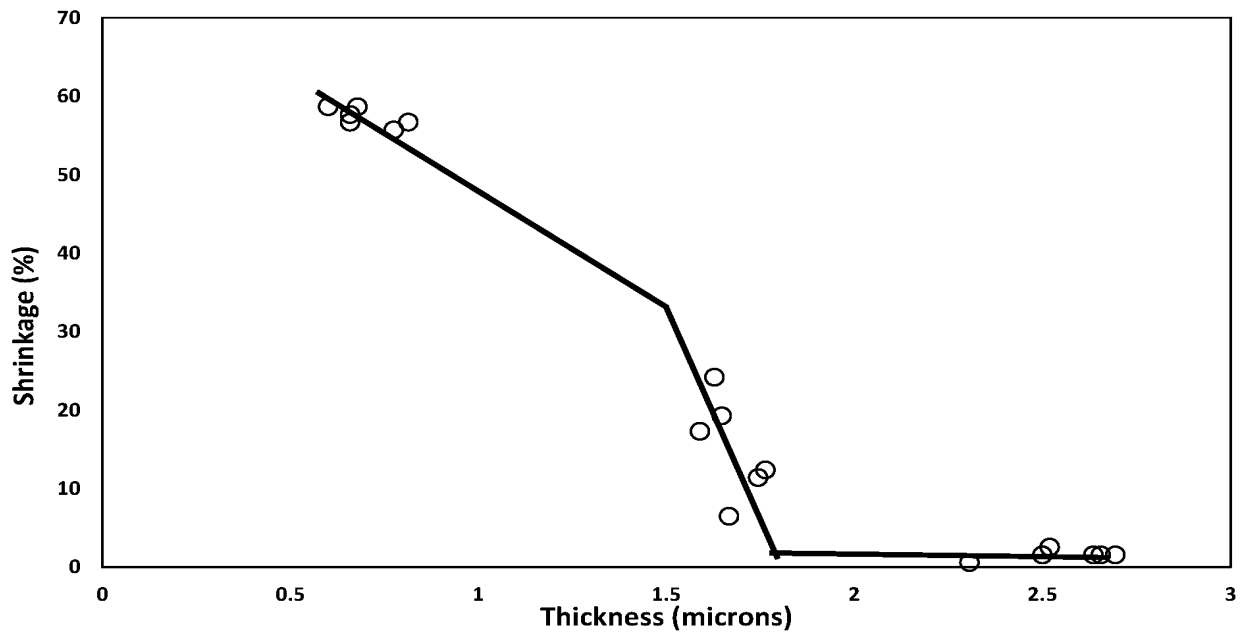


Figure 3

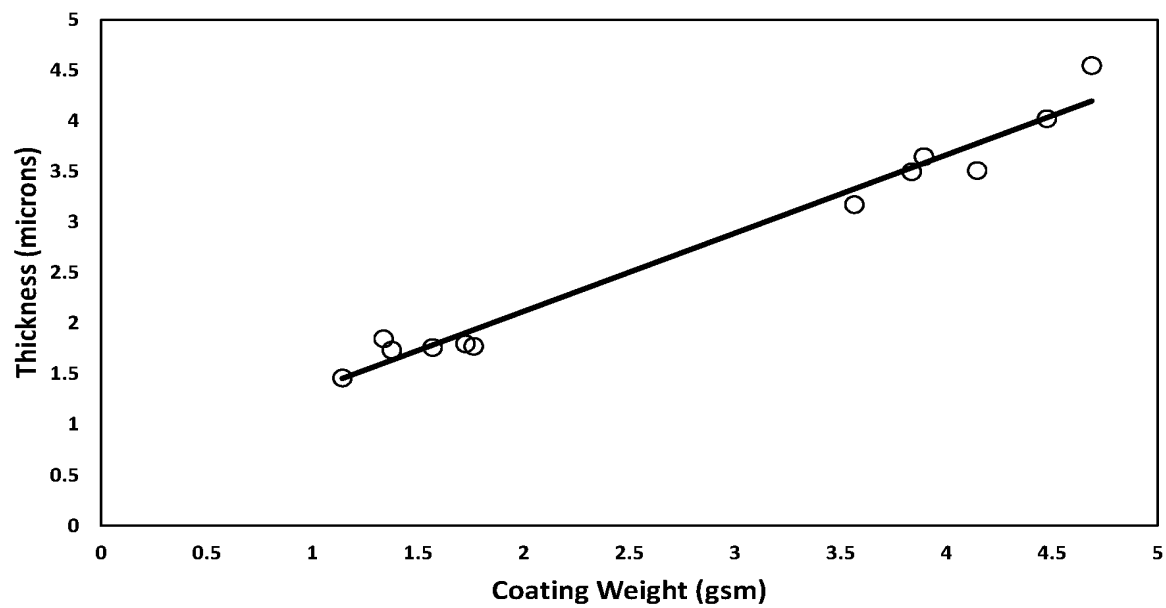


Figure 4

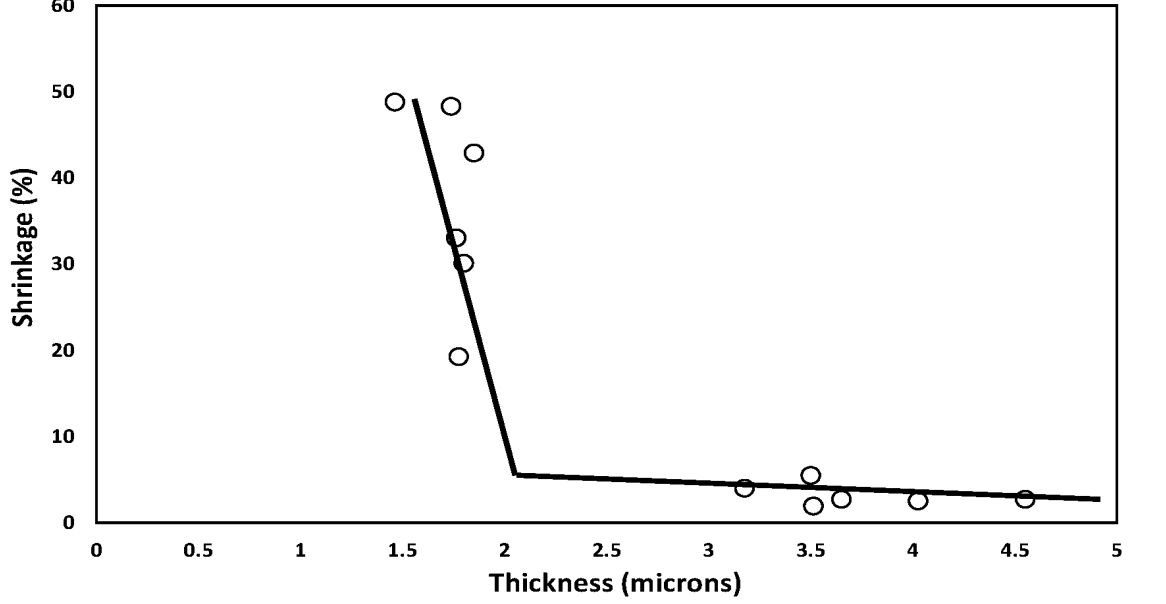


Figure 5

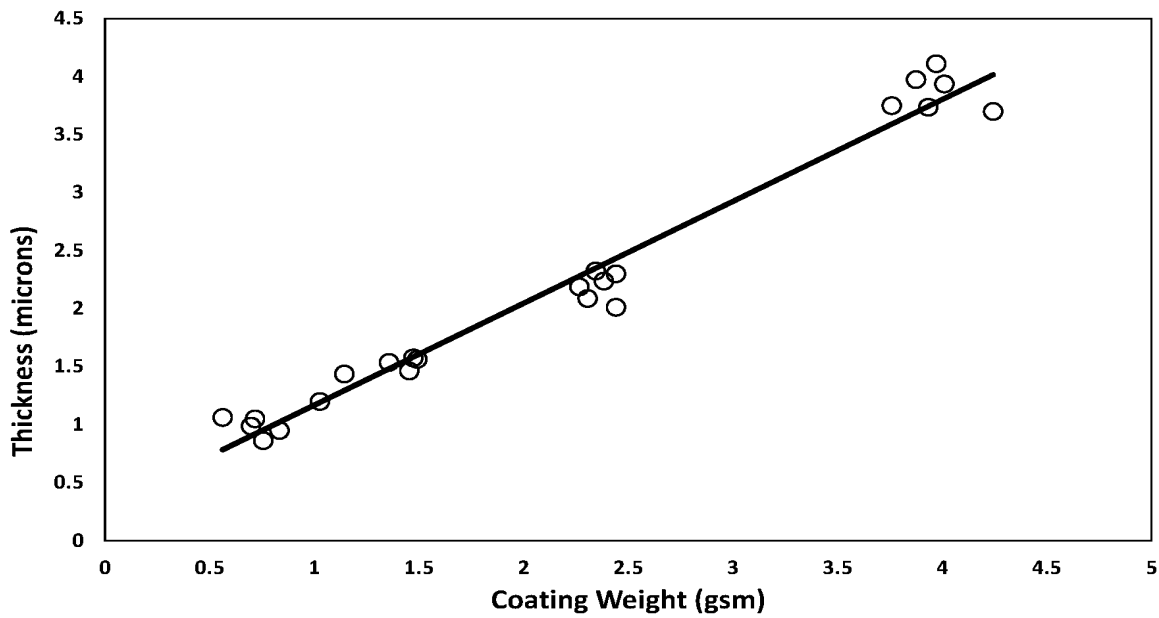


Figure 6

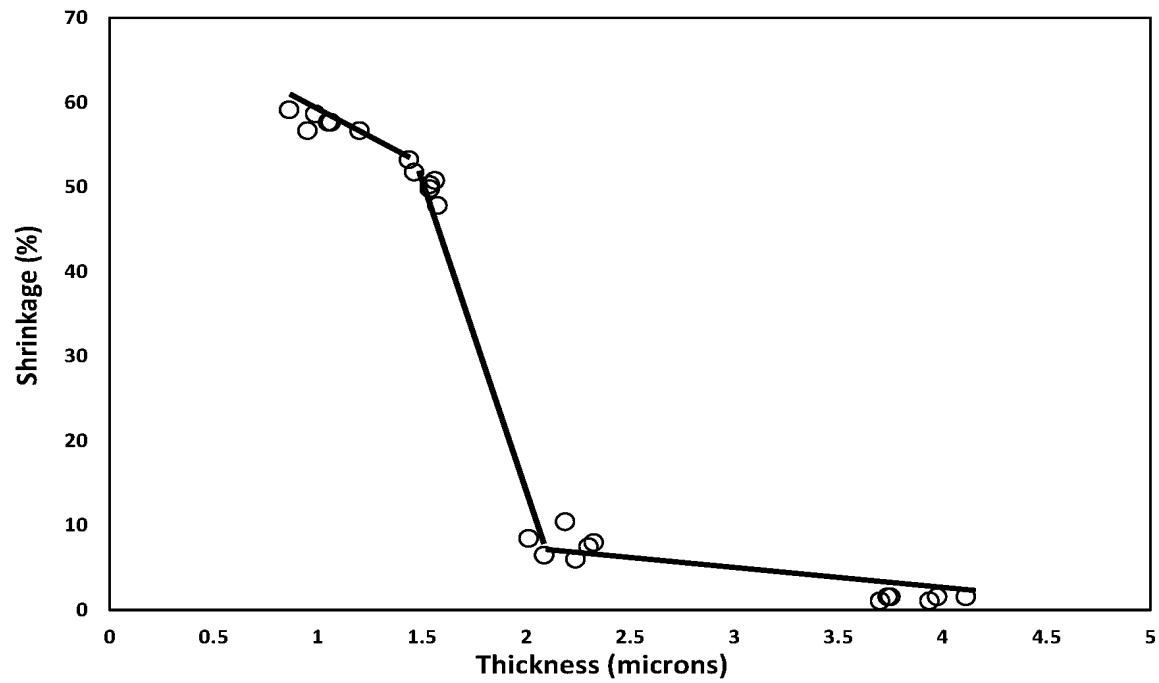


Figure 7

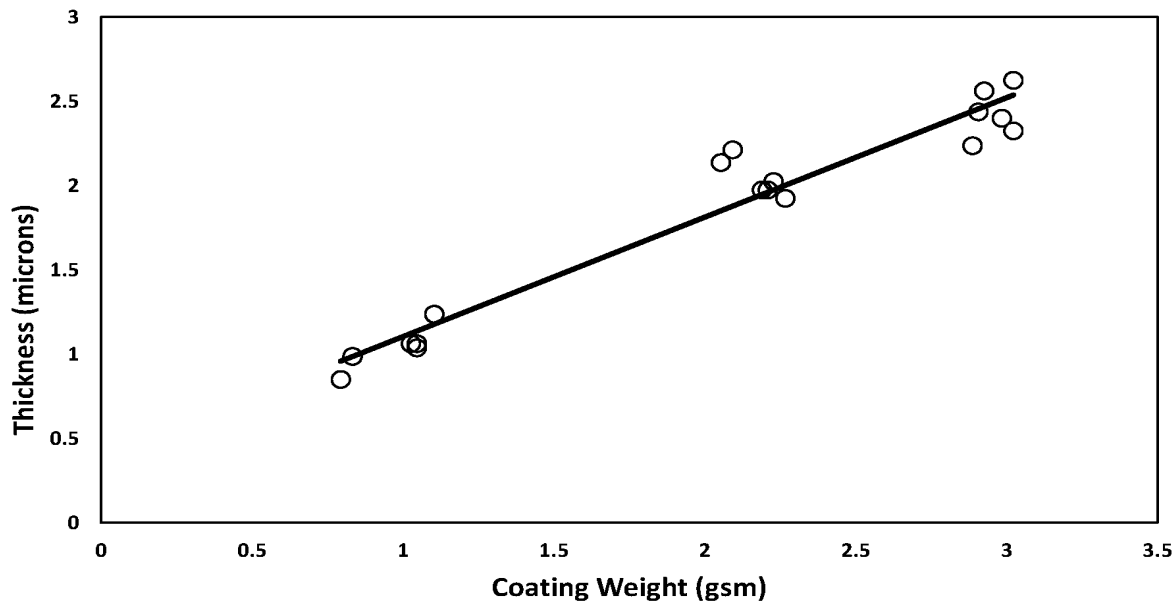


Figure 8

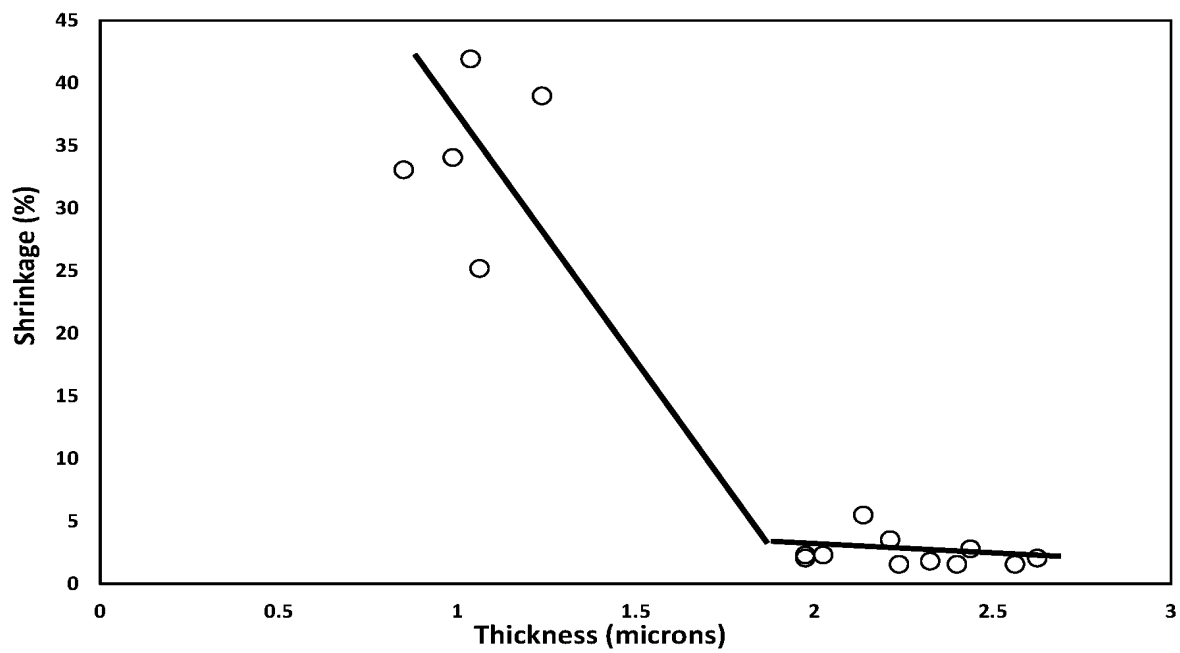


Figure 9

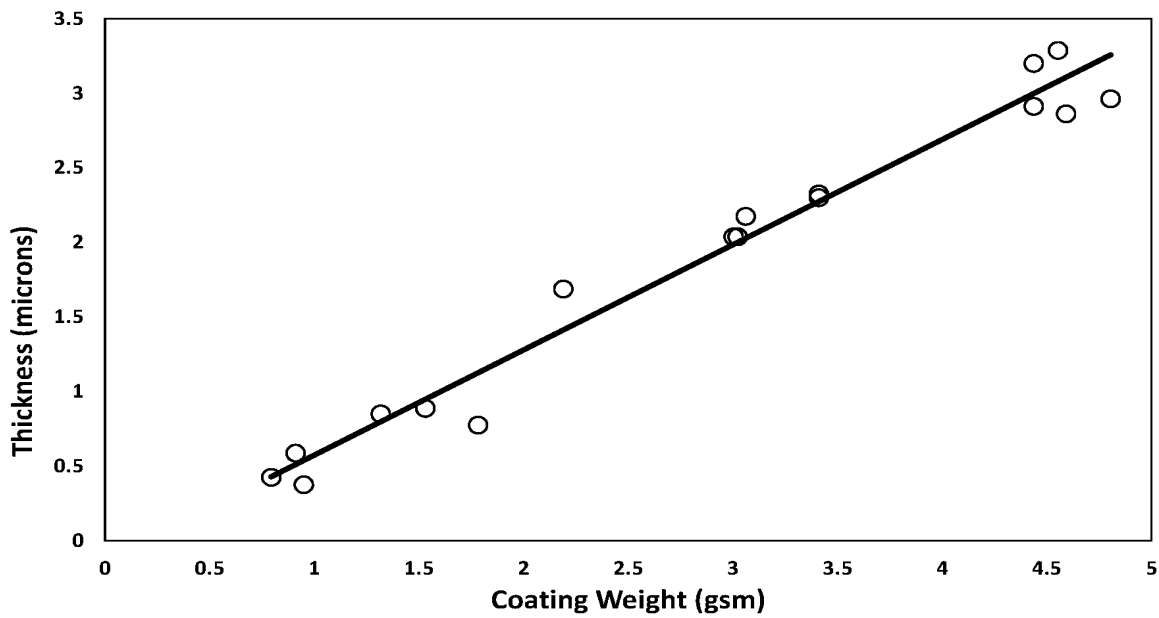
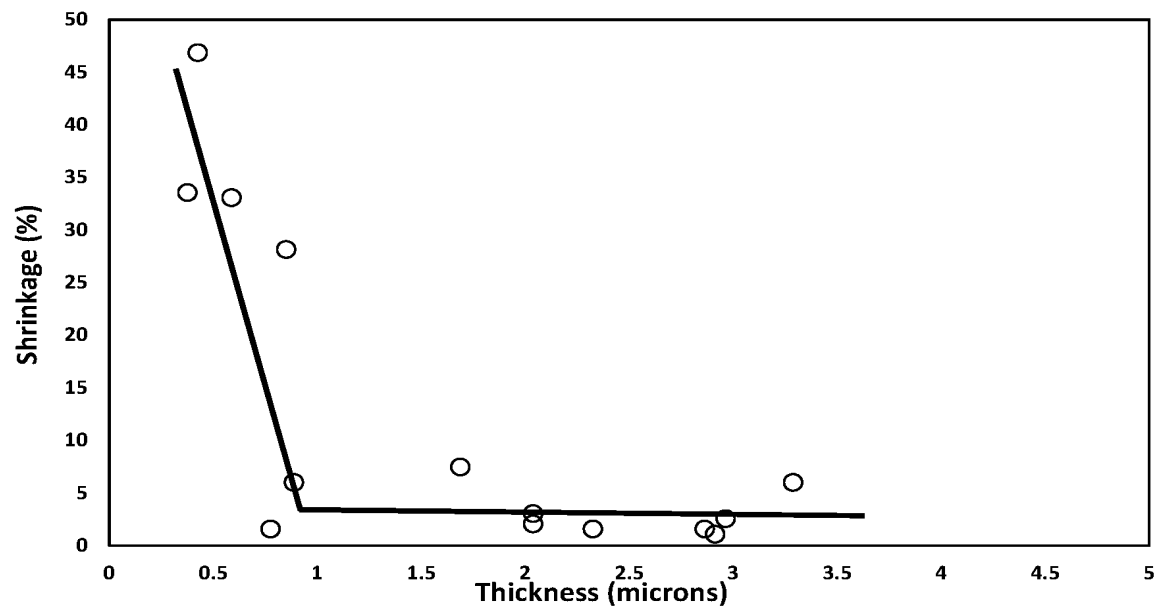


Figure 10



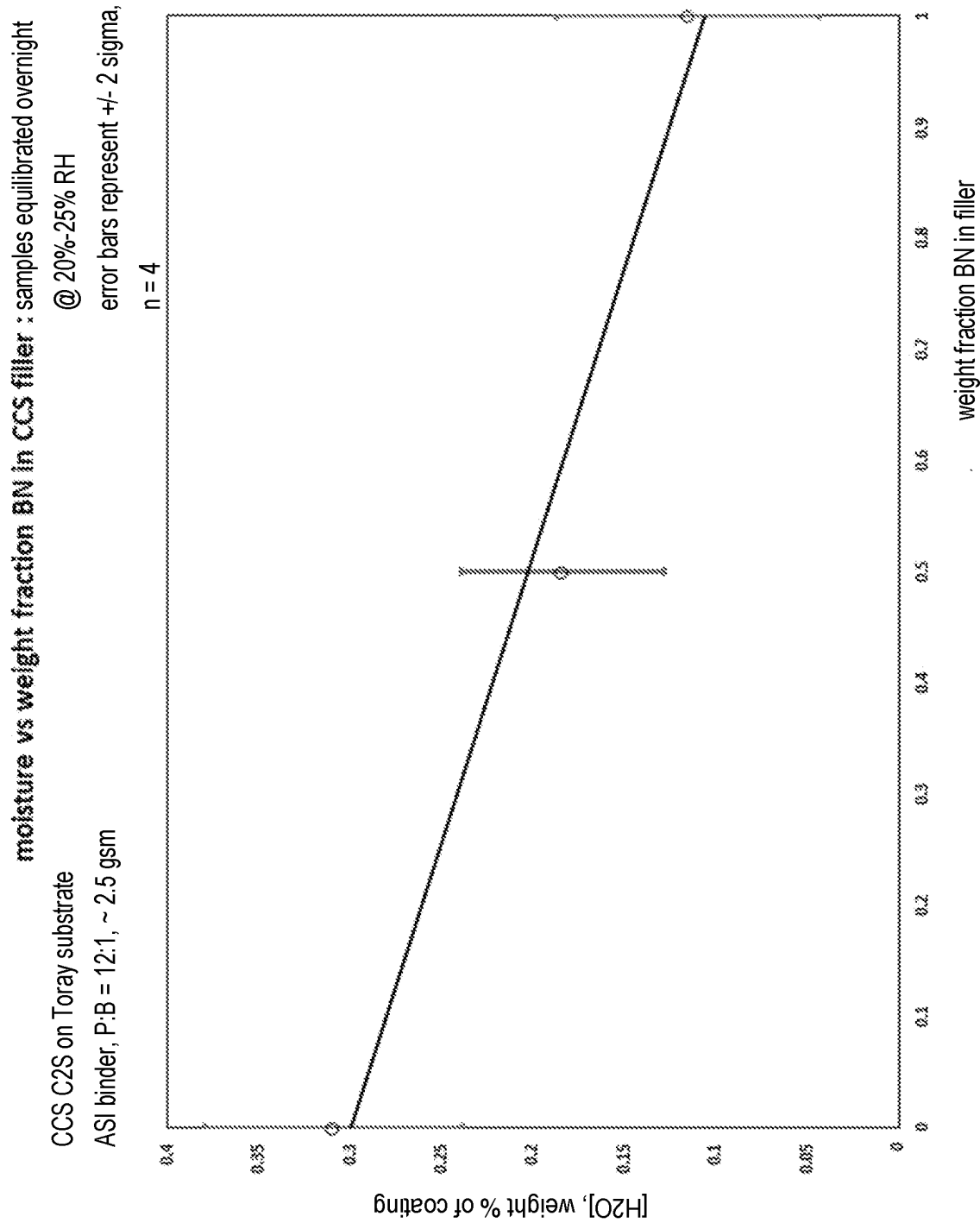


Fig. 11

Docket No. 65213/155

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No. : Not Yet Assigned (Continuation of U.S. Appln. No. 17/223,845)
Filing Date : Herewith
For : MULTILAYER NANOPOROUS SEPARATOR
Customer No. : 1912

**PRELIMINARY AMENDMENT AND
INFORMATION DISCLOSURE STATEMENT**

EFS
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Please amend the above-identified application as set forth below.

EFS

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No. : Not Yet Assigned (Continuation of U.S. Appln. No. 17/223,845)
Filing Date : Herewith
Page 2 of 4

Amendments to the Specification:

On page 1, directly after the title, please amend the "Cross-Reference to Related Applications" section as follows:

[0001] This application is a continuation of U.S. Patent Application No. 17/223,845, filed on April 6, 2021, which is a continuation of U.S. Patent Application No. 16/755,294, filed on April 10, 2020 (now U.S. Patent No. 10,998,536), which is a U.S. national stage entry under 35 U.S.C. §371 of PCT International Patent Application No. PCT/US2018/055862, filed on October 15, 2018, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/572,083, filed on October 13, 2017, the entire contents of which are incorporated by reference herein.

EFS

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No. : Not Yet Assigned (Continuation of U.S. Appln. No. 17/223,845)
Filing Date : Herewith
page 3 of 4

REMARKS

Claims 1-20 are pending in the subject application. By this Preliminary Amendment, the specification has been amended to update the priority data for the subject application.

Applicant maintains that the amendments do not raise an issue of new matter. Entry of the amendments is respectfully requested.

Information Disclosure Statement

In accordance with the duty of disclosure under 37 C.F.R. §1.56, applicant would like to direct the Examiner's attention to the references that are listed on the attached Form PTO/SB/08A. These references were previously submitted to, or cited by, the Patent Office during prosecution of U.S. Patent Application No. 17/223,845, from which the subject application claims priority under 35 U.S.C. §120. Accordingly, pursuant to 37 C.F.R. §1.98(d), copies of previously cited or submitted non-U.S. patent references are not included with this Information Disclosure Statement.

EFS

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No. : Not Yet Assigned (Continuation of U.S. Appln. No. 17/223,845)
Filing Date : Herewith
page 4 of 4

CONCLUSION

The following fees for a large entity are being paid herewith by credit card: 1) the basic filing fee (\$320.00), 2) examination fee (\$800.00), and 3) search fee (\$700.00). No other fee is deemed necessary in connection with the filing of this application, Preliminary Amendment and Information Disclosure Statement. However, if any other fee is required to preserve the pendency of the subject application, authorization is hereby given to charge the amount of any such fee to Deposit Account No. 01-1785. Overpayments may also be credited to Deposit Account No. 01-1785.

Respectfully submitted,

AMSTER, ROTHSTEIN & EBENSTEIN LLP
Attorneys for Applicant
90 Park Avenue
New York, New York 10016
(212) 336-8000

Dated: December 6, 2021
New York, New York

By /Brian J. Amos/
Brian J. Amos, Reg. No. 63,679

DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)Title of
Invention

MULTILAYER NANOPOROUS SEPARATOR

As the below named inventor, I hereby declare that:

This declaration
is directed to:

The attached application, or

United States application or PCT international application number 16/755,294filed on April 10, 2020

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WARNING:

Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.

LEGAL NAME OF INVENTORInventor: David W. Avison

Date (Optional): _____

Signature: David W. Avison

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-5139 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of
Invention

MULTILAYER NANOPOROUS SEPARATOR

As the below named inventor, I hereby declare that:

This declaration
is directed to

☐

The attached application, or

☒

United States application or PCT international application number 16/755,294

filed on April 10, 2020

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

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LEGAL NAME OF INVENTOR

Inventor: Steven A. Carlson

Date (Optional): _____

Signature: Steven A. Carlson

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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**DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN
APPLICATION DATA SHEET (37 CFR 1.76)**Title of
Invention

MULTILAYER NANOPOROUS SEPARATOR

As the below named inventor, I hereby declare that:

This declaration
is directed to:☐

The attached application, or

☒United States application or PCT international application number 16/755,294filed on April 10, 2020

The above-identified application was made or authorized to be made by me.

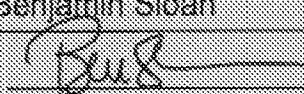
I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001
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LEGAL NAME OF INVENTORInventor: Benjamin Sloan

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Application Number	
Filing Date	
First Named Inventor	David W. Avison
Title	MULTILAYER NANOPOROUS SEPARATOR
Art Unit	
Examiner Name	
Attorney Docket Number	65213/155

SIGNATURE of Applicant or Patent Practitioner			
Signature	/Brian J. Amos/	Date (Optional)	December 6, 2021
Name	Brian J. Amos	Registration Number	63,679
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

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Application Number	Filing Date

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I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

Optodot Corporation

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
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SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature	<i>Steven A. Carlson</i>	Date (Optional)	4/14/2016
Name	Steven A. Carlson		
Title	President, CEO		

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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number		65213/155

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See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

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Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2021-12-06
Name/Print	Brian J. Amos	Registration Number	63,679

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Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number		65213/155

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	9293752	B2	2016-03-22	Murata et al.	
	2	9847519	B2	2017-12-19	Pekalaet et al.	
	3	7709152	B2	2010-05-04	Kim et al.	
	4	7662517	B2	2010-02-16	Lee et al.	
	5	7638241	B2	2009-12-29	Lee et al.	
	6	6432586	B1	2002-08-13	Zhang	
	7	6723467	B2	2004-04-20	Yoshida et al.	
	8	7883799	B2	2011-02-08	Seo et al.	

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Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number	65213/155	

	9	8318360	B2	2012-11-27	Seo et al.	
	10	9484565	B2	2016-11-01	Miyazaki et al.	
	11	9312527	B2	2016-04-12	Muramatsu et al.	
	12	9543077	B2	2017-01-10	Honda et al.	
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	2	20170294638	A1	2017-10-12	Joo et al.	
	3	20180019457	A1	2018-01-18	Alharizah et al.	(U.S. counterpart of WO 2016/123404 A1 listed below)
	4	20180248164	A1	2018-08-30	Pekala et al.	

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First Named Inventor	David W. Avison	
Art Unit		
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Attorney Docket Number	65213/155	

	5	20170301903	A1	2017-10-19	Choi et al.	
	6	20160104876	A1	2016-04-14	Avison et al.	
	7	20110165450	A1	2011-07-07	Nishikawa	
	8	20110003209	A1	2011-01-06	Katayama et al.	
	9	20070190427	A1	2007-08-16	Carlson et al.	
	10	20150155601	A1	2015-06-04	Nakatsutsumi et al.	

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	1	2016123404	WO	A1	2016-08-04	Hercules Incorporated	(WO counterpart of U.S. 2018/0019457 A1 listed above)	

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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
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Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number	65213/155	

	1	F. Sola et al., "A new scanning electron microscopy approach to image aerogels at the nanoscale; A new scanning electron microscopy approach to image aerogels at the nanoscale," Nanotechnology, IOP, BRISTOL, GB, Vol. 22, No. 17, March 16, 2011, page 175704, XP020188888, ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704.	
	2	Chongwen Yang et al., "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries," Journal of Power Sources, Vol. 348, March 3, 2017, pp. 80-86, XP029949881, ISSN: 0378-7753, DOI: 10.1016/J.POWSOUR.2017.02.078.	
	3	International Search Report received for PCT Patent Application No. PCT/US2018/055862, mailed on April 11, 2019, 6 pages,	
	4	Written Opinion received for PCT Patent Application No. PCT/US2018/055862, mailed on April 11, 2019, 13 pages.	

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A new scanning electron microscopy approach to image aerogels at the nanoscale

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Abstract

A new scanning electron microscopy (SEM) technique to image poor electrically conductive aerogels is presented. The process can be performed by non-expert SEM users. We showed that negative charging effects on aerogels can be minimized significantly by inserting dry nitrogen gas close to the region of interest. The process involves the local recombination of accumulated negative charges with positive ions generated from ionization processes. This new technique made possible the acquisition of images of aerogels with pores down to approximately 3 nm in diameter using a positively biased Everhart–Thornley (ET) detector.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Aerogels are highly nanoporous materials that with a thin, three-dimensional (3D) skeletal structure forming the walls of the interconnecting pores. The very high pore volume (often greater than 90%), tortuosity of the pore structure and thinness of the solid struts render the aerogels excellent insulating materials by decreasing both gaseous and solid conduction [1, 2]. Their low optical refractive index properties have been proved to be suitable to detect Cherenkov radiation in high energy physics experiments [3]. Moreover, the highly porous (and low density) structure, exceptional ability to capture hypervelocity dust and optical transparency of aerogels have been exploited to collect cosmic dust in aerospace experiments [4]. Due to their low dielectric constant, aerogels have been used in metal–insulator–semiconductor devices to study current leakage related to high electric fields generated in such devices [5]. It should be noted that important quantities that define the final physical properties of aerogels include: density (or porosity), surface area, dielectric constant, refractive index and thermal conductivity. In some cases, these quantities have a linear dependence with porosity (or density) and thus can be estimated if the porosity content is known [6]. For aerogels to be used as insulators at high temperatures it is important to distinguish thermal exposure

conditions under which the pore structure is maintained as well as to differentiate among aerogel formulations that densify and those that maintain high concentrations of nanopores. Therefore, it is technologically important to find ways to study the porous structure of aerogels.

Both transmission electron microscopy (TEM) and scanning electron microscopy (SEM) techniques are commonly used to estimate the pore size in aerogels [7–9]. One common problem encountered in both techniques during aerogel imaging is the charging effect. These effects are caused by charge accumulation and are due to the dielectric nature and poor electrical properties of the silica and aluminosilicate aerogels of most interest as thermal insulators. Care must be taken during TEM inspection of these materials at the nanoscale, because high electric fields resulting from charge accumulation can cause surface mass migration of SiO_2 species and may form local areas with collapsed porous structures [10]. If a SiO_2 aerogel or aerogel containing SiO_2 is to be investigated by SEM, the most common employed solution to minimize charging effects is to deposit a thin layer of a conducting material (e.g. Au). However, this technique may drastically mask (or even alter) the intrinsic porous structure of the aerogel, which makes the image difficult to interpret [8, 9]. Here we present a new approach to minimize charging effects on aerogels during imaging by *in situ* SEM.

The process involves the local recombination of accumulated charges due to irradiation with charges induced from inert gas injection near the surface of the sample.

2. Experimental details

The aluminosilicate aerogel samples were synthesized by hydrolysis and condensation of a Boehmite powder (Sasol Dispal 25P4, a 7.2 nm crystallite size powder) with tetraethylorthosilicate (TEOS, Sigma Aldrich). The powder was dispersed in acid solution (Jar A). TEOS was added to a 200 proof ethanol solution containing stoichiometric quantities of water (Jar B), and allowed to hydrolyze for 1 h. The contents of the two jars were combined, stirred briefly, and the sol poured into polyethylene molds. The molds were covered with parafilm, and the sol allowed to gel overnight at ambient temperature. The hydrogels were extracted into absolute ethanol, which was exchanged after 24 h. The gels were allowed to age for several days to a few weeks prior to removal of the ethanol solvent by supercritical CO₂ extraction. Surface area and pore size distribution were determined using nitrogen desorption techniques.

For the *in situ* SEM charge compensation experiments we used a Carl Zeiss UltraPlus-Schottky thermal field emission scanning electron microscope, operated at 3 keV and at a working distance of about 6 mm. The instrument was equipped with a charge compensator (CC) device and its movement was driven pneumatically. The CC device used a needle to inject dry nitrogen gas locally, close to the sample surface at about 1 mm. The amount of the injected gas (CC pressure) was regulated between 0% and 100% by a software slider and the corresponding chamber pressure ranged from about 10⁻⁴ to 1 Pa. A positively biased ET detector and line scan at a scan rate of 2 with a line average of 64 and corresponding cycle time of 12.6 s was used for signal collection. It should be noted that the positively biased ET detector is designed to collect both SE and BSE signals. For each collected image, the full three scans were performed after each systematic increase of CC pressure. The aerogel samples were attached to a grounded aluminum disc using conductive silver paste. Finite element (FE) calculations were performed with the Maxwell 2D program from Ansoft Corporation.

3. Results and discussion

When a sample is irradiated with a primary beam of electrons (PE), several signals are generated. A crude estimation for the penetration (or range R) of PE in these aerogels can be obtained using

$$R (\mu\text{m}) = \frac{0.115 E_0^n (\text{keV})}{\rho (\text{g cm}^{-3})} \quad (1)$$

where E_0 is the primary beam energy and is equal to 3 keV and ρ is the density of the aerogels [11]. With $n = 1.35$ (valid for $E_0 < 5$ keV) and average ρ value of 0.06 g cm⁻³, R is approximately 9 μm . Pertinent to this work is the generation of backscattered electrons (BSE) and secondary electron (SE)

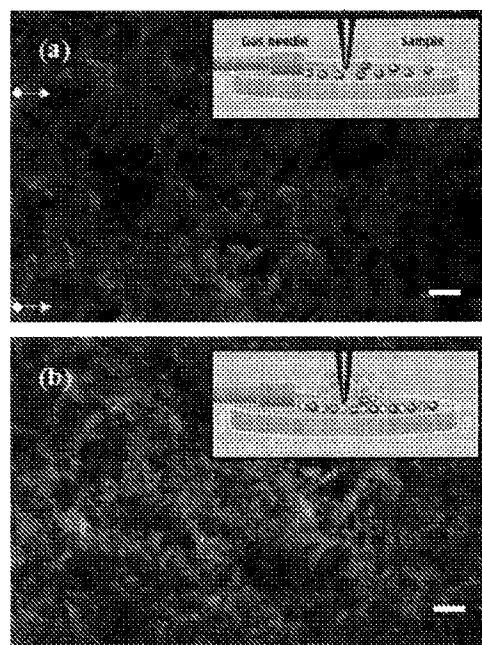


Figure 1. SEM images of aerogels with charging artifacts (gas needle off) (a) and when the charge compensation device is working at 100% (b). Some artifact defocused lines are marked with solid arrows. The scale bar is 100 nm and magnification is $\sim 202\times$. Insets are the corresponding schematic representations of case (a) and (b), respectively.

signals. BSE are produced by multiple elastic scattering events and have relatively high energies, while low energy SE are produced by inelastic scattering events. Associated with the generation of SE and BSE, is the secondary electron (δ) and the backscattering (η) yield coefficients. Generally, during the irradiation process a sample can be charged positively, negatively or be in a neutralized state [12]. The process can be described by using the law of charge conservation, which must hold for all times (t). In general terms

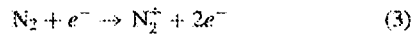
$$I_{PE}[1 - (\delta + \eta)] - I_L = \frac{\partial Q}{\partial t} \quad (2)$$

where I_{PE} is the current injected to the sample by PE, I_L is the leakage current and $\partial Q/\partial t$ is the charge build-up rate [11]. It follows that if the current injected into the sample by PE is greater than the total emission/leakage current, then the sample will have an excess of electrons. This will create a region (of the order of R) of accumulated negative charges on the surface of the sample, and hence the sample will be negatively charged. Typical consequences of this negative charging are the production of image artifacts which include, for instance, image distortion (due to deflection and defocus of PE) and irregular white contrast features.

Figure 1(a) shows an SEM image of an aluminosilicate aerogel sample having a surface area of 256 m² g⁻¹ and 9.45 nm average pore size undergoing charging effects. These artifacts are induced by negative charging. A negative potential

near the surface of the sample defects (and defocuses) the PE to other regions, creating this type of image distortion. For clarity, some distorted lines are marked with white solid arrows in figure 1(a). It is well known that charging is usually not a static problem and may change under scanning events [12]. The defocused lines can be further explained by charging and discharging events during scanning, using capacitor models. Both charging and discharging for a given scanned pixel or set of pixels can be described with a function that has a dominant exponential dependence with the ratio of pixel time to capacitor time constant [12]. The overall trend of such negative surface electric potential function follows a saw-tooth-time dependence with maximum and minimum values that correspond to charging and discharging events, respectively [13]. Thus, the array of pixels in figure 1(a) undergoing charging events with a relatively higher negative surface potential than pixels under discharging will deviate (and defocus) PE more strongly. To minimize these charging problems we used a simple method developed recently by Carl Zeiss NTS, LLC. The process can be performed by non-expert SEM users. The key is to neutralize the negative charges by the recombination of local induced positive charge near the surface of the sample.

For the generation of positive charges (ions), dry nitrogen (N_2) gas was introduced locally (close to the sample surface) in a controlled manner. Details of the process can be found in section 2. Collisions of PE, SE and BSE with the N_2 gas cloud ionizes the nitrogen molecules forming nitrogen positive ions (N_2^+) according to



where the ionization energy is about 15.6 eV [14]. Then N_2^+ are attracted to the surface of the sample by local electric fields and recombine with negative charges, leading to charge neutralization, as shown in schematically in the inset of figure 1(b). In figure 1(b) the CC device was operated at 100%. The porous structure typical of aerogels clearly is visible with porous diameters ranging from ~ 3 to ~ 40 nm. It should be noted that the current leakage term in equation (2) actually has two terms ($I_L = I_G + I_{ion}$), where I_G is the sample leakage current to the ground and I_{ion} is the current leakage due to electron-ion recombination [15]. To a first approximation, it is expected that electron-nitrogen gas molecule collisions with greater cross-sections (σ) will contribute more to the rate of production of ions, because the scattering probability for a given ionization event scales with σ [16]. In figure 2 we present the predicted values for the electron impact ionization σ as a function of the kinetic energy (KE) of the electrons in our experiment, which is for the range of SE above the ionization energy up to PE energy. A close-up of the low KE dependence behavior is plotted in the inset of figure 2. The predicted values are based on the binary-encounter-Bethe (BEB) model, which adds the Mott's cross-section with the high KE factor and Bethe cross-section [17, 18]. Note that for the case of N_2 (black line) the maximum value of σ_{BEB} ($\sim 2.62 \text{ \AA}^2$) is produced with electrons of KE of ~ 100 eV, which corresponds to low energy BSE. For SE with energies of ~ 50 eV, σ_{BEB} is $\sim 2.17 \text{ \AA}^2$.

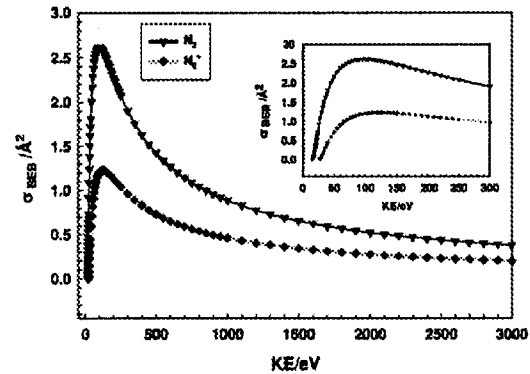


Figure 2. Predicted values for the σ_{BEB} as a function of KE of electrons. The black line indicates results for ionization of N_2 , while the gray line shows the predicted results for N_2^+ .

Moreover, PE and SE (with energy ~ 20.5 eV) produce a σ_{BEB} of $\sim 0.38 \text{ \AA}^2$, and hence should have the same probability to produce N_2^+ during ionization events. However, if we consider the fact that most of the SE ($\sim 90\%$) have KE of 10 eV (or even less), which is below the ionization energy for N_2 , ionization events by SE should be rare. However, there is a possibility that some SE can be accelerated by the positive electric potential at the Faraday cage of the ET detector and may gain sufficient energy to ionize N_2 molecules.

Concerning the production of ions, it is important to consider possible second ionization processes. If N_2^+ produced by the ionization of N_2 molecules does not recombine quickly with the negative charges at the surface of the aerogels, further ionization of N_2^+ may take place and more positive ions will be available for recombination. The σ_{BEB} values for such a situation are presented by the gray line of figure 2, for energies above ~ 27 eV up to the PE energy. The overall trend is to have lower values of σ_{BEB} in contrast with the obtained values for N_2 . Once more, the peak maximum of σ_{BEB} ($\sim 1.23 \text{ \AA}^2$) is produced in the range of BSE (~ 130 eV).

To further explore the effects of N_2 on charging in aerogels we systematically increased the relative amount of nitrogen gas and collected SEM images to see the effects in image contrast. Figures 3(a)–(k) presents a sequence of SEM images showing the effect of changing the percentage of nitrogen, ranging from 0 to 100% in steps of 10%. Due to the highly topographic nature and charging of these aerogels, the overall contrast behavior of the SEM images is expected to be dominated by both topography and voltage contrast. Topography contrast arises due to the relative inclination of the features of the sample with the PE, while voltage contrast is expected to depend on variations of surface potential and hence possible variations in the deflection of the SE due to local electric fields [19]. Although the images of figure 3 are slightly shifted, one can observe the evolution of voltage contrast. Again, for clarity purposes, distorted lines are marked with solid arrows. Note that slight variations in the charging process occur from figure 3(b) (CC operated at 10%) to figure 3(e) (CC operated at 40%), while major effects seem to be observed in figure 3(f) (CC operated at 50%). In figure 3(g) (CC operated at 60%) half

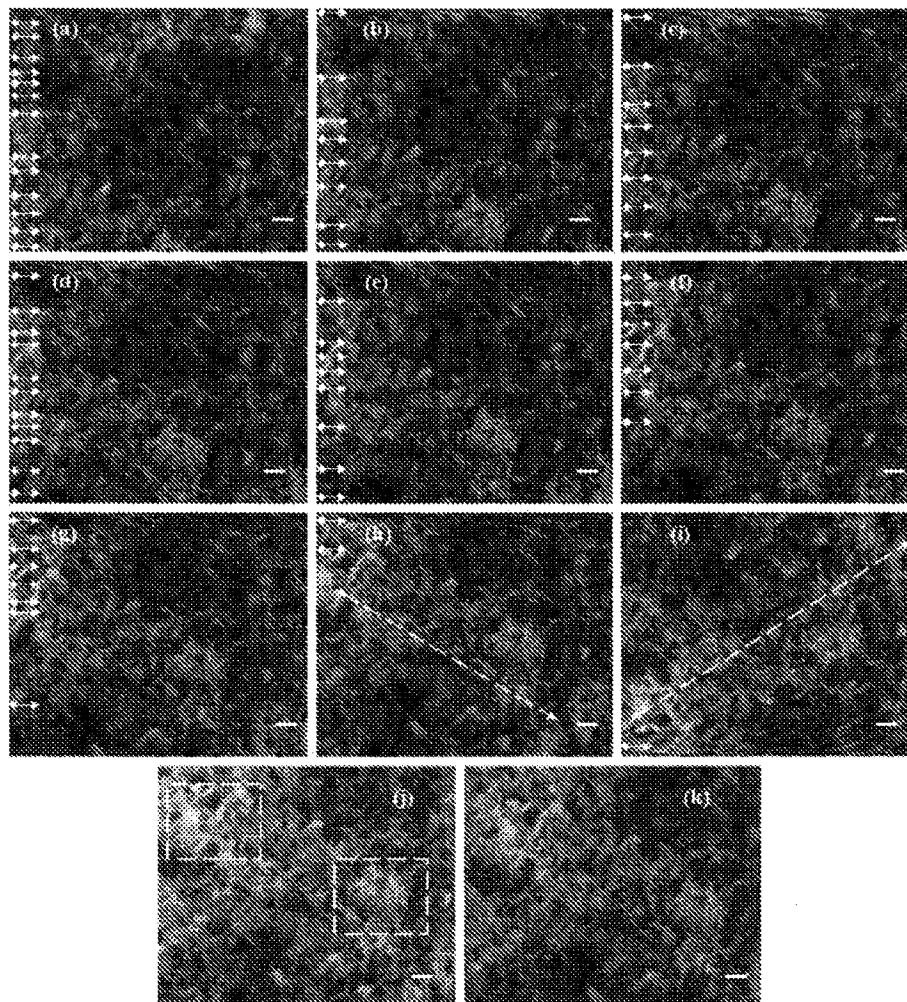


Figure 3. SEM images of aerogels showing the evolution of charging events as a function of relative nitrogen content. CC device off (a), and subsequent increase with steps of 10% ((b)–(k)). Note that positions of artifact lines are marked with solid arrows. For more details refer to the main text. The scale bar is 100 nm and magnification is $\sim 223\times$.

of the image is free of artifact lines due to negative charging. In figure 3(h) (CC operated at 70%) more artifact lines have disappeared but some areas at the top of the image are still not in focus, indicating a relatively strong negative surface potential in that area. In addition, white contrast features (marked with dotted arrows) can be observed distributed along the image from top-left to bottom-right areas, which are also an indication of negative charging. Note that in figure 3(i) (CC operated at 80%) the defocused areas appeared at the lower portion of the image, and the white contrast features (marked with dotted arrows) appear along the image from top-right to bottom-left areas. However, within figure 3(j) (CC operated at 90%) defocused regions were not observed, indicating that strong negative charging has been minimized considerably. Finally, when the CC was operated at 100% (figure 3(k)), the overall bright features observed in figure 3(j) were diminished, indicating that negative charging has been further minimized.

This can be noticed more clearly by comparing the features marked with a dotted box in figure 3(j) and the corresponding areas in figure 3(k). Therefore, to minimize notably negative charging, the CC device needs to be operated at high relative pressures (above 60%). This is expected, since introducing more N_2 gas into the chamber will enhance the generation of positive ions and consequently there will be more positive ions to recombine with negative electrons at the surface of the sample.

At this point we want to indicate that results on imaging of aerogels have been reported previously by Kucheyev *et al* using low vacuum SEM [20]. Although the physics of charge neutralization has similar fundamental principles in both works, we want to briefly clarify the differences and originality of our approach. In regards to the results presented, the SEM images of aerogels in [20] did not show evidence of artifacts and corrections for charging effects, as we clearly

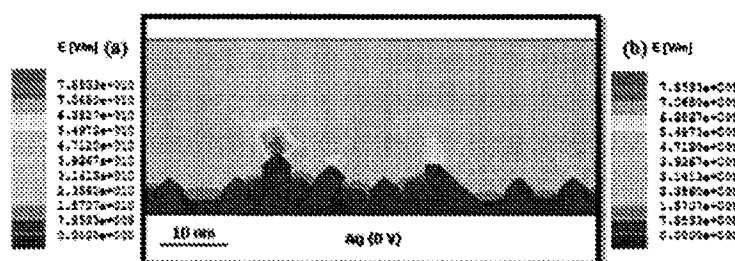


Figure 4. FE model describing the distribution of E for a simplified geometry consisting of an array of silica needles at -1000 V (a) and at -100 V (b), with a grounded silver plate and top electrode, representing the electric potential at the Faraday cage of the ET detector. The direction of the E vector field is from top to bottom.

presented in figures 1 and 3. In terms of the experimental setup, our technique seems to have advantages over the low vacuum SEM technique of Kucheyev *et al* [20]. Firstly, in the low vacuum SEM described in [20] the whole specimen chamber is exposed to a gas (water vapor) and a dedicated gaseous detector (not commonly used) is required for image formation. In addition it has been pointed out that determining the optimum experimental parameters for imaging formation in low vacuum SEM is not trivial, even for ‘experienced’ users [21]. The CC device used in this work is easy to operate and can be installed in conventional SEM equipped with the most common available electron detector (ET).

As pointed out earlier, once the positive ions are generated due to the ionization process, they will be attracted to the surface of the sample by local electric fields (E). This implies that the recombination processes will be affected by such local electric fields. It is important to note that the electric potential at the sample is unknown; therefore the results should be taken as qualitative guidance. Figure 4 depicts the results of the finite element (FE) calculations for a simplified model, which consists of an array of silica needles with different orientations and distances relative to the top electrode (representing the Faraday cage of the ET detector) and a grounded silver electrode. The typical potential at the Faraday cage is ~ 250 V at a distance of ~ 5 cm from the PE impact position. Since in these simulations the distance was decreased to about 30 nm, the potential has been corrected accordingly to be $\sim 1.5 \times 10^{-4}$ V. To solve this electrostatic problem, FE analysis with the Maxwell 2D program was used to solve the Poisson equation using the Neumann boundary conditions and constraining the E field to be continuous across the surface [22]. For a given time, positive ions will preferentially (not exclusively) move to the surface of the aerogels where the magnitude of the local electric field is at its maximum; after that, recombination takes place. In order to explain the differences in defocused lines, and to propose a possible explanation of the evolution process taking into consideration the topography of our aerogels, we have chosen two arbitrary potential values. For instance, consider an area of the sample consisting of an array of silica needles with a local electric potential of -1000 V (which is more than enough to deflect the PE), as described in figure 4(a). Note that in general, the magnitude of E is stronger at the tip-apex area of

the needles, and its maximum value is located at the tip area of the needles closest to the top electrode, and how the E field magnitude diminishes in regions between the needles. This E field behavior is also encountered during the field emission process of rod-like nanostructures [23, 24]. Likewise, if we change the potential of the silica needles to -100 V (which represents an area of the sample with a less severe negative charging) and leave the other parameters unchanged, we notice that the magnitude of the E field is reduced by an order of magnitude but the overall trend is conserved (figure 4(b)). Comparing the results of figures 4(a) and 4(b), we note that the dominant contribution that enhances the local E magnitude is the potential value at the silica needles. Note that needles at -1000 V and further from the top electrode (figure 4(a)) have higher E magnitudes than the needles closest to the top electrode at -100 V potential (figure 4(b)). Based on the FE results one would expect that areas of the aerogels with higher negative charging will attract the positive ions more strongly than areas with relatively lower negative charging. Therefore, for a given time, areas with relatively higher negative electric potentials will present a higher degree of recombination and charging will be minimized faster. Finally, the system will then self-regulate the charging until a relatively uniform electric potential is reached.

4. Conclusions

In summary, we have shown a new SEM approach to image aerogels at relatively high magnifications, where the fine porous structure (down to ~ 3 nm in pore diameter) can be investigated. More importantly, our technique does not use any metal coating to minimize negative charging effects. It uses positive ions which recombine with negative charges at the surface of the sample, revealing the intrinsic porous structure of aerogels. The most common detector (ET) was used for image formation. The results of σ_{FEA} implied that the ionization process and hence the generation of positive ions is governed by low energy BSE. Furthermore, taking into account variations in topography of our aerogels, the simplified FE model employed suggested that areas of the aerogels with the highest local electric potentials should attract positive ions more strongly and hence negative charging will be minimized more rapidly in such areas. To substantially minimize negative

charging considerably, it was required to operate the CC above 60%. It is noteworthy that in this work, the dependence of the PE and pixel scan rates on charging was not considered. Finally, it is important to point out that this technique can be used to image other types of non-conductive nanoporous materials.

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Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries



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HIGHLIGHTS

- AlOOH-modified polyethylene separator gains enhanced thermal stability.
- Interlocking interface structure is formed in the modified separator as PE melts.
- AlOOH coating layer shows appropriate electrolyte wettability.
- Cell performance attains improvement as employing AlOOH-modified separator.

GRAPHICAL ABSTRACT



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ABSTRACT

To exploit high-quality separators for lithium ion batteries, current research activities are mainly focused on the modification of microporous polyolefin membranes by coating them with inorganic particles to achieve comprehensive improvements in their thermal stability, electrochemical compatibility, and overcharge protection. Here, we report a separator made by coating boehmite (AlOOH) particles on microporous polyethylene (PE) membranes. Compared to the commercially applied coating materials, e.g., aluminum oxide (Al₂O₃), AlOOH allows for a substantial reduction in the coating thickness, while ensuring excellent thermal stability of the modified PE membrane. Our study shows that this is due to the formation of an interlocking interface structure that interconnects the PE membrane and AlOOH coating layer as soon as PE melts at about 140 °C, preventing the modified PE membrane from shrinking at subsequently elevated temperatures. The modified PE membrane exhibits suitable electrolyte wettability to facilitate ion transport through it. Thus, the lithium ion batteries employing it as a separator could attain substantially improved electrochemical performance. Furthermore, the AlOOH-coated PE separator was also found to provide an excellent overcharge protection.

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1. Introduction

High-power lithium ion batteries show incomparable application potential in electric vehicles and large energy storage equipment owing to their high energy density and long cycle life, but present safety issues [1–6]. In current lithium ion batteries,

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microporous membranes made of polyolefin such as polyethylene (PE) and polypropylene (PP) are commonly used as separators to isolate the cathode and anode and allow the transport of lithium ions and all degradation products, thereby safeguarding the batteries in working. With softening temperatures of ~130–170 °C, polyolefin membranes can shrink to an extent that causes short circuit and even explosion as the batteries are overheated [7,8]. The polyolefin membranes modified by combination with good heat-resistant organic or inorganic materials can exhibit much lower thermal shrinkage [9–12]. For example, with a coating of polyimide nanoballs, the shrinkage of a PE membrane heated at 140 °C for 0.5 h was found to be reduced from 83.3% to 10% [13]. In general, modification with organic polymers often leads to inferior electrolyte wettability, thereby degrading the electrochemical compatibility of the modified membranes [9]. In contrast, hydrophilic inorganic materials such as silicon dioxide (SiO₂) [14–16] and aluminum oxide (Al₂O₃) [17–19] can offer more favorable modification in terms of both thermal stability and electrochemical compatibility of the modified membranes [7].

A common process currently adopted in industries is to modify the polyolefin membranes with a double-faced coating of Al₂O₃ particles [9]. In this case, thermal shrinkage of the membranes is closely related to the Al₂O₃ coating thickness. Zhao et al. [20] studied the Al₂O₃ coating with thicknesses of 3–9 μm and found that, after exposure at 140 °C for 0.5 h, the corresponding shrinkage of the modified PE membranes decreased from 20% to 0%, while the ion conductivity through them dropped from 1.12 to 1.00 mS cm⁻¹. This indicates that increasing the coating thickness may effectively suppress the thermal shrinkage, but would impart an adverse effect on the electrochemical compatibility for the modified membranes for use as separators. To date, the commercially employed polyolefin separator contains an Al₂O₃ coating that is 3 μm or more thick on each face, which accounts for around 25% of the total thickness of the separator. Therefore, there is room for improvement when it comes to increasing the energy density and cut the cost of batteries by reducing the coating thickness of separators while ensuring good thermal stability [21]. To fulfill this goal, the most feasible strategy is to significantly reinforce the interconnection between the coating layer and polyolefin membranes by means of constructing an interlocking interface structure.

In this study, we report a separator made by coating boehmite (AlOOH) particles on a microporous PE membrane. AlOOH was found to enable the formation of an interlocking interface structure when PE melted at about 140 °C, which significantly strengthened their interconnection. Besides, AlOOH is lighter, softer and cheaper and shows better dispersity in an aqueous slurry compared to Al₂O₃. In this article, we present the study on the thermal stability, electrochemical compatibility, and overcharge protection of the modified PE membrane using a 1.15-μm-thick AlOOH coating.

2. Experimental section

2.1. Material preparation

A slurry composed of 35 wt% AlOOH particles of average size 350 nm (Nabaltec, Germany), 1 wt% polyvinyl alcohol (PVA, Lingfeng Chemistry, China), and distilled water was prepared. Before coating, the slurry was well dispersed by grinding in a 1400 r/min grinding machine for 20 min. Using a roller coating instrument, the slurry was coated on both faces of a 16-μm thick PE membrane (Shuang'ao, Shanghai, China), which was subsequently dried at 60 °C for 24 h. The thickness of the AlOOH particle coating was controlled by the rolling speed.

2.2. Characterization and measurement

Field emission scanning electron microscopy (FE-SEM) was used to observe the microscopic structure and morphology of the PE membranes. The thickness of the PE membrane and AlOOH particle coating was measured by a digital display thickness gauge. The shrinkage of the modified PE membrane was determined by measuring the dimensional change with respect to the original one after the thermal treatment at various temperatures for 0.5 h, and was calculated using the following equation:

$$\text{Shrinkage}(\%) = \frac{S_0 - S_T}{S_0} \times 100\% \quad (1)$$

where S_0 and S_T refer to the area of the membrane before and after the thermal treatment, respectively. The permeability (Gurley number) of the separators was tested using a densometer. Mechanical strength, both in the machine direction (MD) and transverse direction (TD) was measured on a universal testing machine. The electrolyte wettability of the separator was examined by two methods: a contact angle test and an electrolyte immersion-height test. The contact angle was examined using a commercial drop shape analysis system (Powereach JC2000C1, Shanghai Zhongchen Digital Technique Equipment Co. Ltd., China). The electrolyte immersion-height test [22] was carried out by immersing the separator vertically in a liquid electrolyte (EC:DEC = 1:1 v/v containing 1 M LiPF₆) and then measuring the soaked height of the liquid electrolyte after 0.5 h.

The ionic conductivities of the pristine and AlOOH modified PE separators were measured by electrochemical impedance spectroscopy (EIS). The membranes were placed between two stainless steel blocking electrodes, and the EIS spectra were obtained by sweeping from 100 kHz to 0.1 Hz with an AC amplitude of 5 mV at room temperature [23]. The ionic conductivity (σ) was calculated using the following equation:

$$\sigma = \frac{d}{R_b \times S} \quad (2)$$

where R_b is the bulk resistance and d and S are the thickness and area of the fibrous membrane, respectively. To evaluate the electrochemical performance of the separators, CR2016 coin-type half cells were assembled in a glove box filled with argon gas, with a Li₄Ti₅O₁₂ working electrode (Li₄Ti₅O₁₂/acetylene black/PVDF = 8/1/1 by weight), lithium metal counter electrode, and liquid electrolyte (1 M LiPF₆ in EC:DEC = 1:1 v/v). Then, a LAND CT2001A battery test system was used to examine the electrochemical performance within the potential window of 1.0 and 2.5 V (vs. Li⁺/Li) at 25 °C.

3. Results and discussion

The separator was developed by coating a microporous PE membrane with AlOOH particles on a roller coating equipment. A slurry composed of AlOOH particles (35 wt%), PVA (1 wt%), and distilled water was used as the starting material. Here, PVA serves as a binder due to its excellent adhesive and film-forming properties. In addition, it also has high tensile strength and good flexibility. Based on the thermogravimetric (TG) analysis undertaken after drying (Fig. S1), the absorbed water content in the AlOOH layer accounted for less than 1% of the total weight, and it can be estimated to be less than five in ten thousands of overall content in a battery. The coating thickness was accurately controlled by adjusting the rolling speed. Here, we provide an example of a double-faced coating with a thickness of 1.15 μm. Scanning electron microscopy (SEM) observation indicates that the pristine PE

membrane consisted of randomly interconnected dendrites and non-uniformly dispersed micropores (Fig. 1a), and the coating on the membrane is formed from the assembly of AlOOH particles with an average size of 350 nm (Fig. 1b). The pristine PE membrane had higher porosity and wider pore size distribution than the AlOOH particle-modified PE membrane.

Fig. 2a shows the effect of AlOOH coating on the thermal shrinking of the modified PE membrane. Clearly, thermal stability of the modified PE membrane is significantly improved. It showed almost no shrinkage below 140 °C and was found to have only lost less than 3% of the original area after exposure to 180 °C for 0.5 h. In contrast, the pristine PE membrane showed obvious shrinkage at 110 °C and was found to have lost more than 85% of the original area with an increase in the temperature to 150 °C. A clear contrast can be observed in the digital photographs shown in Fig. 2b, which shows that the modified PE membrane maintained appreciable integrity and flattening at high temperatures. Additionally, we also observed that the modified PE membrane turned from opaque to transparent at temperatures higher than 140 °C, implying that the pores in the PE membrane were closed shut, owing to its melting. This transformation is essentially beneficial for the safety of lithium ion batteries in the case of overheating.

The excellent thermal stability of the AlOOH-coated PE membrane was strongly related to the coating interface structure rather than the coating thickness. Fig. 3a shows the cross-sectional SEM image of the modified PE membrane preheated at 150 °C for 0.5 h. The inset in Fig. 3a clearly shows that a portion of PE was embedded in the AlOOH particle assembly forming an interlocking interface structure, because the molten PE has good affinity to AlOOH so that it easily flows into the interspace between the AlOOH particles under a capillary force. The formation of the interlocking interface structure can significantly increase the contact area between the AlOOH particles and PE, thereby improving the thermal stability of the modified PE membrane. This phenomenon can also be witnessed from the thickness variation in the modified PE membrane during the thermal treatment. For a pristine membrane, shrinkage is usually accompanied by thickening. As shown in Fig. 3b, this rule was essentially conformed to until 140 °C. However, when the temperature was elevated further to 150 °C, the modified PE membrane showed an obvious thinning although it shrank slightly. The interlocking interface structure can enhance the mechanical strength of the membrane. Consequently, the area and thickness of the modified PE membrane remained unchanged as the temperature was increased from 150 to 180 °C. Through peeling tests, we confirmed that the formation of this interface structure can result in enhanced bonding strength between the pristine PE membrane and AlOOH particles. As shown in Fig. 3c, the AlOOH particle coating layer could be easily stripped off the newly modified PE membrane with an adhesive tape, and its adhesion strength was estimated to be 35 N/m (Fig. S2). After exposure to temperatures beyond 140 °C, the AlOOH particle coating could not be thus

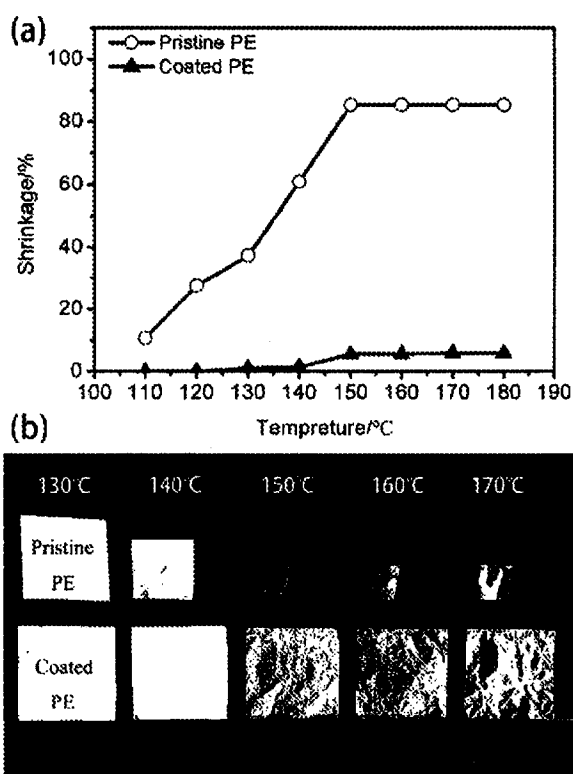


Fig. 2. (a) Temperature dependent shrinkage curves and (b) digital photographs of the pristine PE membrane and AlOOH-coated PE membranes, subjected to thermal treatment between 110 and 180 °C.

stripped off, indicating that the adhesion strength had surpassed 225 N/m.

Besides the thermal stability, some other basic properties of the modified PE membrane were also investigated. As shown in Table 1, the puncture strength and tensile strength of the modified PE membrane were slightly lower than those of the pristine PE membrane, while the permeability was found to decrease considerably by about 20% due to elongation of the gas flow path. Permeability is one of the important factors affecting ion mobility in separators [9]. In general, the thicker the coating, the lower is the permeability and thus the lower is the ion mobility. In this study, the presence of the AlOOH particle coating led to improvement in the wettability to the electrolyte, making up for the decline in the ion mobility resulting from the permeability decrease. As shown in Fig. 4a, the measured contact angle of the electrolyte (1 M LiPF₆ in EC:DEC = 1:1 v/v) on pristine PE membrane was 45.2°, while it was nearly zero on the

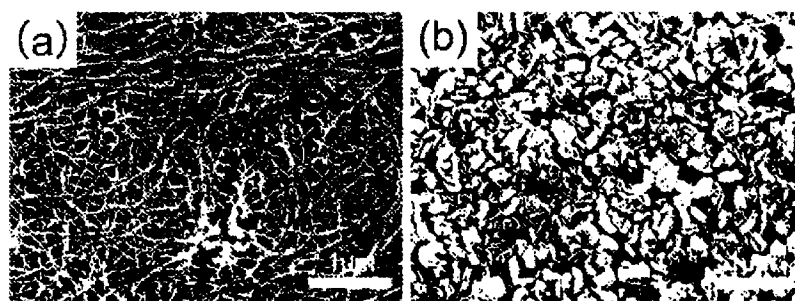


Fig. 3. SEM images showing the surface morphology of the membranes. (a) Pristine PE; (b) AlOOH-coated PE.

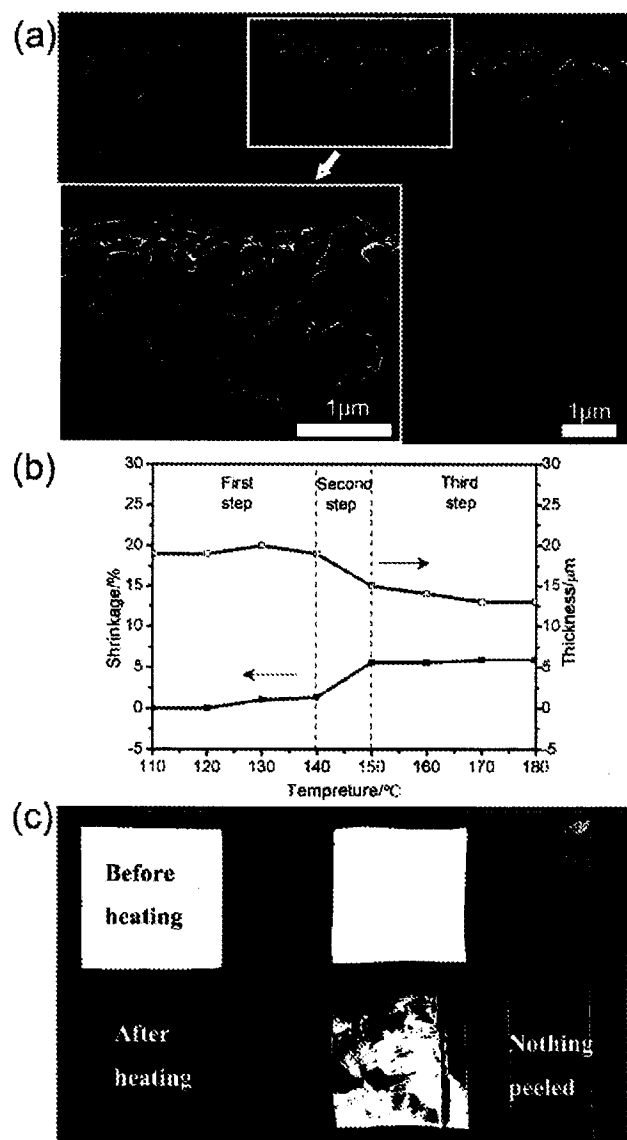


Fig. 3. (a) Cross-sectional SEM image of an ALOOH-coated PE membrane heated at 150 °C for 0.5 h (inset shows an enlarged image of the coating interface); (b) the dependence of the shrinkage and thickness of the ALOOH-coated PE membrane on the heating temperature; (c) digital photographs showing peeling tests on the ALOOH-coated PE membrane before and after heating.

ALOHH-coated PE membrane. Upon simultaneous immersion in a liquid electrolyte, both the pristine and modified PE showed an almost similar soaked height of the electrolyte; however, the modified PE membrane showed stronger capability for electrolyte uptake as shown in Fig. 4b. In summary, the modified PE membrane employed as a separator in lithium ion batteries should facilitate ion transport [22].

The ionic conductivity, an important parameter affecting the

Table 1
Physical properties of the pristine PE separator and the modified PE separator.

Separator	Gurley s/100 ml	Puncture strength/gf	Tensile strength	
			MD/mPa	TD/mPa
Pristine PE	81.0	447.0	105.0	114.5
Modified PE	105.9	430.1	92.7	92.5

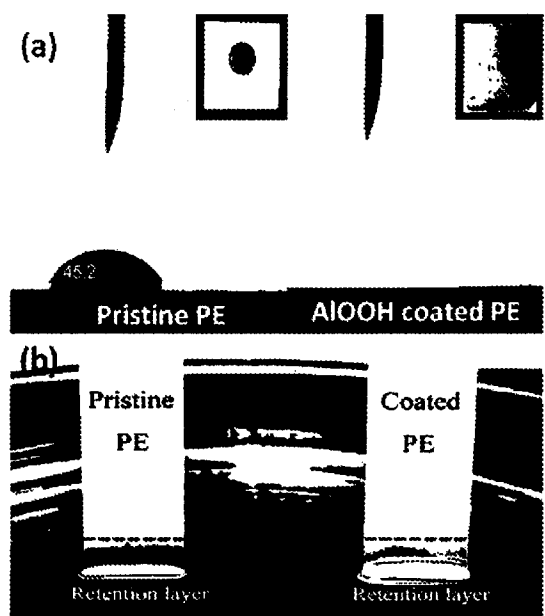


Fig. 4. Digital photographs of (a) contact angle of an electrolyte on the pristine PE membrane (left) and ALOOH coated PE membrane (right); (b) the liquid electrolyte uptake by the pristine PE membrane (left) and the ALOOH-coated PE membrane (right), immersed in a liquid electrolyte, EC: DEC (1:1 v/v containing 1 M LiPF₆) for 0.5 h.

conduction of ionic carriers, was measured. Fig. 5a shows Nyquist plots of the liquid electrolyte-soaked pristine PE membrane and

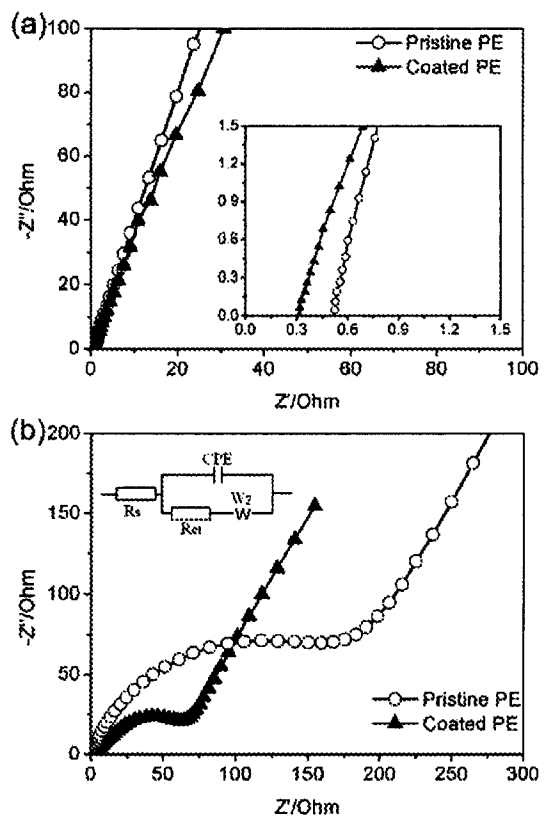


Fig. 5. (a) Ionic conductivities of the liquid electrolyte-soaked separators (inset shows the plots at high-frequency); (b) Nyquist plots for the cells with the equivalent circuit shown in the inset.

modified PE membrane obtained by AC impedance measurements at 25 °C. The intercepts on the real axis of the liquid electrolyte-soaked membranes are 0.51 Ω and 0.31 Ω for the pristine and modified PE membranes (inset in Fig. 5a), respectively. Using Equation (2), ionic conductivity of the modified PE membrane was calculated to be $6.56 \times 10^{-3} \text{ S cm}^{-1}$, while it was $3.48 \times 10^{-3} \text{ S cm}^{-1}$ for the pristine PE membrane. The higher ionic conductivity of the AlOOH-coated membrane may be attributed to its better electrolyte wettability. Furthermore, EIS spectra of the two CR2016 LTO/Li cells that employed the pristine and AlOOH-coated membranes as separators were analyzed to evaluate the interfacial separator resistance in the cells (Fig. 5b). The EIS analysis results were fitted using the equivalent circuit shown in the inset of Fig. 5a [24–26], where R_s denotes the ohmic resistance from the electrode, electrolyte, separator, and current collector, and R_{ct} refers to the charge transfer resistance between the electrolyte and electrode. Based on the fitted data of circuit elements, R_s was calculated to be 2.28 Ω for the AlOOH-coated PE separator, larger than that of the pristine PE separator (0.63 Ω). In contrast, R_{ct} of the AlOOH-coated PE separator was 74.1 Ω , much smaller than that of the pristine PE separator (156.2 Ω), owing to its better electrolyte wettability and membrane-electrode affinity [23,27].

Further evaluation of the modified PE membrane as a separator was done in terms of the C-rate capabilities and cycling performance of two CR2016 coin-type half cells (LTO/Li metal) that employed the pristine and modified PE membranes as separators. Fig. 6 shows the rate performance, where the cells were discharged continuously over current densities ranging from 0.2C to 2C and back to 0.2C for 5 cycles each. For both cells, the discharge capacity

gradually decreased with increase in the discharge current density (Fig. 6a and b). The cell assembled with the modified PE membrane exhibited obviously better rate performance than that of the pristine PE separator at the large current 2C rate (Fig. 6c), which can be attributed to the higher ionic conductivity of the AlOOH-coated PE separator. Cycling performance of the cells, shown in Fig. 6d, was measured with voltages ranging from 1.0 to 2.5 V and a constant charge current density of 1C. In the beginning, the capacity of the cell assembled with the modified PE separator was about 150 mA h g^{-1} , and it retained 96.3% of its initial capacity after 100 cycles. The initial cell capacity of the pristine PE separator was about 142 mA h g^{-1} , and it showed a capacity retention of 94.6% after 100 cycles. The cell employing the modified PE membrane exhibited higher discharge capacity than the one employing a pristine PE membrane for up to 100 cycles.

Moreover, we also found that introduction of the AlOOH particle coating on the PE separator provides excellent overcharge protection. Fig. 7 presents an example of an aluminum-shell full battery consisting of a LiFePO_4 (LFP) cathode, graphite anode, and LiPF_6 electrolyte. The battery that was regularly charged to 4.2 V underwent overcharge up to 10 V under a constant current density of 3 C. It was observed that, with the use of the pristine PE separator, the battery swelled and expelled white smoke at about 15 min after the overcharge (Fig. 7a), and the surface temperature rose to 104.8 °C (Fig. 7b). However, the battery employing an AlOOH-coated PE separator neither deformed nor expelled any smoke (Fig. 7c), thus indicating its superior endurance against overcharge, although the surface temperature rose to 103.4 °C (Fig. 7d). The mechanism of this overcharge protection is proposed to be as

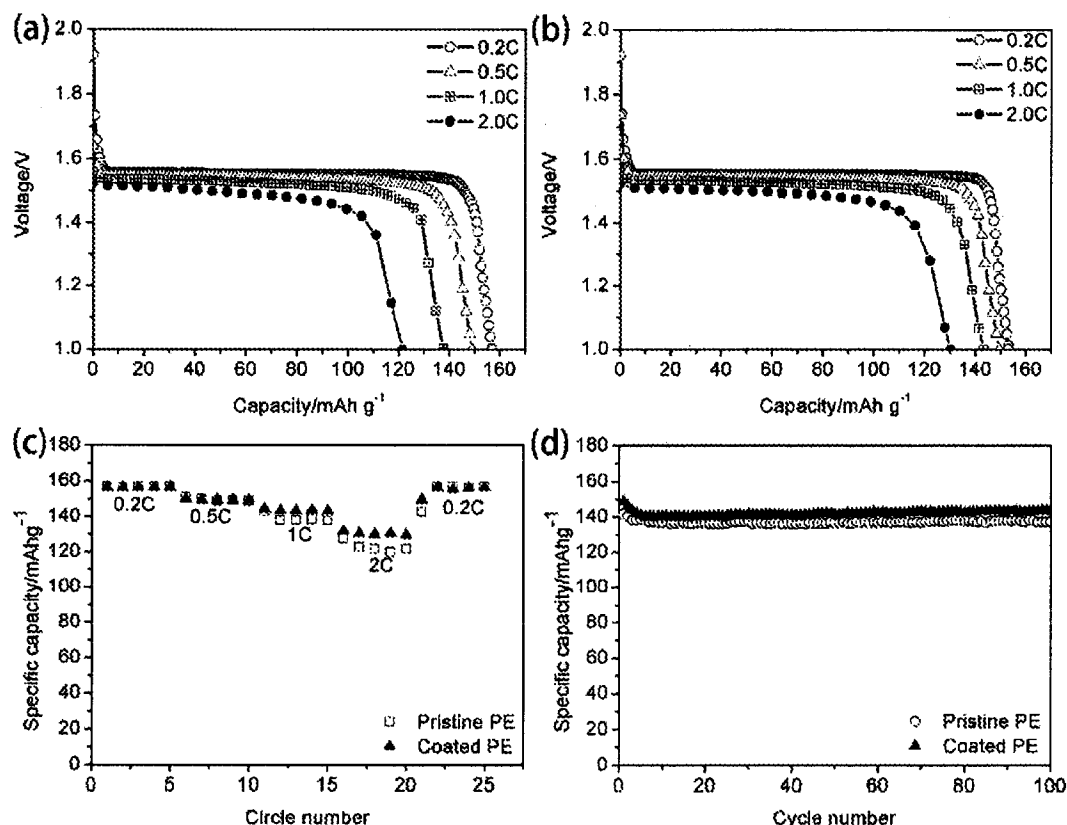


Fig. 6. C-rate charge-discharge profiles of Li/LTO CR2016 coin-type half-cells assembled with (a) a pristine PE separator and (b) an AlOOH-coated PE separator; (c) comparison of the discharge rate capabilities; (d) cycling performances of the two CR2016 coin-type half-cells.

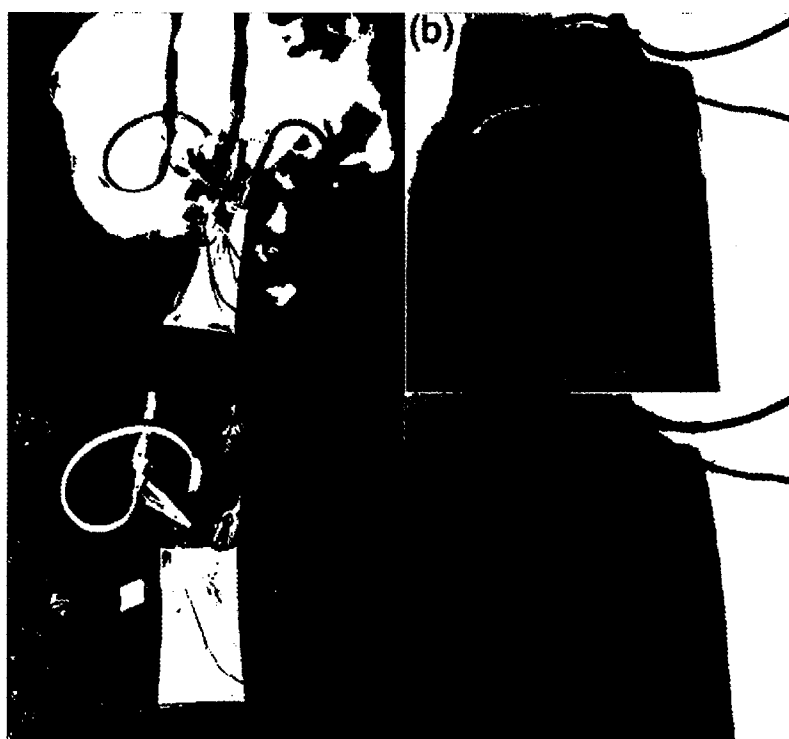


Fig. 7. Demonstration of overcharge in an aluminum-shell full battery employing a pristine PE separator (a, b) and an AlOOH-coated PE separator (c, d).

follows: (i) the strong absorption of LiPF_6 electrolyte on the AlOOH particle coating leads to a reduction in the activity of PF_6^- anions so as to cause an additional overpotential beyond the normal oxidation potential of PF_6^- anions [28]; (ii) the AlOOH particle coating placed in contact with the cathode has a higher oxidation potential and sets the overcharge protection limit to protect the PE from degradation; and (iii) the AlOOH particle coating structurally increases the mechanical strength of the battery. In addition, investigations on the electrochemical performance of LFP/graphite pouch cells (Fig. S3) also show that the AlOOH-coated PE separator can lead to improved C-rate capabilities and cycling performance of cells.

4. Conclusion

In summary, the PE membrane modified by an AlOOH particle coating has been demonstrated to be a promising separator for lithium ion batteries. Due to the formation of an interlocking interface structure with the melting of PE, the modified PE membrane underwent extremely low shrinkage when subjected to thermal treatment up to 180 °C, although the AlOOH coating thickness was only 1.15 μm . Such thin coatings are beneficial in saving space for batteries to gain high energy density. Moreover, the AlOOH-coated PE separator can provide superior electrochemical compatibility and overcharge protection and thus render the batteries more efficient and safer.

Acknowledgement

This study was supported by the Shanghai Science and Technology Committee Project (No. 15dz1201102).

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jpowsour.2017.02.078>.

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PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference 65213-103	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/US2018/055862	International filing date (<i>day/month/year</i>) 15 October 2018 (15-10-2018)	(Earliest) Priority Date (<i>day/month/year</i>) 13 October 2017 (13-10-2017)
Applicant OPTODOT CORPORATION		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 6 sheets.

☒ It is also accompanied by a copy of each prior art document cited in this report.

1. Basis of the report

a. With regard to the language, the international search was carried out on the basis of:

- ☒ the international application in the language in which it was filed
☐ a translation of the international application into _____, which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b))

b. ☐ This international search report has been established taking into account the rectification of an obvious mistake authorized by or notified to this Authority under Rule 91 (Rule 43.6*bis*(a)).

c. ☐ With regard to any nucleotide and/or amino acid sequence disclosed in the international application, see Box No. I.

2. ☐ Certain claims were found unsearchable (See Box No. II)

3. ☒ Unity of invention is lacking (see Box No III)

4. With regard to the title,

- ☒ the text is approved as submitted by the applicant
☐ the text has been established by this Authority to read as follows:

5. With regard to the abstract,

- ☒ the text is approved as submitted by the applicant
☐ the text has been established, according to Rule 38.2, by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority

6. With regard to the drawings,

- a. the figure of the drawings to be published with the abstract is Figure No. _____
☐ as suggested by the applicant
☐ as selected by this Authority, because the applicant failed to suggest a figure
☐ as selected by this Authority, because this figure better characterizes the invention
b. ☒ none of the figures is to be published with the abstract

INTERNATIONAL SEARCH REPORT

International application No.
PCT/US2018/055862

Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:
2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:
3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

see additional sheet

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.
2. ☐ As all searchable claims could be searched without effort justifying an additional fees, this Authority did not invite payment of additional fees.
3. ☒ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

1-10, 13-16
4. ☐ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:

Remark on Protest

- ☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.
- ☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.
- ☒ No protest accompanied the payment of additional search fees.

INTERNATIONAL SEARCH REPORT

International application No
PCT/US2018/055862

A. CLASSIFICATION OF SUBJECT MATTER INV. H01M2/16 H01M10/0525 ADD.		
According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED Minimum documentation searched (classification system followed by classification symbols) H01M		
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched		
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) EPO-Internal, WPI Data		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2016/172705 A1 (HOSHIBA KOJI [JP]) 16 June 2016 (2016-06-16)	1-8,16
Y	par. 83-84, 108, 109 -----	9,10,15
Y	SOLÁ F ET AL: "A new scanning electron microscopy approach to image aerogels at the nanoscale;A new scanning electron microscopy approach to image aerogels at the nanoscale", NANOTECHNOLOGY, IOP, BRISTOL, GB, vol. 22, no. 17, 16 March 2011 (2011-03-16), page 175704, XP020188888, ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704 page 2, second paragraph ----- -/-	9,10,15
☒ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents : "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier application or patent but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone "Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art "&" document member of the same patent family		
Date of the actual completion of the international search 1 April 2019		Date of mailing of the international search report 11/04/2019
Name and mailing address of the ISA/ European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016		Authorized officer Bettio, Andrea

INTERNATIONAL SEARCH REPORT

International application No
PCT/US2018/055862

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	YANG CHONGWEN ET AL: "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries", JOURNAL OF POWER SOURCES, vol. 348, 3 March 2017 (2017-03-03), pages 80-86, XP029949881, ISSN: 0378-7753, DOI: 10.1016/J.JPOWSOUR.2017.02.078	13,14,16
Y	abstract; page 81, left-hand column, 3rd and 4th paragraphs; page 82, figure 2a -----	15
A	US 2017/294638 A1 (JOO DONG JIN [KR] ET AL) 12 October 2017 (2017-10-12) Claim 1 -----	1-10, 13-16

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/US2018/055862

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2016172705 A1	16-06-2016	JP 6399921 B2	03-10-2018
		JP 2016119221 A	30-06-2016
		KR 20160072009 A	22-06-2016
		US 2016172705 A1	16-06-2016

US 2017294638 A1	12-10-2017	CN 107437601 A	05-12-2017
		JP 2017191775 A	19-10-2017
		KR 20170116817 A	20-10-2017
		US 2017294638 A1	12-10-2017

FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

This International Searching Authority found multiple (groups of) inventions in this international application, as follows:

1. claims: 1-10(completely); 16(partially)

A separator for a lithium battery according to present claim
1

2. claims: 11, 12(completely); 16(partially)

A separator for a lithium battery according to present claim
11

3. claims: 13-15(completely); 16(partially)

A separator for a lithium battery according to present claim
13

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

PCT

To:

see form PCT/ISA/220

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY
(PCT Rule 43bis.1)

Date of mailing
(day/month/year) see form PCT/ISA/210 (second sheet)

Applicant's or agent's file reference
see form PCT/ISA/220

FOR FURTHER ACTION
See paragraph 2 below

International application No.
PCT/US2018/055862

International filing date (day/month/year)
15.10.2018

Priority date (day/month/year)
13.10.2017

International Patent Classification (IPC) or both national classification and IPC
INV. H01M2/16 H01M10/0525

Applicant
OPTODOT CORPORATION

1. This opinion contains indications relating to the following items:

- ☒ Box No. I
- Basis of the opinion
- ☐ Box No. II
- Priority
- ☒ Box No. III
- Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☒ Box No. IV
- Lack of unity of invention
- ☒ Box No. V
- Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI
- Certain documents cited
- ☒ Box No. VII
- Certain defects in the international application
- ☐ Box No. VIII
- Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will usually be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA:



European Patent Office
D-80298 Munich
Tel. +49 89 2399 - 0
Fax: +49 89 2399 - 4465

Date of completion of
this opinion

see form
PCT/ISA/210

Authorized Officer

Bettio, Andrea

Telephone No. +49 89 2399-0



**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2018/055862

Box No. I Basis of the opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed.
 - ☐ a translation of the international application into , which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1 (b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43*bis*.1(a))
3. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing:
 - a. ☐ forming part of the international application as filed:
 - ☐ in the form of an Annex C/ST.25 text file.
 - ☐ on paper or in the form of an image file.
 - b. ☐ furnished together with the international application under PCT Rule 13*ter*.1(a) for the purposes of international search only in the form of an Annex C/ST.25 text file.
 - c. ☐ furnished subsequent to the international filing date for the purposes of international search only:
 - ☐ in the form of an Annex C/ST.25 text file (Rule 13*ter*.1(a)).
 - ☐ on paper or in the form of an image file (Rule 13*ter*.1(b) and Administrative Instructions, Section 713).
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that forming part of the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2018/055862

Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability

The questions whether the claimed invention appears to be novel, to involve an inventive step (to be non obvious), or to be industrially applicable have not been examined in respect of

☐ the entire international application

☒ claims Nos. 11, 12

because:

☐ the said international application, or the said claims Nos. relate to the following subject matter which does not require an international search (*specify*):

☐ the description, claims or drawings (*indicate particular elements below*) or said claims Nos. are so unclear that no meaningful opinion could be formed (*specify*):

☐ the claims, or said claims Nos. are so inadequately supported by the description that no meaningful opinion could be formed (*specify*):

☒ no international search report has been established for the whole application or for said claims Nos. 11, 12

☐ a meaningful opinion could not be formed without the sequence listing; the applicant did not, within the prescribed time limit:

☐ furnish a sequence listing in the form of an Annex C/ST.25 text file, and such listing was not available to the International Searching Authority in the form and manner acceptable to it; or the sequence listing furnished did not comply with the standard provided for in Annex C of the Administrative Instructions.

☐ furnish a sequence listing on paper or in the form of an image file complying with the standard provided for in Annex C of the Administrative Instructions, and such listing was not available to the International Searching Authority in the form and manner acceptable to it; or the sequence listing furnished did not comply with the standard provided for in Annex C of the Administrative Instructions.

☐ pay the required late furnishing fee for the furnishing of a sequence listing in response to an invitation under Rule 13ter.1(a) or (b).

☒ See Supplemental Box for further details

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

International application No.
PCT/US2018/055862

Box No. IV Lack of unity of invention

1. ☒ In response to the invitation (Form PCT/ISA/206) to pay additional fees, the applicant has, within the applicable time limit:
- ☒ paid additional fees
 - ☐ paid additional fees under protest and, where applicable, the protest fee
 - ☐ paid additional fees under protest but the applicable protest fee was not paid
 - ☐ not paid additional fees
2. ☐ This Authority found that the requirement of unity of invention is not complied with and chose not to invite the applicant to pay additional fees.
3. This Authority considers that the requirement of unity of invention in accordance with Rule 13.1, 13.2 and 13.3 is
- ☐ complied with
 - ☒ not complied with for the following reasons:
see separate sheet
4. Consequently, this report has been established in respect of the following parts of the international application:
- ☐ all parts.
 - ☒ the parts relating to claims Nos. 1-10, 13-16

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement
- | | | |
|-------------------------------|-------------|------------------------|
| Novelty (N) | Yes: Claims | <u>9, 10, 15</u> |
| | No: Claims | <u>1-8, 13, 14, 16</u> |
| Inventive step (IS) | Yes: Claims | |
| | No: Claims | <u>1-10, 13-16</u> |
| Industrial applicability (IA) | Yes: Claims | <u>1-10, 13-16</u> |
| | No: Claims | |
2. Citations and explanations
- see separate sheet**

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/US2018/055862

Box No. VII Certain defects in the international application

The following defects in the form or contents of the international application have been noted:

see separate sheet

Re Item III

Non-establishment of opinion with regard to novelty, inventive step and industrial applicability

In reply to the objection to lack of unity, the applicant has paid 1 additional search fee for invention 3. The international search report has been established for inventions 1 and 3.

No opinion will be given in respect of subject-matter which is not covered by the search report (Rule 66.1(e) PCT).

Re Item IV

Lack of unity of invention

Reference is made to the following document:

D1:US 2016/172705 A1 (HOSHIBA KOJI [JP]) 16 June 2016 (2016-06-16)

The present application lacks unity of invention (Rule 13.1 PCT) for the following reasons:

The problem underlying the present invention is the provision of battery separators with increased dimensional stability at temperatures such as 150°C or higher (par. 3 of the present application).

As solution to this problem the applicant proposes:

1)A separator for a lithium battery comprising:

(a) a porous polymeric layer, and

(b) a nanoporous layer adjacent at least one of two opposing sides of said porous polymeric layer, said nanoporous layer comprising boehmite particles and one or more polymers, wherein:

a. a volume fraction of said one or more polymers in the solid portion of said nanoporous layer is between about 15% and about 50%, and

b. said boehmite particles have an average crystallite size between about 5 nm and about 90 nm (present claim 1)

2)A separator for a lithium battery comprising:

(a) a porous polyethylene layer, and

(b) a nanoporous layer adjacent both of two opposing sides of said polyethylene layer, said nanoporous layer comprising boehmite particles and one or more polymers, wherein:

- a. a volume fraction of said one or more polymers in the solid portion of said nanoporous layer is between about 15% and about 50%;
- b. a total coating weight of said nanoporous layer on both of said two opposing sides of said polyethylene layer is about 5.5 grams per square meter or less, and
- c. a heat shrinkage of said separator at 150°C for 1 hour is less than 5% (present claim 11).

3) A separator for a lithium battery comprising:

- a. a porous polymeric layer, and
- b. 0.3 gsm or greater of boehmite particles disposed within a plurality of pores of the porous polymeric layer (present claim 13).

The common features linking the different inventions 1)-2)-3) together are:

a separator for a lithium ion battery comprising

- (a) a porous polymeric layer and,
- (b) boehmite particles.

The common features linking the different inventions (1)-(2) together are:

a separator for a lithium ion battery comprising

- (a) a porous polymeric layer and,
- (b) a nanoporous layer adjacent at least one of two opposing sides of said porous polymeric layer, said nanoporous layer comprising boehmite particles and one or more polymers, said boehmite particles have an average crystallite size between about 5 nm and about 90 nm.

Prior art documents D1 (par. 109) teaches a wound electrode assembly for a non-aqueous electrolyte rechargeable battery in which deformation is suppressed and improving heat resistance and energy density are simultaneously improved by improving a handling property of the separator during the manufacturing process.

According to D1 (par. 83-84) said lithium ion battery is provided with a separator comprising a layer consisting of a fluorine resin-containing particulate (which can be considered as a binder), a polyethylene ionomer (a binder) sodium polyacrylate and

boehmite particles having an average particle diameter of 90 nm mixed in a volume ratio of 40:10:0.3: 50 respectively. Said layer is attached to a 12 microns thick layer consisting of a porous polyethylene film.

The idea of a separator for a lithium ion battery comprising

- (a) a porous polymeric layer and,
- (b) boehmite particles.

is known in the prior art, and cannot serve as single general inventive concept linking the individual subjects in which the application has been divided.

The same applies to the idea of

- (a) a porous polymeric layer and,
- (b) a nanoporous layer adjacent at least one of two opposing sides of said porous polymeric layer, said nanoporous layer comprising boehmite particles and one or more polymers, said boehmite particles have an average crystallite size between about 5 nm and about 90 nm.

Starting from said common concepts, invention 1 comprises the additional features of providing a given volume fraction of polymers (present claim 2), a different average particle size of the boehmite particles, (present claim 3), a different heat shrinkage (present claim 4), a different total coating weight of the nanoporous layer on both of the two opposite sides of the porous polymer layer (present claim 5), a different porosity, or pore volume or average pore diameter, or pore diameter (present claim 6), a different material of the porous polymeric layer (present claim 7), a different structure of the nanoporous layer (present claim 8), providing a separator for a lithium battery wherein the boehmite particles have an average crystallite size between about 5 nm and about 8 nm.

The application as filed does not teach which technical problem is solved by said features.

Invention 2 on the other hand comprises the additional features of

- a) porous polyethylene layer
- b) a nanoporous layer adjacent both of two opposite sides of said polyethylene layers wherein
 - a. a volume fraction of said one or more polymers in the solid portion of said nanoporous layer is between about 15% and about 50%.

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING
AUTHORITY (SEPARATE SHEET)**

International application No.

PCT/US2018/055862

b. a total coating weight of said nanoporous layer on both of said two opposing sides of said polyethylene layer is about 5.5 grams per square meter or less, and

c. a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.

The application as filed does not teach which technical problem is solved by said features.

Invention 3 comprises the additional feature of

b. 0.3 gsm or greater of boehmite particles disposed within a plurality of pores of the porous polymeric layer.

The application as filed does not teach which technical problem is solved by said features.

Thus there is no same or corresponding special technical feature in the sense of rule 13.1 PCT.

As there is no other technical feature which could fulfil the role of special technical feature in the sense of rule 13 PCT, the present application lacks unity of invention containing the following subjects:

1. claims: 1-10(completely); 16(partially)

A separator for a lithium battery according to present claim 1

2. claims: 11, 12(completely); 16(partially)

A separator for a lithium battery according to present claim 11

3. claims: 13-15(completely); 16(partially)

A separator for a lithium battery according to present claim 13.

In conclusion, the groups of claims are not linked by common or corresponding special technical features and define 3 different inventions not linked by a single general inventive concept.

The application, hence does not meet the requirements of unity of invention as defined in Rules 13.1 and 13.2 PCT.

Re Item V

Reasoned statement with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

Reference is made to the following documents:

- D1 US 2016/172705 A1 (HOSHIBA KOJI [JP]) 16 June 2016 (2016-06-16)
- D2 SOLÁ F ET AL: "A new scanning electron microscopy approach to image aerogels at the nanoscale;A new scanning electron microscopy approach to image aerogels at the nanoscale",
NANOTECHNOLOGY, IOP, BRISTOL, GB,
vol. 22, no. 17, 16 March 2011 (2011-03-16), page 175704,
XP020188888,
ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704
- D3 YANG CHONGWEN ET AL: "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries",
JOURNAL OF POWER SOURCES,
vol. 348, 3 March 2017 (2017-03-03), pages 80-86, XP029949881,
ISSN: 0378-7753, DOI: 10.1016/J.JPOWSOUR.2017.02.078

Invention 1

Art. 33(2) PCT

D1 (paragraphs 83, 84, 108, 109) teaches a wound electrode assembly for a non-aqueous electrolyte rechargeable battery in which deformation is suppressed and improving heat resistance and energy density are simultaneously improved by improving a handling property of the separator during the manufacturing process.

According to D1 (par. 83-84) said lithium ion battery is provided with a separator comprising a layer consisting of a fluorine resin-containing particulate (which can be considered as a binder), a polyethylene ionomer (a binder) sodium polyacrylate and boehmite particles having an average particle diameter of 90 nm mixed in a volume ratio of 40:10:0.3: 50 respectively. Said layer is attached to a 12 microns thick layer consisting of a porous polyethylene film.

As the structure and composition of the separator of D1 fall under the subject matter of present claim 1, the technical features of present claims 4-7, even if not explicitly taught, are necessarily implicitly disclosed in D1.

Consequently the subject matter of present claims 1-8, 16 is not novel in the sense of Art. 33(2) PCT over D1.

Art. 33(3) PCT

The problem underlying the present invention is the provision of battery separators with increased dimensional stability at temperatures such as 150°C or higher (par. 3 of the present application).

D1 may be considered as the closest prior art for the subject matter of present claims 9-10.

The difference between the subject matter of claims 9 and 10 and D1 is that in the latter the average particle diameter is 90 nm instead of between about 5 nm and about 8 nm.

The data disclosed in table 1 on pages 23-24 of the present application show that a separator produced with a boehmite primary particle (Disperal D60) characterized by a primary particle size of 75 nm has a total coat weight for < 5% shrinkage at 150°C for 1 hour of ≥ 3.9 gsm; on the other hand a separator produced with a boehmite particle (Dispall 25f4) characterized by a primary particle size of 8 nm has a total coat weight for < 5% shrinkage at 150°C for 1 hour of ≥ 3.6 gsm.

There is therefore no technical effect deriving from said difference.

The resulting objective technical problem is the provision of alternative materials for the preparation of the separator of D1.

D2 (page 2, second paragraph) discloses Sasol Dispall 25F4 as a boehmite powder characterized by 7.2 nm crystallite size.

It would be therefore obvious to the skilled person to replace the boehmite particles of D1 with Dispall 25F4, thereby coming to the same subject matter of present claim 9.

As the structure and composition of the separator obtained by the combined teachings of D1 and D2 fall under the subject matter of present claim 10, the subject matter of said claim 10 is also obvious.

The subject matter of present claims 9-10 is therefore not inventive in view of the combined teachings of D1 and D2.

Invention 3

Art. 33(2) PCT

D3 (abstract; page 81, left-hand column, 3rd and 4th paragraphs; page 82, figure 2a) discloses a separator for a lithium ion secondary battery obtained by coating both faces of a 16-micron thick polyethylene porous membrane with a slurry composed of 35% boehmite particles having a 350 nm average particle size, 1% wt polyvinyl alcohol as a binder and distilled water; after drying this results in a 1.15 micron thick boehmite layer on both sides of the PE membrane. The separator shrinkage at 150 °C is about 5 % which, according to the teaching of figures 1-10 of the present application, corresponds to a coating weight significantly higher than 0.36 gsm.

Thus the subject matter of present claims 13, 14 and 16 is not novel in the sense of Art. 33(2) PCT over D3.

Art. 33(3) PCT

D3 represents the closest prior art to the subject matter of present claim 15. No technical effect can be derived from the additional features of claim 15. The resulting objective technical problem may be formulated as the provision of an alternative separator for lithium ion secondary batteries.

By combining D3 with D1 the skilled person would obtain all the features of present claim 15 except the particle size of the boehmite particles.

D2 (page 2, second paragraph) discloses a boehmite powder characterized by 7.2 nm crystallite size.

Thus the subject matter of present claim 15 is not inventive in the sense of Art 33(3) PCT in view of the combined teachings of D3, D1 and D2.

Re Item VII

Certain defects in the international application

The wording "about" in present claims 1, 2, 3, 6, 9, 11-13, 15 renders the subject matter of said claims unclear.

Present claims 1, 3, 9 discloses the parameter "average particle size".

**WRITTEN OPINION OF THE
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The claims do not disclose the method of experimental determination of said parameter. The subject matter of said claims is therefore not clear.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Inventor Information:

Inventor	1				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	David	W.	Avison		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Boxborough	State/Province	MA	Country of Residence	US
Mailing Address of Inventor:					
Address 1	31D Liberty Road				
Address 2					
City	Boxborough	State/Province	MA		
Postal Code	01719	Country	US		
Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Steven	A.	Carlson		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Cambridge	State/Province	MA	Country of Residence	US
Mailing Address of Inventor:					
Address 1	993 Memorial Drive, Apt. 101				
Address 2					
City	Cambridge	State/Province	MA		
Postal Code	02138	Country	US		
Inventor	3				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Benjamin		Sloan		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number		LGES-65213/155	
		Application Number			
Title of Invention		MULTILAYER NANOPOROUS SEPARATOR			
City	Exeter	State/Province	NH	Country of Residence	US
Mailing Address of Inventor:					
Address 1		11 Gardner Street			
Address 2					
City	Exeter	State/Province	NH		
Postal Code	03833	Country i	US		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	1912		
Email Address	ptodocket@arelaw.com	Add Email	Remove Email

Application Information:

Title of the Invention	MULTILAYER NANOPOROUS SEPARATOR				
Attorney Docket Number	LGES-65213/155	Small Entity Status Claimed ☐			
Application Type	Nonprovisional				
Subject Matter	Utility				
Total Number of Drawing Sheets (if any)	6	Suggested Figure for Publication (if any)			

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	1912		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78. When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending		Remove		
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
	Continuation of		17/223845	2021-04-06	
Prior Application Status	Patented		Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
17/223845	Continuation of	16/755294	2020-04-10	10998536	2021-05-04
Prior Application Status	Expired		Remove		
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
16/755294	a 371 of international		PCT/US2018/055862	2018-10-15	
Prior Application Status	Expired		Remove		
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
PCT/US2018/055862	Claims benefit of provisional		62/572083	2017-10-13	
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.					Add

Foreign Priority Information:

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGES-65213/155
	Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR	

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

			Remove
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGES-65213/155
	Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People’s Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO’s Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. ☐ If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. ☐ If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

PTO/AIA/14 (02-18)
Approved for use through 11/30/2020. OMB 0651-0032
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1			Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.				
		Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117		Joint Inventor	
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest		
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:				
Name of the Deceased or Legally Incapacitated Inventor:				
If the Applicant is an Organization check here. ☒				
Organization Name	Optodot Corporation			
Mailing Address Information For Applicant:				
Address 1	31 MacArthur Avenue			
Address 2				
City	Devens	State/Province	MA	
Country	US	Postal Code	01801	
Phone Number		Fax Number		
Email Address				
Additional Applicant Data may be generated within this form by selecting the Add button. Add				

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Assignee

1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

Remove

If the Assignee or Non-Applicant Assignee is an Organization check here.

☐

Prefix	Given Name	Middle Name	Family Name	Suffix

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1				
Address 2				
City		State/Province		
Country		Postal Code		
Phone Number		Fax Number		
Email Address				

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

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Signature:

Remove

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Brian J. Amos/			Date (YYYY-MM-DD)	2021-12-06
First Name	Brian J.	Last Name	Amos	Registration Number	63,679

Additional Signature may be generated within this form by selecting the Add button.

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EFS Web 2.2.13

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal				
Application Number:				
Filing Date:				
Title of Invention:		MULTILAYER NANOPOROUS SEPARATOR		
First Named Inventor/Applicant Name:		David W. Avison		
Filer:		Brian Amos/Joy Chapman		
Attorney Docket Number:		LGES-65213/155		
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	320	320
UTILITY SEARCH FEE	1111	1	700	700
UTILITY EXAMINATION FEE	1311	1	800	800
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1820

File History Content Report

The following content is missing from the original file history record obtained from the United States Patent and Trademark Office. No additional information is available.

Document Date - 2021-12-06

Document Title - Applicant Communication Re: Entity Status Set to Undiscounted
(Initial Default Setting or Status Change)



UNITED STATES PATENT AND TRADEMARK OFFICE

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United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/543,259	12/06/2021		1820	LGES-65213/155	20	2

CONFIRMATION NO. 1607

FILING RECEIPT

1912
AMSTER, ROTHSTEIN & EBENSTEIN LLP
90 PARK AVENUE
NEW YORK, NY 10016



Date Mailed: 12/15/2021

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

David W. Avison, Boxborough, MA;
Steven A. Carlson, Cambridge, MA;
Benjamin Sloan, Exeter, NH;

Applicant(s)

Optodot Corporation, Devens, MA;

Power of Attorney: The patent practitioners associated with Customer Number 1912

Domestic Priority data as claimed by applicant

This application is a CON of 17/223,845 04/06/2021
which is a CON of 16/755,294 04/10/2020 PAT 10998536
which is a 371 of PCT/US2018/055862 10/15/2018
which claims benefit of 62/572,083 10/13/2017

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.
Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/15/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/543,259**

Projected Publication Date: 03/24/2022

Non-Publication Request: No

Early Publication Request: No

Title

MULTILAYER NANOPOROUS SEPARATOR

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/543,259	12/06/2021	David W. Avison	LGES-65213/155

CONFIRMATION NO. 1607

PUBLICATION NOTICE



1912
AMSTER, ROTHSTEIN & EBENSTEIN LLP
90 PARK AVENUE
NEW YORK, NY 10016

Title:MULTILAYER NANOPOROUS SEPARATOR

Publication No.US-2022-0094018-A1
Publication Date:03/24/2022

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <https://portal.uspto.gov/pair/PublicPair>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Managment, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Electronic Acknowledgement Receipt	
EFS ID:	45754237
Application Number:	17543259
International Application Number:	
Confirmation Number:	1607
Title of Invention:	MULTILAYER NANOPOROUS SEPARATOR
First Named Inventor/Applicant Name:	David W. Avison
Customer Number:	1912
Filer:	Brian Amos/Joy Chapman
Filer Authorized By:	Brian Amos
Attorney Docket Number:	00249/0017
Receipt Date:	19-MAY-2022
Filing Date:	06-DEC-2021
Time Stamp:	13:38:47
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment		no			
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	00249-17_IDS_fillable-pdf.pdf	1263331	no	4
			818112b0799dc0b27c82deb8fa7d01849dfb1fda		
Warnings:					

Information:					
2	Non Patent Literature	CN-Office- Action_4-12-2022_CN- Appln-201880079213-4.pdf	499968	no	7
			e082f757d37df590713ab7979aa17e9115e6bac7		
Warnings:					
Information:					
Total Files Size (in bytes):			1763299		
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		17543259
Filing Date		2021-12-06
First Named Inventor	David W. Avison	
Art Unit		1729
Examiner Name		
Attorney Docket Number		00249/0017

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2022-05-19
Name/Print	Brian J. Amos	Registration Number	63,679

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17543259	
	Filing Date		2021-12-06	
	First Named Inventor	David W. Avison		
	Art Unit		1729	
	Examiner Name			
	Attorney Docket Number		00249/0017	

U.S.PATENTS							Remove	
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
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If you wish to add additional U.S. Patent citation information please click the Add button.							Add	
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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
	1	20170012264	A1	2017-01-12	Carlson et al.			
If you wish to add additional U.S. Published Application citation information please click the Add button.							Add	
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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² i	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		17543259
Filing Date		2021-12-06
First Named Inventor	David W. Avison	
Art Unit		1729
Examiner Name		
Attorney Docket Number		00249/0017

	1	Chinese Office Action dated April 12, 2022 issued in Chinese Patent Application No. 201880079213.4.	
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If you wish to add additional non-patent literature document citation information please click the Add button

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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.



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华知识产权代理有限公司
王达佐 (010-58732666) 洪欣 (010-58732666)

发文日:

2022 年 04 月 12 日



申请号或专利号: 201880079213.4

发文序号: 2022040702380310

申请人或专利权人: 奥普图多特公司

发明创造名称: 多层纳米多孔分隔物

第一次审查意见通知书

(进入国家阶段的 PCT 申请)

1. ☒ 应申请人提出的实质审查请求, 根据专利法第 35 条第 1 款的规定, 国家知识产权局对上述发明专利申请进行实质审查。
☐ 根据专利法第 35 条第 2 款的规定, 国家知识产权局决定自行对上述发明专利申请进行实质审查。
2. ☒ 申请人要求以其在:
US 专利局的申请日 2017 年 10 月 13 日为优先权日。
3. ☐ 经审查, 申请人于_____提交的修改文件, 不符合专利法实施细则第 51 条第 1 款的规定, 不予接受。
4. ☐ 审查是针对原始提交的国际申请的中文文本或中文译文进行的。
☒ 审查是针对下列申请文件进行的:
依据专利合作条约第 28 条或者第 41 条提交的修改, 权利要求第 1-21 项;
2020 年 6 月 8 日提交的说明书摘要、说明书第 1-102 段、说明书附图。
5. ☒ 本通知书引用下列对比文献 (其编号在今后的审查过程中继续沿用)

编号	文 件 号 或 名 称	公 开 日 期 (或抵触申请的申请日)
1	CN 105247703A	20160113
2	US 2016172705A1	20160616
3	US 2017012264A1	20170112

6. 审查的结论性意见:

关于说明书:

- ☐ 申请的内容属于专利法第 5 条规定的不授予专利权的范围。
☐ 说明书不符合专利法第 26 条第 3 款的规定。
☐ 说明书不符合专利法第 33 条的规定。
☐ 说明书的撰写不符合专利法实施细则第 17 条的规定。
☐ _____

210402
2021. 6

纸件申请, 回函请寄: 100088 北京市海淀区蓟门桥西土城路 6 号 国家知识产权局专利局受理处收
电子申请, 应当通过电子专利申请系统以电子文件形式提交相关文件。除另有规定外, 以纸件等其他形式提交的文件视为未提交。



国家知识产权局

关于权利要求书：

- ☐ 权利要求_____不符合专利法第 2 条第 2 款的规定。
- ☐ 权利要求_____不符合专利法第 9 条第 1 款的规定。
- ☐ 权利要求_____不具备专利法第 22 条第 2 款规定的新颖性。
- ☒ 权利要求 1-2, 4-21 不具备专利法第 22 条第 3 款规定的创造性。
- ☐ 权利要求_____不具备专利法第 22 条第 4 款规定的实用性。
- ☐ 权利要求_____属于专利法第 25 条规定的不授予专利权的范围。
- ☐ 权利要求_____不符合专利法第 26 条第 4 款的规定。
- ☐ 权利要求_____不符合专利法第 31 条第 1 款的规定。
- ☒ 权利要求 3 不符合专利法第 33 条的规定。
- ☐ 权利要求_____不符合专利法实施细则第 19 条的规定。
- ☐ 权利要求_____不符合专利法实施细则第 20 条的规定。
- ☐ 权利要求_____不符合专利法实施细则第 21 条的规定。
- ☐ 权利要求_____不符合专利法实施细则第 22 条的规定。

- ☐ 申请不符合专利法第 26 条第 5 款或者实施细则第 26 条的规定。
- ☐ 申请不符合专利法第 19 条第 1 款的规定。
- ☐ 分案申请不符合专利法实施细则第 43 条第 1 款的规定。

上述结论性意见的具体分析见本通知书的正文部分。

7. 基于上述结论性意见，审查员认为：

- ☐ 申请人应当按照通知书正文部分提出的要求，对申请文件进行修改。
- ☐ 申请人应当在意见陈述书中论述其专利申请可以被授予专利权的理由，并对通知书正文部分中指出的不符合规定之处进行修改，否则将不能授予专利权。
- ☒ 专利申请中没有可以被授予专利权的实质性内容，如果申请人没有陈述理由或者陈述理由不充分，其申请将被驳回。
- ☐ _____

8. 申请人应注意下列事项：

- (1) 根据专利法第 37 条的规定，申请人应在收到本通知书之日起的 4 个月内陈述意见，如果申请人无正当理由逾期不答复，其申请被视为撤回。
- (2) 申请人对其申请的修改应当符合专利法第 33 条的规定，不得超出原说明书和权利要求书记载的范围，同时申请人对专利申请文件进行的修改应当符合专利法实施细则第 51 条第 3 款的规定，按照本通知书的要求进行修改。
- (3) 申请人的意见陈述书和 / 或修改文本应当邮寄或递交国家知识产权局专利局受理处，凡未邮寄或递交给受理处的文件不具备法律效力。
- (4) 未经预约，申请人和 / 或代理人不得前来国家知识产权局与审查员举行会晤。
- (5) 对进入实质审查阶段的发明专利申请，在第一次审查意见通知书答复期限届满前（已提交答复意见的除外），主动申请撤回的，可以请求退还 50% 的专利申请实质审查费。

9. 本通知书正文部分共有 3 页，并附有下列附件：

- ☐ 引用的对比文件的复印件共_____份_____页。
- ☐ _____

审查员：司慧玲 联系电话：010-53966533 审查部门：专利审查协作北京中心电学发明审查部



国家知识产权局

第一次审查意见通知书

(进入国家阶段的 PCT 申请)

申请号:2018800792134

本申请涉及一种多层纳米多孔分隔物, 及包括此类分隔物的锂蓄电池。经审查, 现提出如下审查意见。

一、权利要求 1-2, 4-21 不具备专利法第 22 条第 3 款规定的创造性。

1. 权利要求 1 要求保护一种用于锂蓄电池的分隔物。对比文件 1 (CN105247703A, 参见说明书第 5-10 段, 第 37-41 段, 附图 5-8) 公开了一种电池组用多孔分隔物, 其主要公开了如下技术特征: 电池组用多孔分隔物 (对应本申请中的纳米多孔层) 包含陶瓷颗粒和聚合物黏合剂 (对应本申请中的纳米多孔层中的一种或多种聚合物), 陶瓷颗粒选自无机氧化物颗粒 (对应本申请中的纳米多孔层中的无机氧化物颗粒) 和无机氮化物颗粒, 多孔分隔物在暴露于 200℃ 的温度至少一小时的情况下表现出小于 1% 的收缩率 (对应本申请中的分隔物在 200℃ 下持续 1 小时的任何收缩均小于 5%); 在实施例中的图示 501 的聚合物分隔物是厚度为 18 μm、Gurley 透气性为 300 秒/100cc 的 Tonen® 三层聚烯烃 (对应本申请中的多孔聚合物层), 图示 503 的分隔物在单侧涂覆有包含 5.5:1 比例的勃姆石 (Disperal® 10SR) 和 Arkema Kynar® 761 的 3.5 μm 层; 图示 505 的分隔物在双侧涂覆有包含 5.5:1 比例的勃姆石 (Disperal® 10SR) 和 Arkema Kynar® 761 的 3.5 μm 层, 对应本申请中的在聚合物层的两个相对侧的至少一侧涂覆纳米多孔层; 另外, 对比文件 1 还公开了一种电化学电池, 包括锂盐的无机电解质和此多孔分隔物, 表明此电池组用多孔分隔物可用于锂蓄电池。权利要求 1 所要求保护的技术方案与对比文件 1 公开的技术内容相比, 区别技术特征是: 纳米多孔层的总涂层重量为 4.5 克/平方米或更小。基于上述区别技术特征, 可以确定权利要求 1 相对于对比文件 1 实际解决的技术问题是如何使纳米多孔层的重量较轻且热收缩率较小。

对于上述区别技术特征, 对比文件 2 (US2016172705A1, 参见说明书第 50, 83-109 段) 公开了一种耐热的隔板, 包括含氟树脂和勃姆石的涂层; 同时采用相同的方法制备了具有不同厚度的涂层的隔板并比较了其收缩率, 实验结果表明当涂层的厚度较小时, 隔板的收缩较大, 且当涂层厚度比较小时, 电池的厚度可更薄且具有更高的能量密度。对比文件 2 给出了隔板的收缩率与涂层厚度有关的启示, 因此, 为了寻求电池稳定性与能量密度之间的平衡, 本领域技术人员经过有限次实验即可确定得到的纳米多孔层的总涂层重量的合适范围为 4.5 克/平方米或更小。

由此可见, 本领域技术人员在对比文件 1 的基础上结合对比文件 2 以及公知常识就可以得到权利要求 1 请求保护的技术方案, 即权利要求 1 请求保护的技术方案对本领域技术人员是显而易见的, 不具有突出的实质性特点。因此, 权利要求 1 不具备专利法第 22 条第 3 款规定的创造性。

2. 权利要求 2 对权利要求 1 作了进一步的限定, 对比文件 1 (参见说明书第 5-10 段, 第 37-41 段, 附图 5-8) 中公开了图示 503 的聚合物分隔物是厚度为 18 μm、Gurley 透气性为 300 秒/100cc 的 Tonen® 三层聚烯烃, 因此, 在引用的权利要求 1 不具备创造性时, 权利要求 2 也不具备专利法第 22 条第 3 款规定的创造性。

3. 权利要求 4, 5 对权利要求 1 作了进一步的限定, 对比文件 1 (参见说明书第 5-10 段, 第 37-41 段, 附图 5-8) 中公开了图示 503 的分隔物包含 5.5:1 比例的勃姆石 (Disperal® 10SR) 和 Arkema Kynar® 761。在对比文件 1 给出的勃姆石与聚合物比例的教导下, 为了在提高能量密度时保证分隔物的高温稳定性, 本领域技术人员经过有限次的实验即可确定得到一种或多种聚合物在纳米多孔层的固体部分中的体积分数合适的范围。因此, 在引用的权利要求 1 不具备创造性时, 权利要求 4 和 5 不具备专利法第 22 条第 3 款规定的创造性。

4. 权利要求 6 对权利要求 1 作了进一步的限定, 对比文件 1 (参见说明书第 5-10 段, 第 37-41 段, 附图 5-8) 中已公开无机陶瓷颗粒包括 Al₂O₃、AlO(OH) 或勃姆石、AlN、BN、SiN、ZnO、ZrO₂、SiO₂ 及其组合。因此, 在引用的权利要求 1 不具备创造性时, 权利要求 6 也不具备专利法第 22 条第 3 款规定的创造性。

5. 权利要求 7 对权利要求 6 作了进一步的限定, 对比文件 2 (参见说明书第 50, 83-109 段) 公开了一种耐热的隔板, 包括含氟树脂和耐热颗粒的涂层, 耐热填料颗粒的平均粒径为约 10nm 至约 100nm; 耐热无机填料颗粒可以是例如陶瓷颗粒或例如金属氧化物颗粒, 金属氧化物颗粒可以是例如氧化铝、勃姆石、二氧化钛、氧化锆、氧化镁、氧化锌、氢氧化铝、氢氧化镁等的微粒。另外, 对比文件 2 中还公开了一种隔板中使用的勃姆石平均粒径是 90nm, 落入了权利要求 7 的保护范围。因此, 在引用的权利要求 6 不具备创造性时, 权利



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要求 7 不具备专利法第 22 条第 3 款规定的创造性。

6. 权利要求 8 对权利要求 7 作了进一步的限定，在对比文件 2（参见说明书第 50，83-109 段）公开的耐热填料颗粒的平均粒径为约 10nm 至约 100nm 的教导下，为了提高耐热性，本领域技术人员经过有限次的实验即可确定得到勃姆石颗粒的平均微晶尺寸。因此，在引用的权利要求 7 不具备创造性时，权利要求 8 也不具备专利法第 22 条第 3 款规定的创造性。

7. 权利要求 9 对权利要求 1 作了进一步的限定，对比文件 3（US 2017012264A1，参见说明书 25-34 段）公开了一种隔板层，包括勃姆石、聚合物及其共混物，其中的勃姆石可以是有机磺酸改性处理形成的改性水合氧化铝（对应于本申请中的疏水勃姆石）。因此，在引用的权利要求 1 不具备创造性时，权利要求 9 也不具备专利法第 22 条第 3 款规定的创造性。

8. 权利要求 10 对权利要求 1 作了进一步的限定，对比文件 1（参见说明书第 5-10 段，第 37-41 段，附图 5-8）公开了图示 505 的分隔物在双侧涂覆有包含 5.5:1 比例的勃姆石(Disperal® 10SR)和 Arkema Kynar® 761 的 3.5 μ m 层，对应于本申请中的多孔聚合物层的两个相对侧的两侧上涂覆有纳米多孔层。对比文件 2（参见说明书第 50，83-109 段）公开了分隔物的收缩率与其厚度有关，分隔物的厚度小，收缩就变大，但电池的能量密度也变大。因此，为了寻求电池稳定性和能量密度之间的平衡，本领域技术人员经过有限次的实验即可确定得到相应的总涂层重量的合适范围。因此，在引用的权利要求 1 不具备创造性时，权利要求 10 也不具备专利法第 22 条第 3 款规定的创造性。

9. 权利要求 11，12 对权利要求 1 作了进一步的限定，对比文件 1（参见说明书第 5-10 段，第 37-41 段，附图 5-8）公开了分隔物的平均孔隙尺寸是 10nm 至 90nm；在一些情况下，少于 1% 的孔具有在 10nm 至 90nm 之外的尺寸，落入了权利要求 11 和 12 的保护范围。因此，在引用的权利要求 1 不具备创造性时，权利要求 11 和 12 也不具备专利法第 22 条第 3 款规定的创造性。

10. 权利要求 13 对权利要求 1 作了进一步的限定，对比文件 2（参见说明书第 83-84 段）公开了将粘合剂层混合浆料涂覆到 12 μ m 厚的电晕处理的多孔聚乙烯隔膜（对应本申请中的聚乙烯）的两侧上并干燥，以制造涂层。因此，在引用的权利要求 1 不具备创造性时，权利要求 13 也不具备专利法第 22 条第 3 款规定的创造性。

11. 权利要求 14 要求保护一种锂蓄电池。对比文件 1（参见说明书第 5-10 段，第 37-41 段，附图 5-8）公开了一种电化学电池，其主要公开了以下技术特征：电化学电池，其包括阳极、阴极、包括锂盐的无机电解质、和包括有机聚合物（对应本申请中的纳米多孔层中的聚合物）和陶瓷材料的多孔分隔物层（对应本申请中的纳米多孔层），其中，多孔分隔物层具有 35% 至 50% 的孔隙率和 10nm 至 90nm 的平均孔隙尺寸，并且在暴露于 200℃ 的温度至少一小时的情况下表现出小于 1% 的收缩率（对应本申请中的分隔物在 200℃ 下持续 1 小时的任何收缩均小于 5%）；无机陶瓷颗粒选自无机氧化物颗粒（对应本申请中的纳米多孔层的无机氧化物颗粒）和无机氮化物颗粒；在实施例中的图示 501 的聚合物分隔物是厚度为 18 μ m、Gurley 透气性为 300 秒/100cc 的 Tonen® 三层聚烯烃（对应本申请中的包含聚烯烃的多孔聚合物层），图示 503 的分隔物在单侧涂覆有包含 5.5:1 比例的勃姆石(Disperal® 10SR)和 Arkema Kynar® 761 的 3.5 μ m 层；图示 505 的分隔物在双侧涂覆有包含 5.5:1 比例的勃姆石(Disperal® 10SR)和 Arkema Kynar® 761 的 3.5 μ m 层，对应本申请中的在聚合物层的两个相对侧的至少一侧涂覆纳米多孔层。权利要求 14 所要求保护的技术方案与对比文件 1 公开的技术内容相比，区别技术特征是：纳米多孔层的总涂层重量为 4.5 克/平方米或更小。基于上述区别技术特征，可以确定权利要求 14 相对于对比文件 1 实际解决的技术问题是如何使纳米多孔层的重量较轻且热收缩率较小。

对于上述区别技术特征，参见权利要求 1 的评述。

由此可见，本领域技术人员在对比文件 1 的基础上结合对比文件 2 以及公知常识就可以得到权利要求 14 请求保护的技术方案，即权利要求 14 请求保护的技术方案对本领域技术人员是显而易见的，不具有突出的实质性特点。因此，权利要求 14 不具备专利法第 22 条第 3 款规定的创造性。

12. 权利要求 15，16 对权利要求 14 作了进一步的限定，二者的附加技术特征的评述参见权利要求 4 和 5 的评述。因此，在引用的权利要求 14 不具备创造性时，权利要求 15 和 16 也不具备专利法第 22 条第 3 款规定的创造性。

210402
2021.6

纸件申请，回函请寄：100088 北京市海淀区蓟门桥西土城路 6 号 国家知识产权局专利局受理处收
电子申请，应当通过电子专利申请系统以电子文件形式提交相关文件。除另有规定外，以纸件等其他形式提交的文件视为未提交。



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13. 权利要求 17 对权利要求 14 作了进一步的限定，对比文件 2（参见说明书第 50，83-109 段）公开了耐热填料颗粒的平均粒径为约 10nm 至约 100nm，且耐热填料颗粒可以是勃姆石。在对比文件 2 给出的勃姆石颗粒的尺寸的教导下，为了提高耐热性，本领域技术人员经过有限次的实验即可确定得到勃姆石颗粒的平均微晶尺寸。因此，在引用的权利要求 14 不具备创造性时，权利要求 17 也不具备专利法第 22 条第 3 款规定的创造性。

14. 权利要求 18-21 对权利要求 14 作了进一步的限定，对权利要求 18-21 的附加技术特征的评述参见权利要求 10-13 的评述。因此，在引用的权利要求 14 不具备创造性时，权利要求 18-21 也不具备专利法第 22 条第 3 款规定的创造性。

二、权利要求 3 不符合专利法第 33 条的规定。

申请人对权利要求 3 的修改超出了原说明书和权利要求书记载的范围，不符合专利法第三十三条的规定。在原说明书和权利要求书中只记载了可以用于本申请的聚合物的实例包括但不限于，诸如羟乙基纤维素的纤维素聚合物、乙烯基吡咯烷酮聚合物和共聚物、诸如壳聚糖的多糖、聚环氧乙烷和聚乙烯醇，优选地，单独或组合的聚合物为纳米多孔层提供内聚强度和粘附强度，其在锂离子电池的电解质中是不溶的，并且在锂离子电池的使用电压范围内是电化学稳定的，如果需要获得这些性质，一种或多种聚合物可以是交联的；并且说明书实施例中记载了用于勃姆石颜料的化学改性的合适的疏水材料包括但不限于，有机磺酸、有机碳酸酯和具有疏水基团和反应性基团（例如羧酸和羟基基团）的聚合物，反应性基团可以与勃姆石或其它陶瓷颗粒的羟基基团反应；由此可见，至少一种包含选自羟基基团和羧酸基团的官能团的聚合物是用于对勃姆石进行化学改性，而不是用于形成分隔物为纳米多孔层提供内聚强度和粘附强度的聚合物，且根据原说明书和权利要求书记载的内容，本领域技术人员也不能直接地、毫无疑义地确定出用于勃姆石改性的聚合物可用于形成分隔物。因此，上述修改超出了原权利要求书和说明书记载的范围，不符合专利法第三十三条的规定。

基于上述理由，本申请不具备授予专利权的前景。如果申请人不能在本通知书规定的答复期限内提出表明本申请具有创造性的充分理由，本申请将被驳回。

对于权利要求书和说明书的修改，申请人应在意见陈述书中指明所作修改在原申请文件中的依据，以及修改不超范围的理由。对于修改，请申请人提交修改对照页，对所作修改做出清楚的标注。并保证对照页与替换页修改内容的一致性。

如您对审查意见存在疑问，可拨打审查员电话 010-53966533，或监督电话 010-53961276。请注意：电话反馈的内容不具备法律效力，如有意见，请将正式的意见陈述书和/或修改文本在规定的期限内提交给专利局受理部门。

审查员姓名:司慧玲
审查员代码:313327



国家知识产权局

检索报告

申请号：2018800792134		申请日：20181015		首次检索	
申请人：奥普图多特公司		最早的优先权日：20171013			
权利要求项数：16		说明书段数：102+11			
审查员确定的 IPC 分类号：H01M2/16,H01M10/0525					
检索记录信息：CN105247703A: CNTXT 语义检索,语义基准:2018800792134					
US2017012264A1: 引文查询					
US2016172705A1: 引文查询					
CN101796668A: 引文关联浏览					
相 关 专 利 文 献					
类型	国别以及代码[11] 给出的文献号	代码[43]或[45] 给出的日期	IPC 分类号	相关的段落 和 / 或图号	涉及的权 利要求
Y	CN105247703A	20160113	H01M2/16	说明书第 5-10 段,37-41 段, 附图 5-8	1-2,4-21
Y	US2016172705A1	20160616	H01M4/04	说明书第 50, 83-109 段	1-2,4-21
Y	US2017012264A1	20170112	H01G11/52	说明书第 25-34 段	9
A	CN101796668A	20100804	H01M2/16	全文	1-21



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相 关 非 专 利 文 献					
类型	书名(包括版本号和卷号)	出版日期	作者姓名和出版者名称	相关页数	涉及的权利要求
类型	期刊或文摘名称 (包括卷号和期号)	发行日期	作者姓名和文章标题	相关页数	涉及的权利要求

表格填写说明事项:

- 1. 审查员实际检索领域的 IPC 分类号应当填写到大组和 / 或小组所在的分类位置。
- 2. 期刊或其它定期出版物的名称可以使用符合一般公认的国际惯例的缩写名称。
- 3. 相关文件的类型说明:
 - X: 一篇文件影响新颖性或创造性;
 - Y: 与本报告中的另外的 Y 类文件组合而影响创造性;
 - A: 背景技术文件;
 - R: 任何单位或个人在申请日向专利局提交的、属于同样的发明创造的专利或专利申请文件。
 - P: 中间文件, 其公开日在申请的申请日与所要求的优先权日之间的文件;
 - E: 抵触申请。

审 查 员: 司慧玲
2022 年 03 月 24 日

审查部门: 专利审查协作北京中心电学发明审查部

Electronic Acknowledgement Receipt	
EFS ID:	46004799
Application Number:	17543259
International Application Number:	
Confirmation Number:	1607
Title of Invention:	MULTILAYER NANOPOROUS SEPARATOR
First Named Inventor/Applicant Name:	David W. Avison
Customer Number:	1912
Filer:	Brian Amos/Joy Chapman
Filer Authorized By:	Brian Amos
Attorney Docket Number:	00249/0017
Receipt Date:	21-JUN-2022
Filing Date:	06-DEC-2021
Time Stamp:	14:25:56
Application Type:	Utility under 35 USC 111(a)

Payment information:					
Submitted with Payment			no		
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00249-17_Second-Preliminary-Amendment.pdf	138901	yes	5
			597120b29102ee59cdb3a9b99364767f8aa1dca5		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Preliminary Amendment	1	1
	Claims	2	4
	Applicant Arguments/Remarks Made in an Amendment	5	5
Warnings:			
Information:			
Total Files Size (in bytes):		138901	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan

Appl. No. : 17/543,259

Filing Date : December 6, 2021

For : MULTILAYER NANOPOROUS SEPARATOR

Examiner : Not Yet Assigned

Art Unit : 1729

Customer No. : 1912

Confirmation No. : 1607

SECOND PRELIMINARY AMENDMENT

EFS
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Please amend the above-identified application as set forth below.

Inventors: David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No.: 17/543,259
Filing Date: December 6, 2021
page 2 of 5

Listing of Claims

Please amend the claims as follows:

1-20. (Canceled)

21. (New) A separator for a lithium battery comprising:

(a) a porous polymeric layer, and

(b) a second porous layer adjacent said porous polymeric layer, wherein

said second porous layer comprises hydrophobic inorganic oxide particles and one or more polymers, and

the total coating weight of said second porous layer is less than 4.5 grams per square meter, and

wherein the amount of water in the separator is no more than 600 parts per million of water.

22. (New) The separator of Claim 21, wherein the hydrophobic inorganic oxide particles comprise boehmite.

23. (New) The separator of Claim 22, wherein the second porous layer comprises an organic carbonate treated boehmite.

24. (New) The separator of Claim 23, wherein the organic carbonate reacts with hydroxyl groups of the boehmite.

Inventors: David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No.: 17/543,259
Filing Date: December 6, 2021
page 3 of 5

25. (New) The separator of Claim 23, wherein the organic carbonate is ethylene carbonate or fluoroethylene carbonate.

26. (New) The separator of Claim 21, wherein the inorganic oxide particles have a crystallite size of about 30 nm to about 90 nm.

27. (New) The separator of Claim 22, wherein the boehmite has a crystallite size of about 30 nm to about 50 nm.

28. (New) The separator of Claim 21, wherein the amount of water in the separator is no more than 300 parts per million of water.

29. (New) The separator of Claim 21, wherein the amount of water in the separator is no more than 100 parts per million of water.

30. (New) The separator of Claim 21, wherein the second porous layer is a nanoporous layer.

31. (New) A separator for a lithium battery comprising:

a porous polyethylene layer, and

a second porous layer adjacent said porous polyethylene layer, said second porous layer comprises a portion that is solid and a portion that is not solid, wherein

a. the solid portion comprises boehmite particles and one or more polymers:

b. a volume fraction of said one or more polymers in the solid portion is between about 15% and about 50%;

c. a total coating weight of said second porous layer on said porous polyethylene layer is about 5.5 grams per square meter or less, and

Inventors: David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No.: 17/543,259
Filing Date: December 6, 2021
page 4 of 5

- d. a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.
32. (New) The separator of claim 31, wherein the volume fraction of said one or more polymers in the solid portion is between 25% and 40%.
33. (New) The separator of claim 31, wherein said boehmite particles have an average crystallite size between 5 nm and 90 nm.
34. (New) The separator of claim 33, wherein the average crystallite size of said boehmite particles is between 30 nm and 50 nm.
35. (New) The separator of claim 31, wherein said second porous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of said pores have a diameter of 80 nm or less.
36. (New) The separator of claim 31, wherein said second porous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of said pores have a diameter of 50 nm or less.
37. (New) The separator of Claim 31, wherein the second porous layer is a nanoporous layer.

EFS

Inventors: David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No.: 17/543,259
Filing Date: December 6, 2021
Page 5 of 5

REMARKS

Claims 1-20 are pending in the subject application. By this Second Preliminary Amendment, applicant has cancelled Claims 1-20 without disclaimer or prejudice to applicant's right to pursue the subject matter of these claims in the future. In addition, applicant has added new Claims 21-37. Support for new Claims 21-37 can be found throughout the application as filed. Applicant maintains that the amendments do not raise an issue of new matter. Entry of the amendments is respectfully requested.

CONCLUSION

No fees are deemed necessary in connection with the filing of this Second Preliminary Amendment. However, if any fee is required, including any fee to preserve the pendency of the subject application, authorization is hereby given to charge the amount of any such fee to Deposit Account No. 01-1785. Overpayments may also be credited to Deposit Account No. 01-1785.

Respectfully submitted,

AMSTER, ROTHSTEIN & EBENSTEIN LLP
Attorneys for Applicant
90 Park Avenue
New York, New York 10016
(212) 336-8000

Dated: June 21, 2022
New York, New York

By /Brian J. Amos/
Brian J. Amos, Reg. No. 63,679

Electronic Acknowledgement Receipt	
EFS ID:	46222533
Application Number:	17543259
International Application Number:	
Confirmation Number:	1607
Title of Invention:	MULTILAYER NANOPOROUS SEPARATOR
First Named Inventor/Applicant Name:	David W. Avison
Customer Number:	1912
Filer:	Brian Amos/Joy Chapman
Filer Authorized By:	Brian Amos
Attorney Docket Number:	00249/0017
Receipt Date:	19-JUL-2022
Filing Date:	06-DEC-2021
Time Stamp:	14:00:23
Application Type:	Utility under 35 USC 111(a)

Payment information:					
Submitted with Payment			no		
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	00249-17_Corrected-ADS_to-update-Applcant.pdf	363272	no	9
			b556c1f32016aa305cb833f2e886e148b9656a17		
Warnings:					

Information:					
This is not an USPTO supplied ADS fillable form					
2	Assignee showing of ownership per 37 CFR 3.73	00249-17_3-73c-Statement.pdf	151532	no	3
			03508991f461422cc133b522d945b216af1a7919		
Warnings:					
Information:					
3	Power of Attorney	00249-17_Executed-General-POA.pdf	313451	no	2
			977c2b6f991a7ea8a92f0e805fccce793a4ba3e8		
Warnings:					
Information:					
Total Files Size (in bytes):			828255		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Inventor Information:

Inventor	1				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	David	W.	Avison		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Boxborough	State/Province	MA	Country of Residence	US
Mailing Address of Inventor:					
Address 1		31D Liberty Road			
Address 2					
City	Boxborough	State/Province	MA		
Postal Code	01719	Country	US		
Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Steven	A.	Carlson		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Cambridge	State/Province	MA	Country of Residence	US
Mailing Address of Inventor:					
Address 1		993 Memorial Drive, Apt. 101			
Address 2					
City	Cambridge	State/Province	MA		
Postal Code	02138	Country	US		
Inventor	3				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Benjamin		Sloan		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number		LGES-65213/155	
		Application Number			
Title of Invention		MULTILAYER NANOPOROUS SEPARATOR			
City	Exeter	State/Province	NH	Country of Residence	US
Mailing Address of Inventor:					
Address 1		11 Gardner Street			
Address 2					
City	Exeter	State/Province	NH		
Postal Code	03833	Country	US		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	1912		
Email Address	ptodocket@arelaw.com	Add Email	Remove Email

Application Information:

Title of the Invention	MULTILAYER NANOPOROUS SEPARATOR				
Attorney Docket Number	LGES-65213/155	Small Entity Status Claimed ☐			
Application Type	Nonprovisional				
Subject Matter	Utility				
Total Number of Drawing Sheets (if any)	6	Suggested Figure for Publication (if any)			

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	1912		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78. When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status		Pending		Remove	
Application Number		Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
		Continuation of		17/223845	2021-04-06
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
17/223845	Continuation of	16/755294	2020-04-10	10998536	2021-05-04
Prior Application Status		Expired		Remove	
Application Number		Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
16/755294		a 371 of international		PCT/US2018/055862	2018-10-15
Prior Application Status		Expired		Remove	
Application Number		Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
PCT/US2018/055862		Claims benefit of provisional		62/572083	2017-10-13
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.					Add

Foreign Priority Information:

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGES-65213/155
	Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR	

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

			Remove
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGES-65213/155
	Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐

A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐

B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1			Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.				
Clear				
● Assignee		Legal Representative under 35 U.S.C. 117		Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest		
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:				
Name of the Deceased or Legally Incapacitated Inventor:				
If the Applicant is an Organization check here. ☒				
Organization Name		Optodot Corporation LG ENERGY SOLUTION, LTD.		
Mailing Address Information For Applicant:				
Address 1		31 MacArthur Avenue Parc.1, 108 Yeoui-daero, Yeongdeungpo-gu		
Address 2				
City		Devens Seoul	State/Province	MA
Country		US KR	Postal Code	01801 07335
Phone Number			Fax Number	
Email Address				
Additional Applicant Data may be generated within this form by selecting the Add button. Add				

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Assignee

1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

Remove

If the Assignee or Non-Applicant Assignee is an Organization check here.

☐

Prefix	Given Name	Middle Name	Family Name	Suffix

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1				
Address 2				
City		State/Province		
Country		Postal Code		
Phone Number		Fax Number		
Email Address				

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

Add

Signature:

Remove

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Brian J. Amos/			Date (YYYY-MM-DD)	2022-07-19
First Name	Brian J.	Last Name	Amos	Registration Number	63,679

Additional Signature may be generated within this form by selecting the Add button.

Add

EFS Web 2.2.13

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES-65213/155
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

STATEMENT UNDER 37 CFR 3.73(c)

Applicant/Patent Owner: LG ENERGY SOLUTION, LTD.

Application No./Patent No.: 17/543,259 Filed/Issue Date: December 6, 2021

Titled: MULTILAYER NANOPOROUS SEPARATOR

LG ENERGY SOLUTION, LTD., a corporation

(Name of Assignee) (Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that, for the patent application/patent identified above, it is (choose one of options 1, 2, 3 or 4 below):

1. ☒ The assignee of the entire right, title, and interest.
2. ☐ An assignee of less than the entire right, title, and interest (check applicable box):

☐ The extent (by percentage) of its ownership interest is % . Additional Statement(s) by the owners holding the balance of the interest must be submitted to account for 100% of the ownership interest.

☐ There are unspecified percentages of ownership. The other parties, including inventors, who together own the entire right, title and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

3. ☐ The assignee of an undivided interest in the entirety (a complete assignment from one of the joint inventors was made). The other parties, including inventors, who together own the entire right, title, and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

4. ☐ The recipient, via a court proceeding or the like (e.g., bankruptcy, probate), of an undivided interest in the entirety (a complete transfer of ownership interest was made). The certified document(s) showing the transfer is attached.

The interest identified in option 1, 2 or 3 above (not option 4) is evidenced by either (choose one of options A or B below):

- A. ☐ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel , Frame , or for which a copy thereof is attached.
- B. ☒ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: David W. Avison, Steven A. Carlson, and Benjamin Sloan To: Optodot Corporation

The document was recorded in the United States Patent and Trademark Office at
Reel 053974 , Frame 0592 , or for which a copy thereof is attached.
2. From: Optodot Corporation To: LG ENERGY SOLUTION, LTD.

The document was recorded in the United States Patent and Trademark Office at
Reel 060000 , Frame 0089 , or for which a copy thereof is attached.

STATEMENT UNDER 37 CFR 3.73(c)

3. From: To:

The document was recorded in the United States Patent and Trademark Office at
Reel , Frame , or for which a copy thereof is attached.

4. From: To:

The document was recorded in the United States Patent and Trademark Office at
Reel , Frame , or for which a copy thereof is attached.

5. From: To:

The document was recorded in the United States Patent and Trademark Office at
Reel , Frame , or for which a copy thereof is attached.

6. From: To:

The document was recorded in the United States Patent and Trademark Office at
Reel , Frame , or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☒ As required by 37 CFR 3.73(c)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

/Brian J. Amos/

Signature

Brian J. Amos

Printed or Typed Name

July 19, 2022

Date

Reg. No. 63,679

Title or Registration Number

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE
REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	17/543,259
Filing Date	December 6, 2021
First Named Inventor	David W. Avison
Title	MULTILAYER NANOPOROUS SEPARATOR
Art Unit	1729
Examiner Name	Not Yet Assigned
Attorney Docket Number	00249/0017

SIGNATURE of Applicant or Patent Practitioner			
Signature	/Brian J. Amos/	Date (Optional)	July 19, 2022
Name	Brian J. Amos	Registration Number	63,679
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☒ *Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number
Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

1912

OR

I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:

The address associated with the above-mentioned Customer Number

OR

The address associated with Customer Number:

OR

Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

LG Energy Solution, Ltd.

Inventor or Joint Inventor (title not required below)

Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)

Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)

Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

June 24, 2022

Name

HAN SUN LEE

Title

Director, IP department

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.





UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/543,259	12/06/2021	1729	1820	00249/0017	20	2

1912
AMSTER, ROTHSTEIN & EBENSTEIN LLP
90 PARK AVENUE
NEW YORK, NY 10016

CONFIRMATION NO. 1607
CORRECTED FILING RECEIPT



Date Mailed: 07/20/2022

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

David W. Avison, Boxborough, MA;
Steven A. Carlson, Cambridge, MA;
Benjamin Sloan, Exeter, NH;

Applicant(s)

LG ENERGY SOULUTION, LTD., Seoul, KOREA, REPUBLIC OF;

Power of Attorney: The patent practitioners associated with Customer Number 1912

Domestic Priority data as claimed by applicant

This application is a CON of 17/223,845 04/06/2021
which is a CON of 16/755,294 04/10/2020 PAT 10998536
which is a 371 of PCT/US2018/055862 10/15/2018
which claims benefit of 62/572,083 10/13/2017

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.
Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/15/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/543,259**

Projected Publication Date: Not Applicable

Non-Publication Request: No

Early Publication Request: No
Title

MULTILAYER NANOPOROUS SEPARATOR

Preliminary Class

429

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific

countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign AssetsControl, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

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technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/543,259	12/06/2021	David W. Avison	00249/0017

1912
AMSTER, ROTHSTEIN & EBENSTEIN LLP
90 PARK AVENUE
NEW YORK, NY 10016

CONFIRMATION NO. 1607
POA ACCEPTANCE LETTER



Date Mailed: 07/20/2022

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 07/19/2022.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/rmohamed/

Electronic Acknowledgement Receipt	
EFS ID:	46234247
Application Number:	17543259
International Application Number:	
Confirmation Number:	1607
Title of Invention:	MULTILAYER NANOPOROUS SEPARATOR
First Named Inventor/Applicant Name:	David W. Avison
Customer Number:	1912
Filer:	Brian Amos/Joy Chapman
Filer Authorized By:	Brian Amos
Attorney Docket Number:	00249/0017
Receipt Date:	20-JUL-2022
Filing Date:	06-DEC-2021
Time Stamp:	15:39:57
Application Type:	Utility under 35 USC 111(a)

Payment information:					
Submitted with Payment			no		
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Corrected Filing Receipt	00249-17_Request-for-Corrected-Filing-Receipt_to-correct-typo-in-Applcant-Name.pdf	98523	no	2
			98cce37f5d968101e317b9c93c6c239e455d4556		
Warnings:					

Information:					
2	Request for Corrected Filing Receipt	00249-17_Marked-Up-Filing-Receipt.pdf	207047	no	4
			23da9a588d1d474477a96cceb7ed0179bf0335e8		
Warnings:					
Information:					
Total Files Size (in bytes):			305570		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Application No. : 17/543,259
Filed : December 6, 2021
For : MULTILAYER NANOPOROUS SEPARATOR
Art Unit : 1729
Examiner : Not Yet Assigned
Confirmation No. : 1607
Customer No. : 1912

REQUEST FOR CORRECTED FILING RECEIPT

VIA EFS

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

A Corrected Filing Receipt was issued by the U.S. Patent Office today, July 20, 2022.
Applicant requests that the following correction be made to the Corrected Filing Receipt:

In the "Applicant(s)" field, there is a typographical error in the Applicant's name. Please change "SOULUTION," to --SOLUTION,--. The Applicant's name was correctly identified on the Corrected Application Data Sheet, Power of Attorney, and 3.73(c) Statement filed on July 19, 2022.

A marked-up Filing Receipt is attached showing the requested correction. A marked-up Application Data Sheet is not required with this filing since the data was correctly indicated on the Corrected Application Data Sheet filed on July 19, 2022.

Please issue a Corrected Filing Receipt to reflect the correction.

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan
Application No. : 17/543,259
Filed : December 6, 2021
Page 2 of 2

No fee is deemed necessary with this request. If any fee is required, authorization is hereby given to withdraw any such fee from Deposit Account No. 01-1785.

Respectfully submitted,

AMSTER, ROTHSTEIN & EBENSTEIN LLP
Attorneys for Applicant
90 Park Avenue
New York, NY 10016
(212) 336-8000

Dated: July 20, 2022
New York, New York

By: /Brian J. Amos/
Brian J. Amos
Registration No. 63,679



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Alexandria, Virginia 22313-1450
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY.DOCKET.NO	TOT CLAIMS	IND CLAIMS
17/543,259	12/06/2021	1729	1820	00249/0017	20	2

1912
AMSTER, ROTHSTEIN & EBENSTEIN LLP
90 PARK AVENUE
NEW YORK, NY 10016

CONFIRMATION NO. 1607
CORRECTED FILING RECEIPT



Date Mailed: 07/20/2022

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

David W. Avison, Boxborough, MA;
Steven A. Carlson, Cambridge, MA;
Benjamin Sloan, Exeter, NH;

Applicant(s)

SOLUTION,
LG ENERGY ~~SOLUTION~~, LTD., Seoul, KOREA, REPUBLIC OF;

Power of Attorney: The patent practitioners associated with Customer Number 1912

Domestic Priority data as claimed by applicant

This application is a CON of 17/223,845 04/06/2021
which is a CON of 16/755,294 04/10/2020 PAT 10998536
which is a 371 of PCT/US2018/055862 10/15/2018
which claims benefit of 62/572,083 10/13/2017

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.
Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/15/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/543,259**

Projected Publication Date: Not Applicable

Non-Publication Request: No

Early Publication Request: No
Title

MULTILAYER NANOPOROUS SEPARATOR

Preliminary Class

429

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific

countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign AssetsControl, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/543,259	12/06/2021	1729	1820	00249/0017	20	2

1912
AMSTER, ROTHSTEIN & EBENSTEIN LLP
90 PARK AVENUE
NEW YORK, NY 10016

CONFIRMATION NO. 1607
CORRECTED FILING RECEIPT



Date Mailed: 07/21/2022

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Inventor(s)

David W. Avison, Boxborough, MA;
Steven A. Carlson, Cambridge, MA;
Benjamin Sloan, Exeter, NH;

Applicant(s)

LG ENERGY SOLUTION, LTD., Seoul, KOREA, REPUBLIC OF;

Power of Attorney: The patent practitioners associated with Customer Number 1912

Domestic Priority data as claimed by applicant

This application is a CON of 17/223,845 04/06/2021
which is a CON of 16/755,294 04/10/2020 PAT 10998536
which is a 371 of PCT/US2018/055862 10/15/2018
which claims benefit of 62/572,083 10/13/2017

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Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/15/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/543,259**

Projected Publication Date: Not Applicable

Non-Publication Request: No

Early Publication Request: No
Title

MULTILAYER NANOPOROUS SEPARATOR

Preliminary Class

429

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

Electronic Acknowledgement Receipt	
EFS ID:	46533742
Application Number:	17543259
International Application Number:	
Confirmation Number:	1607
Title of Invention:	MULTILAYER NANOPOROUS SEPARATOR
First Named Inventor/Applicant Name:	David W. Avison
Customer Number:	1912
Filer:	Brian Amos/Joy Chapman
Filer Authorized By:	Brian Amos
Attorney Docket Number:	00249/0017
Receipt Date:	01-SEP-2022
Filing Date:	06-DEC-2021
Time Stamp:	12:28:26
Application Type:	Utility under 35 USC 111(a)

Payment information:	
Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 100
RAM confirmation Number	E202291C29021873
Deposit Account	011785
Authorized User	Joy Chapman
The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows: 37 CFR 1.16 (National application filing, search, and examination fees) 37 CFR 1.17 (Patent application and reexamination processing fees)	

37 CFR 1.19 (Document supply fees)					
37 CFR 1.20 (Post Issuance fees)					
37 CFR 1.21 (Miscellaneous fees and charges)					
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00249-17_Third-Preliminary-Amendment.pdf	139395	yes	6
			ee24887426164aa4b9d03ccdc59f11f27a0d1a8d		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Claims		2	5	
	Applicant Arguments/Remarks Made in an Amendment		6	6	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	37978	no	2
			cbb9d06c35dd58126ed2e925f8195fc16d0970cb		
Warnings:					
Information:					
Total Files Size (in bytes):			177373		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventors : David W. Avison, Steven A. Carlson, and Benjamin Sloan

Appl. No. : 17/543,259

Filing Date : December 6, 2021

For : MULTILAYER NANOPOROUS SEPARATOR

Examiner : Not Yet Assigned

Art Unit : 1729

Customer No. : 1912

Confirmation No. : 1607

THIRD PRELIMINARY AMENDMENT

EFS
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Please amend the above-identified application as set forth below.

Inventors: David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No.: 17/543,259
Filing Date: December 6, 2021
page 2 of 6

Listing of Claims

Please amend the claims as follows:

1-20. (Canceled)

21. (Previously Presented) A separator for a lithium battery comprising:

(a) a porous polymeric layer, and

(b) a second porous layer adjacent said porous polymeric layer, wherein

said second porous layer comprises hydrophobic inorganic oxide particles and one or more polymers, and

the total coating weight of said second porous layer is less than 4.5 grams per square meter, and

wherein the amount of water in the separator is no more than 600 parts per million of water.

22. (Previously Presented) The separator of Claim 21, wherein the hydrophobic inorganic oxide particles comprise boehmite.

23. (Previously Presented) The separator of Claim 22, wherein the second porous layer comprises an organic carbonate treated boehmite.

24. (Previously Presented) The separator of Claim 23, wherein the organic carbonate reacts with hydroxyl groups of the boehmite.

Inventors: David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No.: 17/543,259
Filing Date: December 6, 2021
page 3 of 6

25. (Previously Presented) The separator of Claim 23, wherein the organic carbonate is ethylene carbonate or fluoroethylene carbonate.

26. (Previously Presented) The separator of Claim 21, wherein the inorganic oxide particles have a crystallite size of about 30 nm to about 90 nm.

27. (Previously Presented) The separator of Claim 22, wherein the boehmite has a crystallite size of about 30 nm to about 50 nm.

28. (Previously Presented) The separator of Claim 21, wherein the amount of water in the separator is no more than 300 parts per million of water.

29. (Previously Presented) The separator of Claim 21, wherein the amount of water in the separator is no more than 100 parts per million of water.

30. (Previously Presented) The separator of Claim 21, wherein the second porous layer is a nanoporous layer.

31. (Previously Presented) A separator for a lithium battery comprising:

a porous polyethylene layer, and

a second porous layer adjacent said porous polyethylene layer, said second porous layer comprises a portion that is solid and a portion that is not solid, wherein

a. the solid portion comprises boehmite particles and one or more polymers:

b. a volume fraction of said one or more polymers in the solid portion is between about 15% and about 50%;

c. a total coating weight of said second porous layer on said porous polyethylene

Inventors: David W. Avison, Steven A. Carlson, and Benjamin Sloan
Appl. No.: 17/543,259
Filing Date: December 6, 2021
page 4 of 6

layer is about 5.5 grams per square meter or less, and

d. a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.

32. (Previously Presented) The separator of claim 31, wherein the volume fraction of said one or more polymers in the solid portion is between 25% and 40%.

33. (Previously Presented) The separator of claim 31, wherein said boehmite particles have an average crystallite size between 5 nm and 90 nm.

34. (Previously Presented) The separator of claim 33, wherein the average crystallite size of said boehmite particles is between 30 nm and 50 nm.

35. (Previously Presented) The separator of claim 31, wherein said second porous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of said pores have a diameter of 80 nm or less.

36. (Previously Presented) The separator of claim 31, wherein said second porous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of said pores have a diameter of 50 nm or less.

37. (Previously Presented) The separator of Claim 31, wherein the second porous layer is a nanoporous layer.

38. (New) A process for making a separator for a lithium battery comprising:

(a) treating boehmite with an organic carbonate so as to make hydrophobic boehmite particles;

(b) providing a porous polymeric layer;

(c) providing a second porous layer, wherein said second porous layer comprises the hydrophobic boehmite particles and one or more polymers, and wherein the total coating weight

Inventors: David W. Avison, Steven A. Carlson, and Benjamin Sloan
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page 5 of 6

of said second porous layer is less than 4.5 grams per square meter, and

(d) bringing the second porous layer adjacent the porous polymeric layer, so as to make a separator with no more than 600 parts per million of water.

39. (New) The process of Claim 38, wherein the organic carbonate comprises ethylene carbonate.

40. (New) The process of Claim 38, wherein the organic carbonate comprises fluoroethylene carbonate.

41. (New) The process of Claim 38, wherein the organic carbonate reacts with hydroxyl groups of the boehmite.

EFS

Inventors: David W. Avison, Steven A. Carlson, and Benjamin Sloan
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Page 6 of 6

REMARKS

Claims 21-37 are pending in the subject application. By this Third Preliminary Amendment, applicant has added new Claims 38-41. Support for new Claims 38-41 can be found throughout the application as filed. Applicant maintains that the amendments do not raise an issue of new matter. Entry of the amendments is respectfully requested.

CONCLUSION

Applicant will be paying the excess claims fee herewith. However, if any additional fee is required, including any fee to preserve the pendency of the subject application, authorization is hereby given to charge the amount of any such fee to Deposit Account No. 01-1785. Overpayments may also be credited to Deposit Account No. 01-1785.

Respectfully submitted,

AMSTER, ROTHSTEIN & EBENSTEIN LLP
Attorneys for Applicant
90 Park Avenue
New York, New York 10016
(212) 336-8000

Dated: September 1, 2022
New York, New York

By /Brian J. Amos/
Brian J. Amos, Reg. No. 63,679

Electronic Patent Application Fee Transmittal				
Application Number:		17543259		
Filing Date:		06-Dec-2021		
Title of Invention:		MULTILAYER NANOPOROUS SEPARATOR		
First Named Inventor/Applicant Name:		David W. Avison		
Filer:		Brian Amos/Joy Chapman		
Attorney Docket Number:		00249/0017		
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	1202	1	100	100
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				100



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/543,259	12/06/2021	David W. Avison	00249/0017	1607
1912	7590	02/28/2023		
AMSTER, ROTHSTEIN & EBENSTEIN LLP			EXAMINER	
90 PARK AVENUE			PARK, LISA S	
NEW YORK, NY 10016				
ART UNIT		PAPER NUMBER		
1729				
MAIL DATE		DELIVERY MODE		
02/28/2023		PAPER		

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

2. Restriction to one of the following inventions is required under 35 U.S.C. 121:
 - I. Claims 21-37, drawn to a separator for a lithium battery, classified in H01M50/489.
 - II. Claims 38-41, drawn to a process of forming a separator for a lithium battery, classified in H01M50/446.
3. Inventions II and I are related as process of making and product made. The inventions are distinct if either or both of the following can be shown: (1) that the process as claimed can be used to make another and materially different product or (2) that the product as claimed can be made by another and materially different process (MPEP § 806.05(f)). In the instant case the separator of the battery could be formed by using a method using inorganic oxide or boehmite particles of a specific crystallite size, and the method of making the separator could be used to make an article having much higher water content than what is claimed in the product.

The examiner has required restriction between product or apparatus claims and process claims. Where applicant elects claims directed to the product/apparatus, and all product/apparatus claims are subsequently found allowable, withdrawn process claims that include all the limitations of the allowable product/apparatus claims should be considered for rejoinder. All claims directed to a nonelected process invention must include all the limitations of an allowable product/apparatus claim for that process invention to be rejoined.

In the event of rejoinder, the requirement for restriction between the product/apparatus claims and the rejoined process claims will be withdrawn, and the rejoined process claims will be fully examined for

patentability in accordance with 37 CFR 1.104. Thus, to be allowable, the rejoined claims must meet all criteria for patentability including the requirements of 35 U.S.C. 101, 102, 103 and 112. Until all claims to the elected product/apparatus are found allowable, an otherwise proper restriction requirement between product/apparatus claims and process claims may be maintained. Withdrawn process claims that are not commensurate in scope with an allowable product/apparatus claim will not be rejoined. See MPEP § 821.04. Additionally, in order for rejoinder to occur, applicant is advised that the process claims should be amended during prosecution to require the limitations of the product/apparatus claims. **Failure to do so may result in no rejoinder.** Further, note that the prohibition against double patenting rejections of 35 U.S.C. 121 does not apply where the restriction requirement is withdrawn by the examiner before the patent issues. See MPEP § 804.01.

4. Restriction for examination purposes as indicated is proper because all these inventions listed in this action are independent or distinct for the reasons given above and there would be a serious search and/or examination burden if restriction were not required because at least the following reason(s) apply:
- (a) the inventions have acquired a separate status in the art in view of their different classification; (as set forth above)
 - (b) the inventions have acquired a separate status in the art due to their recognized divergent subject matter; (as the method and product are not commensurate
 - (c) the inventions require a different field of search (for example, searching different classes/subclasses or electronic resources, or employing different search queries); (as the method and structure would require different searches) and
 - (d) the prior art applicable to one invention would not likely be applicable to another invention. (in light of the differences)

Applicant is advised that the reply to this requirement to be complete must include (i) an election of an invention to be examined even though the requirement may be traversed (37 CFR 1.143) and (ii) identification of the claims encompassing the elected invention.

The election of an invention may be made with or without traverse. To reserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the restriction requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable upon the elected invention.

Should applicant traverse on the ground that the inventions are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing the inventions to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the inventions unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103(a) of the other invention.

5. FURTHER, IF GROUP I (CLAIMS 21-17) IS ELECTED:

This application contains claims directed to the following **patentably distinct species**:

Species 1: The separator is described with respect to water content and hydrophobicity of boehmite particles (**Claims 21-30**)

Species 2: The separator is described with respect to heat shrinkage, pore diameter, and volume fraction of polymer (**Claims 31-37**)

The species are independent or distinct because the properties described in Species 1 and 2 are mutually exclusive. In addition, these species are not obvious variants of each other based on the current record.

6. Applicant is required under 35 U.S.C. 121 to elect a single disclosed species, or a single grouping of patentably indistinct species, for prosecution on the merits to which the claims shall be restricted if no generic claim is finally held to be allowable. The generic claim(s) in the instant case is/are NONE.

There is a search and/or examination burden for the patentably distinct species as set forth above because at least the following reason(s) apply: a separator having hydrophobic boehmite particles and a specific water content will require a special search that would not overlap with a search concerned with standard boehmite, shrinkage requirements, and polymer content by volume.

Applicant is advised that the reply to this requirement to be complete must include (i) an election of a species to be examined even though the requirement may be traversed (37 CFR 1.143) **and (ii) identification of the claims encompassing the elected species or grouping of patentably indistinct species**, including any claims subsequently added. An argument that a claim is allowable or that all claims are generic is considered nonresponsive unless accompanied by an election.

The election may be made with or without traverse. To preserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the election of species requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable on the elected species or grouping of patentably indistinct species.

Should applicant traverse on the ground that the species, or groupings of patentably indistinct species from which election is required, are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing them to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the species unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103 or pre-AIA 35 U.S.C. 103(a) of the other species.

Upon the allowance of a generic claim, applicant will be entitled to consideration of claims to additional species which depend from or otherwise require all the limitations of an allowable generic claim as provided by 37 CFR 1.141.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached on M-Th 6-3; Alt Fri 6-3.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Tavares-Crockett can be reached on 5712721481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LISA S PARK/

Primary Examiner, Art Unit 1729

Electronic Acknowledgement Receipt	
EFS ID:	47921029
Application Number:	17543259
International Application Number:	
Confirmation Number:	1607
Title of Invention:	MULTILAYER NANOPOROUS SEPARATOR
First Named Inventor/Applicant Name:	David W. Avison
Customer Number:	1912
Filer:	Brian Amos/Joy Chapman
Filer Authorized By:	Brian Amos
Attorney Docket Number:	00249/0017
Receipt Date:	28-APR-2023
Filing Date:	06-DEC-2021
Time Stamp:	13:38:06
Application Type:	Utility under 35 USC 111(a)

Payment information:					
Submitted with Payment			no		
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00249-17_Response-to-Restriction-Req.pdf	117536	yes	4
			6e78bb0af3a27e9241da60c2988669b7774c5ea6		

	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Response to Election / Restriction Filed		1	1	
	Applicant Arguments/Remarks Made in an Amendment		2	4	
Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	00249-17_IDS_fillable-pdf.pdf	1263411	no	4
			09b2a586a58bab7e9dc96ca4ed1692c2902ffd83		
Warnings:					
Information:					
Total Files Size (in bytes):			1380947		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Docket No. 00249.17/BJA/AGN

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant(s) : LG Energy Solution, Ltd.
U.S. Serial No. : 17/543,259 Examiner : Lisa S. Park
Filing Date : December 6, 2021 Art Unit : 1729
Confirmation No. : 1607
For : MULTILAYER NANOPOROUS SEPARATOR
Customer No. : 1912

405 Lexington Avenue
48th Floor
New York, NY 10174

BY EFS

Commissioner for Patents
Alexandria, VA 22313-1450

COMMUNICATION IN RESPONSE TO FEBRUARY 28, 2023 OFFICE ACTION
(RESTRICTION REQUIREMENT)

This Communication is submitted in response to the February 28, 2023 Office Action (Restriction Requirement) issued in connection with the above-identified application. A response to the February 28, 2023 Office Action (Restriction Requirement) is due April 28, 2023. Accordingly, this Communication is being timely filed.

Remarks begin on page 2 of this paper.

Applicant(s) : LG Energy Solution, Ltd.
Serial No. : 17/543,259
Filed : December 6, 2021
Page 2 of 4 : Communication in Response to February 28, 2023 Office
Action (Restriction Requirement)

REMARKS

Restriction Requirement

On page 2 of the February 28, 2023 Office Action, the Examiner required "Restriction to one of the following inventions [...] under 35 U.S.C. 121:"

Group I Claims 21-37, drawn to a separator for a lithium battery, classified in H01M50/489.

Group II Claims 38-41, drawn to a process of forming a separator for a lithium battery, classified in H01M50/446.

Applicant's Response

In response, the Applicant elects Group I, claims 21-37, drawn to a separator for a lithium battery, classified in H01M50/489.

Species Election

On page 4 of the February 28, 2023 Office Action, the Examiner alleged that the application "contains claims directed to the following patentably distinct species:"

Species 1 The separator is described with respect to water content and hydrophobicity of boehmite particles **(Claims 21-30)**

Species 2 The separator is described with respect to heat shrinkage, pore diameter, and volume fraction of polymer **(Claims 31-37)**

The Examiner required the Applicant to elect a single disclosed species, or a single grouping of patentably indistinct species under 35 U.S.C. §121, for prosecution on the merits to which the claims shall be restricted if no generic claim is finally held to be allowable.

Applicant(s) : LG Energy Solution, Ltd.
Serial No. : 17/543,259
Filed : December 6, 2021
Page 3 of 4 : Communication in Response to February 28, 2023 Office
Action (Restriction Requirement)

Applicant's Response

The Applicant respectfully disagrees with the Examiner's characterization of species I and II as "mutually exclusive".

Nonetheless, without conceding the correctness of the Examiner's characterization thereof, the Applicant elects "Species I - the separator is described with respect to water content and hydrophobicity of boehmite particles."

Claims 21-30 read on the elected species.

Applicant(s) : LG Energy Solution, Ltd.
Serial No. : 17/543,259
Filed : December 6, 2021
Page 4 of 4 : Communication in Response to February 28, 2023 Office
Action (Restriction Requirement)

If a telephone conference would be of assistance in advancing prosecution of the subject application, applicants' undersigned attorneys invite the Examiner to telephone them at the number provided below.

No fee is deemed necessary in connection with the filing of this Communication. However, if any fee is required, authorization is hereby given to charge the amount of any such fee to Deposit Account No. 01-1785.

Respectfully submitted,

/Brian J. Amos/

Dated: April 28, 2023
New York, New York

Brian J. Amos
Registration No. 63,679
Amster, Rothstein & Ebenstein LLP
405 Lexington Avenue
48th Floor
New York, NY 10174
(212) 336-8000

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number	17543259
Filing Date	2021-12-06
First Named Inventor	David W. Avison
Art Unit	1729
Examiner Name	Lisa S. Park
Attorney Docket Number	00249/0017

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2023-04-28
Name/Print	Brian J. Amos	Registration Number	63,679

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17543259	
	Filing Date		2021-12-06	
	First Named Inventor	David W. Avison		
	Art Unit		1729	
	Examiner Name	Lisa S. Park		
	Attorney Docket Number		00249/0017	

U.S.PATENTS							Remove	
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
	1	4024323	A	1977-05-17	Versteegh			
If you wish to add additional U.S. Patent citation information please click the Add button.							Add	
U.S.PATENT APPLICATION PUBLICATIONS							Remove	
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
	1	20170373295	A1	2017-12-28	Kim et al.			
	2	20160043370	A1	2016-02-11	Hatta et al.			
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

		Application Number	17543259
		Filing Date	2021-12-06
		First Named Inventor	David W. Avison
		Art Unit	1729
		Examiner Name	Lisa S. Park
		Attorney Docket Number	00249/0017

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1		

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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/543,259	12/06/2021	David W. Avison	00249/0017	1607
1912 7590 06/15/2023 AMSTER, ROTHSTEIN & EBENSTEIN LLP 405 LEXINGTON AVENUE FLOOR 48 NEW YORK, NY 10174			EXAMINER PARK, LISA S	
			ART UNIT	PAPER NUMBER
			1729	
			MAIL DATE	DELIVERY MODE
			06/15/2023	PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

2. Applicant's election without traverse of Group I (Claims 21-37) in the reply filed on 4/28/2023 is acknowledged. Claims 38-41 are withdrawn from further consideration pursuant to 37 CFR 1.142(b), as being drawn to a nonelected Group II, there being no allowable generic or linking claim.

Applicant's election with traverse of Species 1 in the reply filed on 4/28/2023 is acknowledged. The traversal is on the ground(s) that the species are not mutually exclusive.

The Office has considered this argument and respectfully disagrees. The properties claimed in Species 1 and 2 have no apparent overlap, as explained in the requirement for restriction, and therefore these species will require separate and extensive searches to properly address the patentability of each species. To search both species would cause undue burden on the Examiner.

The requirement is still deemed proper and is therefore made FINAL.

Therefore, **the claims under consideration are 21-30.**

Applicant timely traversed the restriction (election) requirement in the reply filed on 4/28/2023.

Priority

3. Applicant's claim for the benefit of a prior-filed application under 35 U.S.C. 119(e) or under 35 U.S.C. 120, 121, or 365(c) is acknowledged.

Information Disclosure Statement

4. Information disclosure statements (IDS), submitted November 30, 2011 and December 8, 2001, have been received and considered by the examiner.

Claim Interpretation

5. All “wherein” clauses are given patentable weight unless otherwise noted. Please see MPEP 2111.04 regarding optional claim language.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

6. Claim 21-22 and 26-29 are rejected under 35 U.S.C. 103 as being unpatentable over Katayama US PG Publication 2009/0067119 (hereinafter “Katayama ‘119”) in view of Katayama US PG Publication 2011/0003209 (hereinafter “Katayama ‘209”) and Nishikawa US PG Publication 2011/0165450.

Regarding Claims 21-22 and 28, Katayama ‘119 discloses a separator for a lithium battery comprising a porous polymeric layer (porous first separator layer including as a main ingredient at least one kind of resin/polymer such as PE fine particles) and a second porous layer (second separator layer) including an inorganic oxide particle filler such as boehmite plate-like particles and one or more polymers such as PP and SBR (styrene butadiene rubber) (paras 0014-0018, 0022-0025, 0119-0135, esp 0126). Katayama ‘119 does not specifically disclose wherein the inorganic oxide particle filler is a hydrophobic inorganic oxide filler or that the water content in the separator is no more than 600 ppm of water. However, in the same field of endeavor of composite, porous multilayer separators for nonaqueous lithium batteries including inorganic fillers, Katayama ‘209 teaches that plate-like or flake-like boehmite particles used as filler in a separator layer should be treated to be hydrophobic because this reduces the water content to a desirable level, such as below 500 ppm (para 0027) and preferably below 300 ppm (meeting **Claim 28**) and recites that reducing water content in the separator to the recited level using hydrophobic inorganic filler such as boehmite improves charge-discharge cycle characteristics, reliability, and storage characteristics of the battery (see e.g. paras 0027-0035, 0043-0043, 0049-0050, 0122-0123). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to treat the inorganic oxide (boehmite) particles of Katayama ‘119 such that they are hydrophobic and such that the water content of the resulting separator is less than 600 ppm and less than 300 ppm because Kayatama ‘209 teaches that reducing water content to this level using hydrophobically treated boehmite as the filler improves charge-discharge cycle characteristics, reliability, and storage characteristics of the battery.

Katayama '119 modified by Katayama '209 does not specifically disclose wherein the total coating weight of said second porous layer is less than 4.5 g/m². However, in the same field of endeavor of endeavor of composite, porous multilayer separators for nonaqueous lithium batteries including inorganic fillers, Nishikawa discloses that in a multilayer configuration where a porous layer containing an inorganic filler is coated on a base porous layer, the coating amount/weight and resulting thickness of the layer is a result-effective variable such as that is optimized to be below 3 g/m² to control heat shrinkage of the separator and porosity (see e.g. paras 0063-0072, 0106, 0123-0124, 0131-0132). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the separator of Katayama '119 modified by Katayama '209 such that the total coating weight of said second porous layer is less than 4.5 g/m² because Nishikawa teaches that such a coating layer should be coated at less than 3 g/m² to control heat shrinkage of the separator and porosity and that this characteristic (coating weight) is a result-effective variable. "[W]here the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." See *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). The discovery of an optimum value of a known result effective variable, without producing any new or unexpected results, is within the ambit of a person of ordinary skill in the art. See *In re Boesch*, 205 USPQ 215 (CCPA 1980) (see MPEP § 2144.05, II.).

Regarding Claims 26 and 27, Katayama '119 discloses in para 0060 that the boehmite particle flakes have a crystallite size (i.e. average particle diameter) of greater than 10 nm and is optimizable as long as it is smaller than the thickness of the separator (para 0060), a range that overlaps the claimed range of about 30 nm to about 90 nm or about 30 nm to about 50 nm. In the case where the claimed ranges "overlap or lie inside ranges disclosed by the prior art" a *prima facie* case of obviousness exists. *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990) (The prior art taught

carbon monoxide concentrations of "about 1-5%" while the claim was limited to "more than 5%." The court held that "about 1-5%" allowed for concentrations slightly above 5% thus the ranges overlapped.); *In re Geisler*, 116 F.3d 1465, 1469-71, 43 USPQ2d 1362, 1365-66 (Fed. Cir. 1997) (Claim reciting thickness of a protective layer as falling within a range of "50 to 100 Angstroms" considered *prima facie* obvious in view of prior art reference teaching that "for suitable protection, the thickness of the protective layer should be not less than about 10 nm [i.e., 100 Angstroms].") The court stated that "by stating that 'suitable protection' is provided if the protective layer is 'about' 100 Angstroms thick, [the prior art reference] directly teaches the use of a thickness within [applicant's] claimed range."). Similarly, a *prima facie* case of obviousness exists where the claimed ranges and prior art ranges do not overlap but are close enough that one skilled in the art would have expected them to have the same properties. *Titanium Metals Corp. of America v. Banner*, 778 F.2d 775, 227 USPQ 773 (Fed. Cir. 1985) (Court held as proper a rejection of a claim directed to an alloy of "having 0.8% nickel, 0.3% molybdenum, up to 0.1% iron, balance titanium" as obvious over a reference disclosing alloys of 0.75% nickel, 0.25% molybdenum, balance titanium and 0.94% nickel, 0.31% molybdenum, balance titanium.).

Regarding Claim 29, since Katayama '209 teaches that water concentration should be minimized as much as possible, where 0 ppm is the most preferable and that the usual lower limit is usually "about 100 ppm" (para 0123), e.g. exemplifying 95 ppm in Example 12 (Table 2) and the skilled artisan would understand that "about" would include values below 100 ppm. In the case where the claimed ranges "overlap or lie inside ranges disclosed by the prior art" a *prima facie* case of obviousness exists. *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990) (The prior art taught carbon monoxide concentrations of "about 1-5%" while the claim was limited to "more than 5%." The

court held that "about 1-5%" allowed for concentrations slightly above 5% thus the ranges overlapped.); *In re Geisler*, 116 F.3d 1465, 1469-71, 43 USPQ2d 1362, 1365-66 (Fed. Cir. 1997) (Claim reciting thickness of a protective layer as falling within a range of "50 to 100 Angstroms" considered *prima facie* obvious in view of prior art reference teaching that "for suitable protection, the thickness of the protective layer should be not less than about 10 nm [i.e., 100 Angstroms].") The court stated that "by stating that 'suitable protection' is provided if the protective layer is 'about' 100 Angstroms thick, [the prior art reference] directly teaches the use of a thickness within [applicant's] claimed range."). Similarly, a *prima facie* case of obviousness exists where the claimed ranges and prior art ranges do not overlap but are close enough that one skilled in the art would have expected them to have the same properties. *Titanium Metals Corp. of America v. Banner*, 778 F.2d 775, 227 USPQ 773 (Fed. Cir. 1985) (Court held as proper a rejection of a claim directed to an alloy of "having 0.8% nickel, 0.3% molybdenum, up to 0.1% iron, balance titanium" as obvious over a reference disclosing alloys of 0.75% nickel, 0.25% molybdenum, balance titanium and 0.94% nickel, 0.31% molybdenum, balance titanium.).

7. Claims 23-25 are rejected under 35 U.S.C. 103 as being unpatentable over Katayama US PG Publication 2009/0067119 (hereinafter "Katayama '119") in view of Katayama US PG Publication 2011/0003209 (hereinafter "Katayama '209"), as applied to Claim 22, and further in view of Carlson US PG Publication 2007/0190427.

Regarding Claims 23-25, Katayama '119 modified by Katayama '209 discloses the claimed separator for use in a lithium battery as described above in the rejection of Claim 22, which is incorporated herein in its entirety, including the use of hydrophobic boehmite particles. Katayama '119 modified by Katayama '209 does not specifically disclose that the hydrophobic boehmite particles comprise a reaction product of an organic carbonate (e.g. ethylene carbonate or

fluorethylene carbonate) with boehmite. However, Carlson discloses a separator for a battery wherein boehmite particles are treated with an organic carbonate such as ethylene carbonate (meeting **Claims 24 and 25**) since a reaction of the boehmite particles with the polymer would necessarily react with hydroxyl groups of boehmite to provide organically-modified boehmite filler in order to provide high ionic conductivity and improved strength (see at least paras 0017, 0033, 0045). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to include in the boehmite particles of Katayama '119 modified by Katayama '209 the organic carbonate-modified hydrophobic boehmite (specifically ethylene carbonate-modified) particles of Carlson because these particles are known to improve the ionic conductivity and strength of a battery separator. The selection of a known material, which is based upon its suitability for the intended use, is within the ambit of one of ordinary skill in the art. See *In re Leshin*, 125 USPQ 416 (CCPA 1960) (see MPEP § 2144.07).

8. Claim 30 is rejected under 35 U.S.C. 103 as being unpatentable over Katayama US PG Publication 2009/0067119 (hereinafter “Katayama ‘119”) in view of Katayama US PG Publication 2011/0003209 (hereinafter “Katayama ‘209”), as applied to Claim 21, and further in view of Ying US Patent 6,194,098.

Regarding Claim 30, Katayama '119 modified by Katayama '209 discloses the claimed separator for use in a lithium battery as described above in the rejection of Claim 21, which is incorporated herein in its entirety, including the use of hydrophobic boehmite particles. Katayama '119 modified by Katayama '209 does not specifically disclose wherein the second porous layer is a nanoporous layer. However, in the same field of endeavor of boehmite-based separator layers in lithium ion batteries, Ying discloses wherein the inorganic oxide-containing layer is nanoporous because such porosity enables the separator to readily uptake and retain electrolyte liquids while

also functioning as an ultrafiltration membrane, blocking all particles and insoluble materials to inhibit diffusion of soluble materials below a particular cutoff level (see e.g. col 14, lines 28-39, col 15, line 40- col 16, line 19). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the second porous layer of Katayama '119 modified by Katayama '209 as a nanoporous layer because Ying teaches that this size of pore enables the separator to readily uptake and retain electrolyte liquids while also functioning as an ultrafiltration membrane, blocking all particles and insoluble materials to inhibit diffusion of soluble materials below a particular cutoff level.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached on M-Th 6 or 7 to 3 or 4; Alt Fri afternoons.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached on 5712721481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer

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Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LISA S PARK/
Primary Examiner, Art Unit 1729

<i>Notice of References Cited</i>	Application/Control No. 17/543,259		Applicant(s)/Patent Under Reexamination Avison et al.	
	Examiner LISA S PARK		Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS						
*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20090067119-A1	03-2009	Katayama; Hideaki	H01M50/403	361/523
*	B	US-6194098-B1	02-2001	Ying; Qicong	H01M50/443	429/251
	C					
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	E					
	F					
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number	65213/155	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

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☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2021-12-06
Name/Print	Brian J. Amos	Registration Number	63,679

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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PTO/SB/08a (02-18)

Approved for use through 11/30/2020. OMB 0651-0031

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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number		65213/155

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	9293752	B2	2016-03-22	Murata et al.	
	2	9847519	B2	2017-12-19	Pekalaet et al.	
	3	7709152	B2	2010-05-04	Kim et al.	
	4	7662517	B2	2010-02-16	Lee et al.	
	5	7638241	B2	2009-12-29	Lee et al.	
	6	6432586	B1	2002-08-13	Zhang	
	7	6723467	B2	2004-04-20	Yoshida et al.	
	8	7883799	B2	2011-02-08	Seo et al.	

Receipt date: 12/06/2021

17/543,259 - GAU: 1729

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STATEMENT BY APPLICANT
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Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number	65213/155	

	9	8318360	B2	2012-11-27	Seo et al.	
	10	9484565	B2	2016-11-01	Miyazaki et al.	
	11	9312527	B2	2016-04-12	Muramatsu et al.	
	12	9543077	B2	2017-01-10	Honda et al.	
	13	9478784	B2	2016-10-25	Iwase	

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	1	20160172705	A1	2016-06-16	Hoshiba	
	2	20170294638	A1	2017-10-12	Joo et al.	
	3	20180019457	A1	2018-01-18	Alharizah et al.	(U.S. counterpart of WO 2016/123404 A1 listed below)
	4	20180248164	A1	2018-08-30	Pekala et al.	

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Attorney Docket Number	65213/155	

	5	20170301903	A1	2017-10-19	Choi et al.	
	6	20160104876	A1	2016-04-14	Avison et al.	
	7	20110165450	A1	2011-07-07	Nishikawa	
	8	20110003209	A1	2011-01-06	Katayama et al.	
	9	20070190427	A1	2007-08-16	Carlson et al.	
	10	20150155601	A1	2015-06-04	Nakatsutsumi et al.	

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	1	2016123404	WO	A1	2016-08-04	Hercules Incorporated	(WO counterpart of U.S. 2018/0019457 A1 listed above)	

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Receipt date: 12/06/2021

17/543,259 - GAU: 1729

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number	65213/155	

	1	F. Sola et al., "A new scanning electron microscopy approach to image aerogels at the nanoscale; A new scanning electron microscopy approach to image aerogels at the nanoscale," Nanotechnology, IOP, BRISTOL, GB, Vol. 22, No. 17, March 16, 2011, page 175704, XP020188888, ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704.	
	2	Chongwen Yang et al., "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries," Journal of Power Sources, Vol. 348, March 3, 2017, pp. 80-86, XP029949881, ISSN: 0378-7753, DOI: 10.1016/J.POWSOUR.2017.02.078.	
	3	International Search Report received for PCT Patent Application No. PCT/US2018/055862, mailed on April 11, 2019, 6 pages,	
	4	Written Opinion received for PCT Patent Application No. PCT/US2018/055862, mailed on April 11, 2019, 13 pages.	

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17543259
	Filing Date		2021-12-06
	First Named Inventor	David W. Avison	
	Art Unit		1729
	Examiner Name		
	Attorney Docket Number		00249/0017

CERTIFICATION STATEMENT

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That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐

That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.
The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2022-05-19
Name/Print	Brian J. Amos	Registration Number	63,679

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Privacy Act Statement

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Receipt date: 05/19/2022

17/543,259 - GAU: 1729

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (01-22)
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number	17543259
Filing Date	2021-12-06
First Named Inventor	David W. Avison
Art Unit	1729
Examiner Name	
Attorney Docket Number	00249/0017

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
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	1	20170012264	A1	2017-01-12	Carlson et al.			
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17/543,259 - GAU: 1729

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17543259
	Filing Date		2021-12-06
	First Named Inventor	David W. Avison	
	Art Unit		1729
	Examiner Name		
	Attorney Docket Number		00249/0017

1	Chinese Office Action dated April 12, 2022 issued in Chinese Patent Application No. 201880079213.4.		
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Examiner Signature	/LISA S PARK/		Date Considered 06/14/2023
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.			

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Application Number	17543259
Filing Date	2021-12-06
First Named Inventor	David W. Avison
Art Unit	1729
Examiner Name	Lisa S. Park
Attorney Docket Number	00249/0017

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Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2023-04-28
Name/Print	Brian J. Amos	Registration Number	63,679

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Receipt date: 04/28/2023

17/543,259 - GAU: 1729

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (01-22)
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number	17543259
Filing Date	2021-12-06
First Named Inventor	David W. Avison
Art Unit	1729
Examiner Name	Lisa S. Park
Attorney Docket Number	00249/0017

U.S.PATENTS

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	1	20170373295	A1	2017-12-28	Kim et al.	
	2	20160043370	A1	2016-02-11	Hatta et al.	

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Receipt date: 04/28/2023

17/543,259 - GAU: 1729

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Examiner Name	Lisa S. Park	
Attorney Docket Number		00249/0017

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Electronic Acknowledgement Receipt	
EFS ID:	48717012
Application Number:	17543259
International Application Number:	
Confirmation Number:	1607
Title of Invention:	MULTILAYER NANOPOROUS SEPARATOR
First Named Inventor/Applicant Name:	David W. Avison
Customer Number:	1912
Filer:	Brian Amos/Joy Chapman
Filer Authorized By:	Brian Amos
Attorney Docket Number:	00249/0017
Receipt Date:	13-OCT-2023
Filing Date:	06-DEC-2021
Time Stamp:	13:26:34
Application Type:	Utility under 35 USC 111(a)

Payment information:	
Submitted with Payment	yes
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RAM confirmation Number	E20230CD27216434
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3		00249-0017_Reply-to-OA.pdf	183068	yes	8
			40440d504a2091d43a5c310c1f8161eeac690abc		
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	Claims		2	5	
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4	Fee Worksheet (SB06)	fee-info.pdf	40631	no	2
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National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		17543259
Filing Date		2021-12-06
First Named Inventor	David W. Avison	
Art Unit		1729
Examiner Name	Lisa S. Park	
Attorney Docket Number		00249/0017

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	Filing Date		2021-12-06	
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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17543259
	Filing Date		2021-12-06
	First Named Inventor	David W. Avison	
	Art Unit		1729
	Examiner Name	Lisa S. Park	
	Attorney Docket Number		00249/0017

	1	Communication Pursuant to Article 94(3) EPC dated September 8, 2023 issued in European Patent Application No. 18 799 922.2.	☐
If you wish to add additional non-patent literature document citation information please click the Add button			
EXAMINER SIGNATURE			
Examiner Signature			Date Considered
*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.			
¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.			



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Questions about this communication?
Contact Customer Services at
www.epo.org/contact

Primary Examiner
Name: Bettio, Andrea
Tel: +49 89 2399 - 8346

Application No. 18 799 922.2 - 1108	Ref. OPTO-003-EP	Date 08.09.2023
Applicant LG Energy Solution, Ltd.		

Communication pursuant to Article 94(3) EPC

The examination of the above-identified application has revealed that it does not meet the requirements of the European Patent Convention for the reasons enclosed herewith. If the deficiencies indicated are not rectified the application may be refused pursuant to Article 97(2) EPC.

You are invited to file your observations and insofar as the deficiencies are such as to be rectifiable, to correct the indicated deficiencies within a period

of 4 months

from the notification of this communication, this period being computed in accordance with Rules 126(2) and 131(2) and (4) EPC. One set of amendments to the description, claims and drawings is to be filed within the said period on separate sheets (R. 50(1) EPC).

If filing amendments, you must identify them and indicate the basis for them in the application as filed. Failure to meet either requirement may lead to a communication from the Examining Division requesting that you correct this deficiency (R. 137(4) EPC).

Failure to comply with this invitation in due time will result in the application being deemed to be withdrawn (Art. 94(4) EPC).

Registered Letter

EPO Form 2001 11.16AUWF (05.09.2023)



Bettio, Andrea
Primary Examiner
For the Examining Division

Enclosure(s): 2 page/s reasons (EPO Form 2906)

The examination is being carried out on the **following application documents**

Description, Pages

1-26	filed in electronic form on	01-06-2023
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Claims, Numbers

1-9	filed in electronic form on	01-06-2023
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Drawings, Sheets

1/6-6/6	as published
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Reference is made to the following documents:

- | | |
|----|---|
| D1 | US 2016/172705 A1 (HOSHIBA KOJI [JP]) 16 June 2016 (2016-06-16) |
| D2 | SOLÁ F ET AL: "A new scanning electron microscopy approach to image aerogels at the nanoscale;A new scanning electron microscopy approach to image aerogels at the nanoscale",
NANOTECHNOLOGY, IOP, BRISTOL, GB,
vol. 22, no. 17, 16 March 2011 (2011-03-16), page 175704, XP020188888,
ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704 |
| D3 | YANG CHONGWEN ET AL: "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries",
JOURNAL OF POWER SOURCES,
vol. 348, 3 March 2017 (2017-03-03), pages 80-86, XP029949881,
ISSN: 0378-7753, DOI: 10.1016/J.JPOWSOUR.2017.02.078 |

Art. 84 EPC

Claim 1 is not supported by the description as required by Article 84 EPC, as its scope is broader than justified by the description. Claim 1 discloses the wording "one or more polymer binders that provide adhesive and cohesive strength to the nanoporous layer". Whilst the chemical nature of said polymers is not disclosed by the claim's wording, the

teaching of par. 37 of the present application is that only some specific classes of polymers may be used for said purpose, namely "cellulosic polymers...vinyl pyrrolidone polymers and copolymers, polysaccharides, polyethylene oxides and polyvinyl alcohol.". The subject matter of present Claim 1 therefore does not comply with the requirements of ART. 84 EPC.

30384

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant(s) : LG Energy Solution, Ltd.
U.S. Serial No. : 17/543,259 Examiner : Lisa S. Park
Filing Date : December 6, 2021 Art Unit : 1729
Confirmation No. : 1607
For : MULTILAYER NANOPOROUS SEPARATOR

405 Lexington Avenue
48th Floor
New York, NY 10174
October 13, 2023

BY EFS
Commissioner for Patents
Alexandria, VA 22313-1450

**RESPONSE TO JUNE 15, 2023 OFFICE ACTION AND PETITION FOR ONE-MONTH
EXTENSION OF TIME**

This Amendment is submitted in response to the June 15, 2023 Office Action issued in connection with the above-identified application. A response to the June 15, 2023 Office Action was due September 15, 2023. Applicant petitions herewith for a one-month extension of time and is paying the required fee. Accordingly, this Amendment is being timely filed.

Amendments to the Claims begin on page 2 of this paper.

Remarks begin on page 6 of this paper.

Applicant(s) : LG Energy Solution, Ltd.
Serial No. : 17/543,259
Filed : December 6, 2021
Page 2 of 8 : Response to June 15, 2023 Office Action AND PETITION
FOR ONE-MONTH EXTENSION OF TIME

AMENDMENTS TO THE CLAIMS

Please amend the claims by replacing all prior versions of the claims with the listing of claims below pursuant to 37 C.F.R. §1.121.

Listing of Claims

1-20. (Canceled)

21. (Currently Amended) A separator for a lithium battery comprising:

(a) a porous polymeric layer, and

(b) a second porous layer adjacent said porous polymeric layer,
wherein said second porous layer comprises hydrophobic inorganic
oxide particles and one or more polymers, ~~and~~

wherein 0.27 grams, or greater, per square meter of hydrophobic
inorganic oxide particles are disposed within the pores of the porous
polymeric layer, and

~~the total coating weight of said second porous layer is
less than 4.5 grams per square meter, and~~

wherein the amount of water in the separator is no more than 600 parts
per million of water.

22. (Previously Presented) The separator of Claim 21, wherein the
hydrophobic inorganic oxide particles comprise boehmite.

23. (Previously Presented) The separator of Claim 22, wherein the
second porous layer comprises an organic carbonate treated boehmite.

24. (Previously Presented) The separator of Claim 23, wherein the
organic carbonate reacts with hydroxyl groups of the boehmite.

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FOR ONE-MONTH EXTENSION OF TIME

25. (Previously Presented) The separator of Claim 23, wherein the organic carbonate is ethylene carbonate or fluoroethylene carbonate.

26. (Previously Presented) The separator of Claim 21, wherein the inorganic oxide particles have a crystallite size of about 30 nm to about 90 nm.

27. (Previously Presented) The separator of Claim 22, wherein the boehmite has a crystallite size of about 30 nm to about 50 nm.

28. (Previously Presented) The separator of Claim 21, wherein the amount of water in the separator is no more than 300 parts per million of water.

29. (Previously Presented) The separator of Claim 21, wherein the amount of water in the separator is no more than 100 parts per million of water.

30. (Previously Presented) The separator of Claim 21, wherein the second porous layer is a nanoporous layer.

31. (Withdrawn) A separator for a lithium battery comprising:
a porous polyethylene layer, and
a second porous layer adjacent said porous polyethylene layer, said second porous layer comprises a portion that is solid and a portion that is not solid, wherein
a. the solid portion comprises boehmite particles and one or more polymers:
b. a volume fraction of said one or more polymers in the solid portion is between about 15% and about 50%;

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c. a total coating weight of said second porous layer on said porous polyethylene layer is about 5.5 grams per square meter or less, and
d. a heat shrinkage of said separator at 150°C for 1 hour is less than 5%.

32. (Withdrawn) The separator of claim 31, wherein the volume fraction of said one or more polymers in the solid portion is between 25% and 40%.

33. (Withdrawn) The separator of claim 31, wherein said boehmite particles have an average crystallite size between 5 nm and 90 nm.

34. (Withdrawn) The separator of claim 33, wherein the average crystallite size of said boehmite particles is between 30 nm and 50 nm.

35. (Withdrawn) The separator of claim 31, wherein said second porous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of said pores have a diameter of 80 nm or less.

36. (Withdrawn) The separator of claim 31, wherein said second porous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of said pores have a diameter of 50 nm or less.

37. (Withdrawn) The separator of Claim 31, wherein the second porous layer is a nanoporous layer.

38. (Withdrawn) A process for making a separator for a lithium battery comprising:

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(a) treating boehmite with an organic carbonate so as to make hydrophobic boehmite particles;

(b) providing a porous polymeric layer;

(c) providing a second porous layer, wherein said second porous layer comprises the hydrophobic boehmite particles and one or more polymers, and wherein the total coating weight of said second porous layer is less than 4.5 grams per square meter, and

(d) bringing the second porous layer adjacent the porous polymeric layer, so as to make a separator with no more than 600 parts per million of water.

39. (Withdrawn) The process of Claim 38, wherein the organic carbonate comprises ethylene carbonate.

40. (Withdrawn) The process of Claim 38, wherein the organic carbonate comprises fluoroethylene carbonate.

41. (Withdrawn) The process of Claim 38, wherein the organic carbonate reacts with hydroxyl groups of the boehmite.

42. (New) The separator of Claim 21, wherein the total coating weight of said second porous layer is less than 4.5 grams per square meter.

43. (New) The separator of Claim 21, wherein the wherein an average crystallite size of said inorganic particles disposed within the pores of said second porous polymeric layer is between 5 nm and 90 nm.

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FOR ONE-MONTH EXTENSION OF TIME

REMARKS

Claims 21-41 are pending in the subject application, with claims 31-41 withdrawn from examination. Independent claim 21 has been amended herein. Support for the amendment to claim 21 can be found in the specification at paragraph [0079]. Dependent claims 42 and 43 have been added. Support for claims 42 and 43 can be found in the specification at, *inter alia*, [0043], [0044], and example 3. Applicant maintains that the amendments to the claims raise no issue of new matter and entry of these amendments is respectfully requested.

Rejection under 35 U.S.C. §103

In the June 15, 2023 office action, the Examiner alleged, *inter alia*, that claim 21 was obvious over Katayama US PG Publication 2009/0067119 (hereinafter "Katayama `119") in view of Katayama US PG Publication 2011/0003209 (hereinafter "Katayama `209") and Nishikawa US PG Publication 2011/0165450. The Examiner alleged Katayama `119 discloses a porous polymeric layer, and a ceramic layer; Katayama `209 discloses using plate- or flake-like hydrophobically treated boehmite particles to reduce water content below 300 ppm; and Nishikawa teaches that low coating weight is a result effective variable.

The Examiner further alleged that claims 23-25 are obvious over Katayama `119 in view of Katayama `209 in addition to Carlson US PG Publication 2007/0190427 (hereinafter "Carlson") because Katayama `119 modified by Katayama `209 discloses the claimed separator for use in a lithium battery, including the use of hydrophobic boehmite particles. The Examiner noted that Katayama `119 modified by Katayama `209 does not specifically disclose that the hydrophobic boehmite particles comprise a reaction product of an organic carbonate, but alleged Carlson discloses a separator for a battery, wherein boehmite particles are treated with an organic carbonate.

Finally the Examiner alleged that claim 30 was obvious over Katayama `119 modified by Katayama `209 which discloses the claimed

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separator for use in a lithium battery, and which is incorporated herein in its entirety, including the use of hydrophobic boehmite particles. The Examiner noted that Katayama `119 modified by Katayama `209 does not specifically disclose wherein the second porous layer is a nanoporous layer, but alleged Ying US Patent 6,194,098 (hereinafter "Ying") discloses wherein the inorganic oxide-containing layer is nanoporous.

Applicant's response

In response, without conceding the correctness of the Examiner's rejection, and solely to advance prosecution, applicant has amended claim 21 to recite in relevant part "wherein 0.27 grams, or greater, per square meter of hydrophobic inorganic oxide particles are disposed within the pores of the porous polymeric layer." As discussed below, the separators that are claimed are non-obvious over the prior art.

None of Katayama `119 and `209 and Nishikawa discloses a separator containing "0.27 grams, or greater, per square meter of hydrophobic inorganic oxide particles are disposed within the pores of the porous polymeric layer" let alone the same "wherein the amount of water in the separator is no more than 600 parts per million of water." Furthermore, nothing in these references suggest that it would have been obvious to a person of ordinary skill in the art before the effective filing date to arrive at the claimed invention.

As the remainder of the rejected claims are dependent from claim 21, applicant respectfully submits that, since they incorporate the non-obvious limitations of claim 21, they are also not obvious at least for this reason.

Accordingly, applicant respectfully requests the withdrawal of the rejections.

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FOR ONE-MONTH EXTENSION OF TIME

If a telephone conference would be of assistance in advancing prosecution of the subject application, applicants' undersigned attorneys invite the Examiner to telephone them at the number provided below.

No fee, other than the one-month extension fee and the fee for two (2) extra total claims in excess of twenty (20) paid herewith, is deemed necessary in connection with the filing of this Amendment. However, if any fee is required, authorization is hereby given to charge the amount of any such fee to Deposit Account No. 01-1785.

Respectfully submitted,

AMSTER, ROTHSTEIN & EBENSTEIN LLP
Attorneys for Applicant
405 Lexington Ave.
New York, New York 10174
(212) 336-8000

Dated: October 13, 2023
New York, New York

By /Brian J. Amos/
Brian J. Amos, Reg. No. 63,679

Electronic Patent Application Fee Transmittal				
Application Number:		17543259		
Filing Date:		06-Dec-2021		
Title of Invention:		MULTILAYER NANOPOROUS SEPARATOR		
First Named Inventor/Applicant Name:		David W. Avison		
Filer:		Brian Amos/Joy Chapman		
Attorney Docket Number:		00249/0017		
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	1202	2	100	200
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Extension - 1 month with \$0 paid	1251	1	220	220
Miscellaneous:				
Total in USD (\$)				420

Examiner-Initiated Interview Summary	Application No. 17/543,259	Applicant(s) Avison et al.		
	Examiner LISA S PARK	Art Unit 1729	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
LISA S PARK	Primary Examiner	Telephonic
Brian Amos	Attorney of Record	

Date of Interview: 11 December 2023

Issues Discussed:

Proposed Amendment(s)

Examiner proposed an Examiner's amendment to cancel non-elected claims to place the application in condition for allowance.

/LISA S PARK/ Primary Examiner, Art Unit 1729	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions:It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

Notice of Allowability	Application No. 17/543,259	Applicant(s) Avison et al.	
	Examiner LISA S PARK	Art Unit 1729	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. [X] This communication is responsive to amendment filed 10/13/2023.
[] A declaration(s)/affidavit(s) under 37 CFR 1.130(b) was/were filed on ____.

2. [] An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

3. [X] The allowed claim(s) is/are 21-30 and 42-43 . As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information , please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. [] Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:
a) [] All b) [] Some* c) [] None of the:
1. [] Certified copies of the priority documents have been received.
2. [] Certified copies of the priority documents have been received in Application No. ____ .
3. [] Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).
* Certified copies not received: ____ .

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. [] CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
[] including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____ .
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. [] DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)
1. [] Notice of References Cited (PTO-892)
2. [X] Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date 20231013.
3. [] Examiner's Comment Regarding Requirement for Deposit of Biological Material ____.
4. [X] Interview Summary (PTO-413), Paper No./Mail Date. 20231211.
5. [X] Examiner's Amendment/Comment
6. [X] Examiner's Statement of Reasons for Allowance
7. [] Other ____.

/LISA S PARK/ Primary Examiner, Art Unit 1729	
--	--

DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Elections/Restriction

2. This application is in condition for allowance except for the presence of claims 31-41 directed to Species 2 and Group II, non-elected on 4/28/2023. Accordingly, claims 31-41 have been cancelled.

Response to Amendment

3. In response to the amendment received on 10/13/2023:
 - a. Claims 21-41 are pending in the current application. Claim 21 is amended, Claims 31-41 stand withdrawn, Claims 1-20 are cancelled, and Claims 42-43 are newly added.
 - b. The previous prior art rejections are overcome in light of the amendment.

Examiner's Amendment

4. An Examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee. This examiner's amendment to cancel non-elected claims was approved by Applicant's representative, Brian Amos, by telephone on 12/11/2023. The Examiner's amendment to Claim 21 is provided herein as a matter not affecting the scope of the claim.

- Cancel non-elected claims 31-41.
- 21. (Currently Amended) A separator for a lithium battery comprising: (a) a porous polymeric layer, and (b) a second porous layer adjacent said porous polymeric layer, wherein said second porous layer comprises hydrophobic inorganic oxide particles and one or more polymers, wherein 0.27 grams, or greater, per square meter of hydrophobic inorganic oxide particles are disposed within the pores of the porous polymeric layer, and wherein the amount of water in the separator is no more than 600 parts per million ~~of water~~.

Allowable Subject Matter

5. Claims 21-30 and 42-43 are allowed.

The following is an examiner's statement of reasons for allowance: none of the prior art of record, alone or in combination, appears to teach, suggest, or render obvious the invention of Claims 21-30 and 42-43.

Independent Claim 21 recites a separator comprising a porous polymeric layer and a second porous layer adjacent to the porous polymer layer, said layer comprising polymer and also comprising hydrophobic inorganic oxide particles disposed in pores of the porous polymer layer in a quantity of 0.27 g/m² and wherein the amount of water in the separator is no more than 600 ppm. Related US application 17/223,845 has previously been allowed based largely on this same concept, i.e. inorganic oxide loading level in pores of a polymeric separator, but the claims scopes are sufficiently different as to not cause an issue of double patenting. There is no teaching in the prior art in this application (or related applications) that would lead the skilled artisan to arrive at the claimed invention such that a separator would specifically feature a porous polymeric layer and a second porous layer adjacent to the porous polymer layer, said layer

comprising polymer and also comprising hydrophobic inorganic oxide particles disposed in pores of the porous polymer layer in a quantity of 0.27 g/m² and wherein the amount of water in the separator is no more than 600 ppm. Therefore, the references of record fail to teach or suggest the particulars of the independent Claim 21, and it's not obvious to modify these teachings to give the instant claimed invention. Thus none of the prior art of the record, alone or in combination, appears to teach, suggest, or render obvious the invention of independent Claim 21. Since Claims 22-30 and 42-43 depend on Claim 21, they are allowable for the same reason.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached M-Th 5:30a to 3p Eastern Time.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Tavares-Crockett can be reached on 5712721481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LISA S PARK/
Primary Examiner, Art Unit 1729

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		17543259
Filing Date		2021-12-06
First Named Inventor	David W. Avison	
Art Unit		1729
Examiner Name	Lisa S. Park	
Attorney Docket Number		00249/0017

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

- ☐ See attached certification statement.
- ☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.
- ☐ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2023-10-13
Name/Print	Brian J. Amos	Registration Number	63,679

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these record s.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Receipt date: 10/13/2023

17/543,259 -- GAU: 1729

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (01-22)
Approved for use through 05/31/2024. OMB 0651-0031
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE
Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17543259	
	Filing Date		2021-12-06	
	First Named Inventor	David W. Avison		
	Art Unit		1729	
	Examiner Name	Lisa S. Park		
	Attorney Docket Number		00249/0017	

U.S.PATENTS								
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
	1							
If you wish to add additional U.S. Patent citation information please click the Add button.								
U.S.PATENT APPLICATION PUBLICATIONS								
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
	1							
If you wish to add additional U.S. Published Application citation information please click the Add button.								
FOREIGN PATENT DOCUMENTS								
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐
If you wish to add additional Foreign Patent Document citation information please click the Add button								
NON-PATENT LITERATURE DOCUMENTS								
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.						T ⁵

17/543,259 -- GAU: 1729

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17543259
	Filing Date		2021-12-06
	First Named Inventor	David W. Avison	
	Art Unit	1729	
	Examiner Name	Lisa S. Park	
	Attorney Docket Number	00249/0017	

1	Communication Pursuant to Article 94(3) EPC dated September 8, 2023 issued in European Patent Application No. 18 799 922.2.	☐
If you wish to add additional non-patent literature document citation information please click the Add button		
EXAMINER SIGNATURE		
Examiner Signature	/LISA S PARK/	Date Considered 10/18/2023
*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.		
¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.		



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

1912 7590 12/28/2023
AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

EXAMINER
PARK, LISA S

ART UNIT PAPER NUMBER

1729

DATE MAILED: 12/28/2023

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/543,259	12/06/2021	David W. Avison	00249/0017	1607

TITLE OF INVENTION: MULTILAYER NANOPOROUS SEPARATOR

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/28/2024

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.
If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.
If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".
For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

1912759012/28/2023

AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

	(Typed or printed name)
	(Signature)
	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/543,259	12/06/2021	David W. Avison	00249/0017	1607

TITLE OF INVENTION: MULTILAYER NANOPOROUS SEPARATOR

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/28/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
PARK, LISA S	1729	429-145000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	--

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____Date _____

Typed or printed name _____Registration No. _____



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UNITED STATES DEPARTMENT OF COMMERCE
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/543,259	12/06/2021	David W. Avison	00249/0017	1607
1912	7590	12/28/2023	EXAMINER	
AMSTER, ROTHSTEIN & EBENSTEIN LLP			PARK, LISA S	
405 LEXINGTON AVENUE			ART UNIT	
FLOOR 48			PAPER NUMBER	
NEW YORK, NY 10174			1729	
DATE MAILED: 12/28/2023				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b)
(Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.



UNITED STATES
PATENT AND TRADEMARK OFFICE

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
17/543,259	03/26/2024 04:28:12 PM Z ET	00249/0017

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	1607	FILED BY	Joy Chapman
PATENT CENTER #	64844324	FILING DATE	12/06/2021
CUSTOMER #	1912	FIRST NAMED INVENTOR	David W. Avison
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Brian Amos

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
00249-17_Issue-Fee-Transmittal.pdf	1	Issue Fee Payment (PTO-85B)	164 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
00249-17_Issue-Fee-Transmittal.pdf	FDAAB047295EC78A84A3C38528C84264989B5EF89D434ABE9096C5FCEE16D1630F4AB154183938AC38AF112D1DB25E5C961E663E5F0D1EFB623EE5E5326F4638

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

1912 7590 12/28/2023
AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

Brian J. Amos	(Typed or printed name)
/Brian J. Amos/	(Signature)
March 26, 2024	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/543,259	12/06/2021	David W. Avison	00249/0017	1607

TITLE OF INVENTION: MULTILAYER NANOPOROUS SEPARATOR

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/28/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
PARK, LISA S	1729	429-145000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.
	1 _____ 2 Amster, Rothstein & Ebenstein LLP 3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE
LG ENERGY SOLUTION, LTD.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)
Seoul, Korea

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 01-1785

5. Change in Entity Status (from status indicated above)
- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Brian J. Amos/

Date March 26, 2024

Typed or printed name Brian J. Amos

Registration No. 63,679



UNITED STATES
PATENT AND TRADEMARK OFFICE

ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
17/543,259	03/26/2024 04:28:12 PM Z ET	00249/0017

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	1607	FILED BY	Joy Chapman
PATENT CENTER #	64844324	AUTHORIZED BY	Brian Amos
CUSTOMER #	1912	FILING DATE	12/06/2021
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	David W. Avison

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 1077	E20243PG29095744	Joy Chapman
PRE-AUTHORIZED ACCOUNT	PRE-AUTHORIZED CATEGORY	
011785	37 CFR 1.16 (National application filing, search, and examination fees); 37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.19 (Document supply fees); 37 CFR 1.20 (Post Issuance fees); 37 CFR 1.21 (Miscellaneous fees and charges)	

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1200.00	1	1200.00
			TOTAL AMOUNT:	\$1,200.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement

Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

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United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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17/543,259	04/30/2024	11973181	00249/0017	1607
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1912	7590	04/10/2024
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AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO’s Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b)
(application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

David W. Avison, Boxborough, MA;
Steven A. Carlson, Cambridge, MA;
Benjamin Sloan, Exeter, NH;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

LG ENERGY SOLUTION, LTD., Seoul, KOREA, REPUBLIC OF;

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United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/543,259	12/06/2021	David W. Avison	00249/0017	1607
1912 7590 05/02/2024 AMSTER, ROTHSTEIN & EBENSTEIN LLP 405 LEXINGTON AVENUE FLOOR 48 NEW YORK, NY 10174			EXAMINER PARK, LISA S	
			ART UNIT	PAPER NUMBER
			1729	
			MAIL DATE	DELIVERY MODE
			05/02/2024	PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

APPLICATION NO.	ISSUE DATE	PATENT NO.
17/543,259	30-Apr-2024	11973181

AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
NEW YORK, NY 10174

EGRANT NOTIFICATION

Your electronic patent grant (eGrant) is now available, which can be accessed via Patent Center at <https://patentcenter.uspto.gov>

The electronic patent grant is the official patent grant under 35 U.S.C. 153. For more information, please visit <https://www.uspto.gov/electronicgrants>

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/543,259		Filing Date 12/06/2021		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)		(Column 2)					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)	
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A			
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A			
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A			
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		x \$100 =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		x \$480 =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL			
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT	06/21/2022	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 17	Minus	** 20	= 0	x \$ 100 =		0	
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0	x \$ 480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						SLIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/KIMBERLY D WILLIAMS/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875						Application or Docket Number 17/543,259				
APPLICATION AS FILED - PART I										
(Column 1)		(Column 2)		SMALL ENTITY		OR	OTHER THAN SMALL ENTITY			
FOR	NUMBER FILED	NUMBER EXTRA		RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)		
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A		N/A			N/A	320		
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A		N/A			N/A	700		
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A		N/A			N/A	800		
TOTAL CLAIMS (37 CFR 1.16(j))	20	minus 20 =	*			OR	x 100 =	0.00		
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2	minus 3 =	*				x 480 =	0.00		
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							0.00		
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))								0.00		
* If the difference in column 1 is less than zero, enter "0" in column 2.				TOTAL			TOTAL	1820		
APPLICATION AS AMENDED - PART II										
(Column 1)		(Column 2)		(Column 3)		SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.										

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/543,259		Filing Date 12/06/2021		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)		(Column 2)					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)	
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A			
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A			
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A			
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		x \$100 =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		x \$480 =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL			
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT	09/01/2022	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 21	Minus	** 20	= 1	x \$ 100 =		100	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$ 480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		100	
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						SLIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/SHARON M WEST/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/543,259		Filing Date 12/06/2021		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)		(Column 2)					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)	
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A			
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A			
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A			
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		x \$100 =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		x \$480 =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL			
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT	10/13/2023	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 23	Minus	** 21	= 2	x \$ 100 =		200	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$ 480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		200	
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/VERONICA EVERETTE/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

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Patent Export, 2026-06-18 02:26:22 +0000

Search results for: PNS=(US11973181);

Collections searched: DWPI; ARIPO grant; Argentina UM; Argentina app; Armenia grant; Australia UM; Australia app; Australia grant; Austria UM; Austria app; Austria grant; Belarus UM; Belarus grant; Belgium app; Belgium grant; Brazil UM; Brazil app; Brazil grant; Bulgaria UM; Bulgaria app; Bulgaria grant; Canada app; Canada grant; China, Mainland UM; China, Mainland app; China, Mainland grant; Colombia app; Costa Rica UM; Costa Rica app; Croatia UM; Croatia app; Croatia grant; Cuba app; Cuba grant; Czech Republic UM; Czech Republic app; Czech Republic grant; Czechoslovakia grant; Denmark UM; Denmark app; Denmark grant; Estonia UM; Estonia app; Estonia grant; Eurasian (EAPO) grant; Europe (EP) app; Europe (EP) grant; Finland app; Finland grant; France app; France grant; Georgia grant; German Democratic Republic app; German Democratic Republic grant; Germany UM; Germany app; Germany grant; Greece app; Greece grant; Gulf Cooperation Council grant; Hong Kong UM; Hong Kong app; Hong Kong grant; Hungary grant; Iceland grant; India app; India grant; Indonesia UM; Indonesia app; Ireland UM; Ireland app; Ireland grant; Israel grant; Italy grant; Italy grant; Italy grant; Italy grant; Japan UM; Japan app; Japan grant; Korea UM; Korea app; Korea grant; Latvia grant; Lithuania grant; Luxembourg app; Luxembourg grant; Malaysia app; Malaysia grant; Mexico app; Mexico grant; Moldova UM; Moldova grant; Monaco grant; Mongolia UM; Mongolia grant; Morocco app; Morocco grant; Netherlands app; Netherlands grant; New Zealand grant; Norway app; Norway grant; OAPI grant; PCT (WO) app; Philippines UM; Philippines app; Philippines grant; Poland UM; Poland grant; Portugal UM; Portugal app; Portugal grant; Romania UM; Romania app; Romania granted; Russia UM; Russia app; Russia grant; Saudi Arabia app; Saudi Arabia grant; Serbia UM; Serbia app; Serbia grant; Singapore app; Singapore grant; Slovakia UM; Slovakia app; Slovakia grant; Slovenia UM; Slovenia grant; South Africa app; Spain UM; Spain app; Spain grant; Sweden app; Sweden grant; Switzerland app; Switzerland grant; Taiwan UM; Taiwan appl; Taiwan grant; Thailand UM; Thailand app; Thailand grant; Tunisia app; Turkey UM; Turkey app; UK app; UK grant; USSR grant; Ukraine UM; Ukraine grant; United States app; United States grant; Uruguay app; Vietnam UM; Vietnam app; Vietnam grant; Other

Table of Contents

-
1. US11973181B2 Multilayer nanoporous separator
-

Family 1/1

1 record(s) per family

Record 1/1 US11973181B2 Multilayer nanoporous separator

Title: Multilayer nanoporous separator

Title - DWPI: Separator used for lithium battery, comprises porous polymeric layer and nanoporous layer which comprises boehmite particles having preset average crystallite size, and polymers

Priority Number: US62572083P | WO2018US55862A | US16755294A | US17223845A

Priority Number - DWPI: US2017572083P | WO2018US55862A | US16755294A | US16755294A | US2021223845A | US2021543259A | AU2023200585A | US2024631783A | AU2024205059A | EP2024199383A | US202521913A | US202519022365A | US202519023905A | US2025299871A | US202619453852A

Priority Date: 2017-10-13 | 2018-10-15 | 2020-04-10 | 2021-04-06

Priority Date - DWPI: 2017-10-13 | 2018-10-15 | 2020-04-10 | 2020-04-10 | 2021-04-06 | 2021-12-06 | 2023-02-03 | 2024-04-10 | 2024-07-24 | 2024-09-10 | 2025-01-15 | 2025-01-15 | 2025-01-16 | 2025-08-14 | 2026-01-20

Application Number: US17543259A

Application Date: 2021-12-06

Publication Date: 2024-04-30

IPC Class Table:

IPC	Section	Class	Subclass	Class Group	Subgroup
H01M0050446	H	H01	H01M	H01M0050	H01M0050446
H01M0010052	H	H01	H01M	H01M0010	H01M0010052
H01M00100525	H	H01	H01M	H01M0010	H01M00100525
H01M0050417	H	H01	H01M	H01M0050	H01M0050417
H01M0050451	H	H01	H01M	H01M0050	H01M0050451
H01M0050457	H	H01	H01M	H01M0050	H01M0050457
H01M0050489	H	H01	H01M	H01M0050	H01M0050489
H01M0050491	H	H01	H01M	H01M0050	H01M0050491

IPC Class Table - DWPI:

IPC - DWPI	Section - DWPI	Class - DWPI	Subclass - DWPI	Class Group - DWPI	Subgroup - DWPI
H01M0010052	H	H01	H01M	H01M0010	H01M0010052
H01M00100525	H	H01	H01M	H01M0010	H01M00100525
H01M0050403	H	H01	H01M	H01M0050	H01M0050403
H01M0050406	H	H01	H01M	H01M0050	H01M0050406
H01M0050409	H	H01	H01M	H01M0050	H01M0050409
H01M0050411	H	H01	H01M	H01M0050	H01M0050411
H01M0050414	H	H01	H01M	H01M0050	H01M0050414
H01M0050417	H	H01	H01M	H01M0050	H01M0050417
H01M0050429	H	H01	H01M	H01M0050	H01M0050429
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H01M0050443	H	H01	H01M	H01M0050	H01M0050443
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H01M0050497	H	H01	H01M	H01M0050	H01M0050497

Assignee/Applicant: LG ENERGY SOLUTION LTD.,Seoul,KR

Assignee - Current US: LG ENERGY SOLUTION LTD.

JP F Terms:

JP FI Codes:

Item Number: 1

Assignee - Original: LG ENERGY SOLUTION LTD.

Any CPC Table:

Type	CPC - Current - Invention	CPC - Current - Additional	Version	Office
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Current	H01M 10/0525	-	20130101	EP; US
Current	H01M 10/052	-	20130101	CN
Current	H01M 50/403	-	20210101	CN
Current	H01M 50/434		20210101	CN
Current	H01M 50/497		20210101	CN
Current	H01M 50/417	-	20210101	CN; EP; KR; US
Current	H01M 50/446		20210101	CN; EP; KR; US
Current	H01M 50/431	-	20210101	KR
Current	H01M 50/443		20210101	KR
Current	H01M 50/451	-	20210101	CN; EP; US
Current	H01M 50/489		20210101	CN; EP; US
Current	H01M 50/457	-	20210101	EP; KR; US
Current	H01M 50/491		20210101	EP; KR; US
Current	-	Y02E 60/10	20130101	EP; KR
Original	H01M 10/0525	-	20130101	US
Original	H01M 50/417	-	20210101	US
Original	H01M 50/446		20210101	US
Original	H01M 50/451		20210101	US
Original	H01M 50/457		20210101	US
Original	H01M 50/489		20210101	US
Original	H01M 50/491		20210101	US

ECLA:

Abstract:

A separator for a lithium battery having (a) a porous polymeric layer, such as a polyethylene layer; and (b) a nanoporous inorganic particle/polymer layer on both sides of the polymeric layer, the nanoporous layer having an inorganic oxide and one or more polymers; the volume fraction of the polymers in the nanoporous layer is about 15% to about 50%, and the crystallite size of the inorganic oxide is 5 nm to 90 nm.

Language of Publication: EN

INPADOC Legal Status Table:

Gazette Date	Code	INPADOC Legal Status Impact
2024-04-10	STCF	-
Description: INFORMATION ON STATUS: PATENT GRANT PATENTED CASE		
2024-03-27	STPP	-

Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL PUBLICATIONS -- ISSUE FEE PAYMENT VERIFIED		
2023-12-28	ZAAB	-
Description: NOTICE OF ALLOWANCE MAILED ORIGINAL CODE: MN/=.		
2023-12-27	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL NOTICE OF ALLOWANCE MAILED -- APPLICATION RECEIVED IN OFFICE OF PUBLICATIONS		
2023-10-17	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL RESPONSE TO NON-FINAL OFFICE ACTION ENTERED AND FORWARDED TO EXAMINER		
2023-06-15	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL NON FINAL ACTION MAILED		
2022-05-24	AS	-
Description: ASSIGNMENT LG ENERGY SOLUTION, LTD., KOREA, REPUBLIC OF NUNC PRO TUNC ASSIGNMENT; ASSIGNOR:OPTODOT CORPORATION; REEL/FRAME:060000/0089 2022-05-23		
2021-12-15	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL DOCKETED NEW CASE - READY FOR EXAMINATION		
2021-12-06	FEPP	-
Description: FEE PAYMENT PROCEDURE ENTITY STATUS SET TO UNDISCOUNTED (ORIGINAL EVENT CODE: BIG.); ENTITY STATUS OF PATENT OWNER: LARGE ENTITY		

Post-Issuance (US):

Reassignment (US) Table:

Assignee	Assignor	Date Signed	Reel/Frame	Date
LG ENERGY SOLUTION LTD.,SEOUL,KR	OPTODOT CORPORATION	2022-05-23	060000/0089	2022-05-24
Conveyance: NUNC PRO TUNC ASSIGNMENT				

Corresponent: AMSTER, ROTHSTEIN & EBENSTEIN LLP 90 PARK AVENUE NEW YORK, NY 10016

Maintenance Status (US):

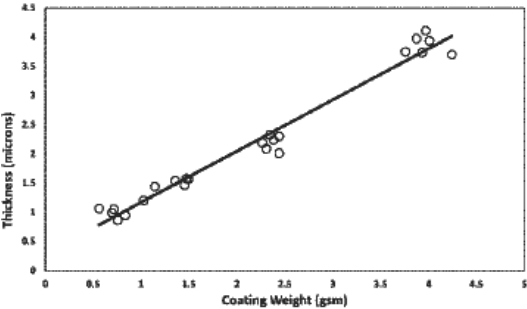
Litigation (US):

Opposition (EP):

License (EP):

EPO Procedural Status:

Front Page Drawing:





United States Patent and Trademark Office

Office of the Commissioner for Patents

MULTILAYER NANOPOROUS SEPARATOR

PATENT #

11973181

APPLICATION #

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12/06/2021

ISSUE DATE

04/30/2024

Payment Window Status

WINDOW		STATUS		FEES	
3.5 Year		Not Open		Not Due	
Window	First Day to Pay	Surcharge Starts	Last Day to Pay	Status	Fees
3.5 Year	04/30/2027	11/02/2027	05/01/2028	Not Open	Not Due
7.5 Year	04/30/2031	10/31/2031	04/30/2032	Not Open	Not Due
11.5 Year	04/30/2035	10/31/2035	04/30/2036	Not Open	Not Due

No maintenance fees are due at this time. 3.5 year window opens on 04/30/2027.

Patent Holder Information

Customer #

1912

Entity Status

UNDISCOUNTED

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